

# Bell-Scale Growth of OEIS A088714

A rigorous  $n$ th-root asymptotic, quantitative bounds,  
and a separately identified finer conjecture

Part II: golden-ratio moment laws for A088714 and A088713

Part III: local estimates for the finer Bell comparison

Part IV: densities, the correction profile, and the companion sequence

Part V: the Bell normalization of A088714 and A088713

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Part I: September 20, 2026   Part II: October 1, 2026   Part III: October 2, 2026  
Parts IV and V: October 4, 2026 (UTC)

## Abstract

Let  $A(x) = \sum_{n \geq 0} a_n x^n$  be the unique formal power series satisfying  $A(x) = 1 + xA(x)^2A(xA(x))$ . Its coefficients form OEIS A088714. We prove

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right), \quad a_n^{1/n} \sim \frac{n}{e \log n}.$$

The proof is based on an exact triangular recurrence obtained from composition iterates. A simple path through that recurrence supplies the lower bound. The upper bound comes from a coefficientwise inequality for shifted Poisson moments, proved here by a positivity argument involving Stirling numbers. More explicitly, for every  $n \geq 1$ ,

$$\max_{1 \leq h \leq n+1} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda),$$

where  $T_n$  is the  $n$ th Touchard polynomial. Consequently the ordinary generating function has radius zero, whereas the exponential generating function is entire. Exact coefficient computations accompany the proof. They suggest a considerably sharper Bell-number comparison, but that comparison and its multiplicative constant are not claimed as established results.

*Part II* (added October 1, 2026) realizes A088714 and its companion A088713 as the moment sequences of unique probability measures  $\mu, \nu$  on  $[0, \infty)$ , proves strict positivity of all Hankel minors, exact Stieltjes-transform equations, golden-ratio endpoint laws  $F(s) \sim s^{-(3-\sqrt{5})/2}$  and  $G(s) \sim s^{-(\sqrt{5}-1)/2}$  with leading constant one, the ratio laws  $a_{n+1}/a_n \sim c_{n+1}/c_n \sim n/\log n$ , and logarithmic far-tail laws. It uses the theorem of Part I and does not settle the finer conjecture.

*Part III* (added October 2, 2026) proves Part I's formal near-endpoint coefficient expansion as a theorem, uniformly in a logarithmic window, and combines it with Part II's log-convexity to prove the unconditional one-sided correction  $\liminf_{n \rightarrow \infty} (a_n/a_{n-1} - n/W(n)) \geq 3/2$ ; the finer conjecture needs the correction 2. An exact Poisson-weighted endpoint identity reduces the conjecture to a single scalar estimate, or to an averaged local-curvature estimate together with a complementary-endpoint bound, and a tilted-variance estimate is shown to suffice for the former. Exact computations through  $n = 1600$  test each missing estimate. Part III leaves the finer conjecture open.

*Part IV* (added October 4, 2026) proves that both measures have strictly positive real-analytic densities on  $(0, \infty)$  and supports  $[0, \infty)$ , with density and derivative equivalents at zero; that the correction to  $F(s) \sim s^{-(3-\sqrt{5})/2}$  is governed by a nonzero real-analytic antiperiodic profile in  $\log_\varphi \log s$ ,  $\varphi = (1 + \sqrt{5})/2$ , with a convergent expansion in powers of  $1/\log s$ ; and that, with  $\gamma_j = [z^j](1 - zA(z))^{-2}$ ,  $c_n = \sum_{j \leq K} \gamma_j a_{n-1-j} + O(a_{n-K-2})$  for every fixed  $K$ , so that  $c_n \sim a_{n-1}$ .

*Part V* (added October 4, 2026) proves the complementary endpoint estimate that Part III left open, to every fixed order, and with it the finer conjecture:  $a_n = C_* \mathcal{N}_n (1 + O(W(n)^5/n))$  with  $0 < C_* < \infty$ , so  $a_n \sim C_* \mathcal{B}_n \exp(W(n)^2 + 3W(n))$ , and  $c_n \sim C_* (W(n)/n) \mathcal{N}_n$  with the same constant. The value of  $C_*$  is not determined. This proof has not yet been independently reviewed.

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## Part I

# Bell-scale growth

## 1 The problem and the precise result

The OEIS entry A088714 defines the sequence through the formal functional equation [1]

$$A(x) = 1 + xA(x)^2A(xA(x)), \quad A(0) = 1. \quad (1.1)$$

Its beginning is

$$1, 1, 3, 13, 69, 419, 2809, 20353, 157199, 1281993, 10963825, \dots$$

The index convention throughout is  $a_0 = 1$ ,  $a_1 = 1$ ,  $a_2 = 3$ . All logarithms are natural. Formal composition is legitimate because the series substituted into  $A$  has zero constant term.

The composition in (1.1) makes this a different problem from ordinary algebraic generating-function asymptotics. In fact, the answer rules out a positive radius of convergence. One must not assume that the formal identity initially defines a holomorphic function near the origin.

**Theorem 1.1** (Main asymptotic). *For the coefficients of (1.1),*

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (1.2)$$

*Equivalently, at the level of  $n$ th-root asymptotics,*

$$a_n^{1/n} = \frac{n}{e \log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right), \quad \lim_{n \rightarrow \infty} \frac{e \log n}{n} a_n^{1/n} = 1. \quad (1.3)$$

*In particular,*

$$a_n = \left(\frac{n}{e \log n}\right)^n \exp\left(O\left(\frac{n \log \log n}{\log n}\right)\right). \quad (1.4)$$

The  $-n$  in (1.2) is important: it identifies the constant  $1/e$  in (1.3). The weaker assertion  $\log a_n \sim n \log n$  would miss both the logarithmic denominator and this constant.

**What the theorem does not say.** Equation (1.4) is not the multiplicative equivalent  $a_n \sim (n/(e \log n))^n$ . The error in its logarithm is  $o(n)$ , but it need not tend to zero. Likewise, an  $n$ th-root equivalent does not by itself imply an equivalent for  $a_n/a_{n-1}$ . (2026-10-01: Part II proves  $a_{n+1}/a_n \sim n/\log n$  through the Stieltjes-moment representation and log-convexity, Theorem 12.3.) Section 8 discusses a finer numerical pattern without confusing it with the theorem.

A useful feature of the argument is that it gives bounds for every index, not merely an eventual limiting statement.

**Theorem 1.2** (Finite-index comparison). *Define the Touchard polynomials by*

$$T_n(\lambda) = \sum_{j=0}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \lambda^j, \quad (1.5)$$

where  $\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}$  denotes a Stirling number of the second kind, and let  $T'_n(\lambda)$  mean differentiation with respect to  $\lambda$ . For every integer  $n \geq 1$ ,

$$\max_{h \in \{1, \dots, n+1\}} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (1.6)$$

For every  $r > 0$  and  $\lambda > 0$  this implies the entirely explicit bound

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n \frac{n!}{r^n} (e^r - 1) \exp(\lambda(e^r - 1)). \quad (1.7)$$

The remainder of the proof is self-contained. The OEIS supplies the sequence identification and related formal identities. Standard Bell number and Lambert  $W$  information is used for interpretation and numerical comparison, not to assume the answer for A088714.

## 2 Formal identities and the exact coefficient recurrence

### 2.1 Existence, uniqueness, and direct extraction

In (1.1), the coefficient of  $x^n$  on the right depends only on  $a_0, \dots, a_{n-1}$ . Starting with  $a_0 = 1$  therefore determines the coefficients successively and proves existence and uniqueness in  $\mathbb{Z}[[x]]$ . The same observation proves their nonnegativity. Expanding the outer occurrence of  $A$  gives the exact identity

$$a_n = \sum_{k=0}^{n-1} a_k [x^{n-1-k}] A(x)^{k+2}, \quad n \geq 1. \quad (2.1)$$

This recurrence is useful for independently checking computed coefficients. It also explains the finer heuristic in Section 8, but it is not the most convenient recurrence for our rigorous upper bound.

### 2.2 Passage to composition iterates

Put  $B(x) = xA(x)$ . Equation (1.1) becomes

$$B(x) = x + B(x)B(B(x)). \quad (2.2)$$

Since  $B(x) = x + O(x^2)$ , it has a formal compositional inverse. Substitution of that inverse into (2.2) yields

$$B^{(-1)}(x) = x - xB(x) = x - x^2A(x). \quad (2.3)$$

This is the reversion identity also recorded in the OEIS entry [1].

For each integer  $k \geq 0$ , define

$$F_k(x) = \frac{B^{\circ k}(x)}{x}, \quad F_0(x) = 1, \quad F_1(x) = A(x). \quad (2.4)$$

Here  $B^{\circ 0}(x) = x$  and  $B^{\circ k}$  denotes  $k$ -fold composition, not an ordinary power. Composing (2.2) with  $B^{\circ(k-1)}$  and dividing by  $x$  gives

$$F_k(x) = F_{k-1}(x) + xF_k(x)F_{k+1}(x), \quad k \geq 1. \quad (2.5)$$

The first few equations are the infinite system displayed in the OEIS entry; the derivation here fixes the indexing and proves the general identity [1].

Write

$$F_k(x) = \sum_{n \geq 0} c_{n,k} x^n.$$

Then

$$c_{0,k} = 1 \quad (k \geq 0), \quad c_{n,0} = 0 \quad (n \geq 1), \quad a_n = c_{n,1}. \quad (2.6)$$

Coefficient extraction in (2.5) gives

$$c_{n,k} - c_{n,k-1} = \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1}. \quad (2.7)$$

Equivalently,

$$c_{n,k} = \sum_{j=1}^k \sum_{i=0}^{n-1} c_{i,j} c_{n-1-i,j+1}. \quad (2.8)$$

Every term on the right has first index less than  $n$ . Consequently this is a genuine triangular recurrence despite the occurrence of  $k+1$ . It proves positivity and monotonicity in  $k$ , and is the main structural tool of the article.

For illustration, the first three coefficient polynomials are

$$\begin{aligned} c_{1,k} &= k, \\ c_{2,k} &= k^2 + 2k, \\ c_{3,k} &= k^3 + 5k^2 + 7k. \end{aligned}$$

Evaluating at  $k=1$  gives 1, 3, 13 as required. These are not ordinary Touchard polynomials; the Poisson polynomials introduced later are comparison objects, not an exact formula for the sequence.

### 2.3 Relation to the signed reversion sequence

For completeness, set

$$g(x) = -B^{(-1)}(-x) = x + x^2 A(-x).$$

The inverse identity implies

$$g(g(x)) = (1+x)g(x).$$

For  $n \geq 2$  its coefficient of  $x^n$  is  $(-1)^{n-2} a_{n-2}$ . Thus the signed series in A067145 has the same coefficient magnitudes, with this explicit index shift [2]. This identity is an additional verification route, not an analytic-convergence argument.

## 3 A lower bound from an increasing-index path

**Lemma 3.1.** *For every  $n \geq 0$  and every integer  $k \geq 1$ ,*

$$c_{n,k} \geq k^n. \quad (3.1)$$

*Proof.* The case  $n = 0$  is immediate. Suppose the statement holds for smaller first indices. By (2.8),

$$\begin{aligned} c_{n,k} &\geq \sum_{j=1}^k \sum_{i=0}^{n-1} j^i (j+1)^{n-1-i} \\ &= \sum_{j=1}^k ((j+1)^n - j^n) = (k+1)^n - 1 \geq k^n. \end{aligned}$$

The inner sum is a difference of two  $n$ th powers, and the outer sum then telescopes. This completes the induction.  $\square$

The  $i = 0$  summand of (2.7) and positivity imply

$$c_{n,k} \geq c_{n-1,k+1} \quad (n \geq 1).$$

Iterating this inequality  $h - 1$  times, for  $1 \leq h \leq n + 1$ , gives

$$a_n = c_{n,1} \geq c_{n-h+1,h} \geq h^{n-h+1}. \quad (3.2)$$

This proves the lower half of Theorem 1.2.

### 3.1 Optimizing the lower bound

Let  $N = n + 1$  and temporarily allow  $h$  to be real. The function

$$\phi_N(h) = (N - h) \log h$$

has derivatives

$$\phi'_N(h) = \frac{N}{h} - 1 - \log h, \quad \phi''_N(h) = -\frac{N}{h^2} - \frac{1}{h} < 0.$$

Its unique maximum is attained at

$$h_*(1 + \log h_*) = N, \quad h_* = \frac{N}{W(eN)}. \quad (3.3)$$

The positive real branch of Lambert  $W$ , characterized by  $W(z)e^{W(z)} = z$ , is intended [4]. The exact maximizing integer is among  $\lfloor h_* \rfloor$  and  $\lceil h_* \rceil$ . If  $w_* = W(eN)$ , the continuous maximum is

$$\phi_N(h_*) = N \left( w_* - 2 + \frac{1}{w_*} \right). \quad (3.4)$$

Rounding to a nearest integer changes this value by  $O((\log N)^2/N)$ : on an interval of fixed length around  $h_*$ ,  $|\phi''_N(h)| = O((\log N)^2/N)$ , and  $\phi'_N(h_*) = 0$ . This provides an accurate, explicit lower bound even when the leading logarithmic expression is numerically crude.

For the main theorem, a simpler choice suffices. Put

$$L = \log n, \quad \ell = \log \log n, \quad h = \left\lfloor \frac{n}{L} \right\rfloor.$$

As  $n \rightarrow \infty$ ,

$$\log h = L - \ell + O(L/n), \quad h = n/L + O(1).$$

Hence (3.2) yields

$$\log a_n \geq n(L - \ell - 1) + \frac{n\ell}{L} - O(L). \quad (3.5)$$

In particular the lower bound has the same  $n \log n$ ,  $-n \log \log n$ , and  $-n$  terms as the claimed answer.

## 4 The Poisson-moment comparison inequality

The upper bound requires a comparison family that respects the increase from  $k$  to  $k + 1$  in (2.7). Ordinary exponential bounds are too small, while a rough factorial bound loses the logarithmic factor. Shifted Poisson moments provide the needed intermediate scale.

Let  $X$  have the Poisson distribution with mean  $\lambda > 0$ , and define

$$M_n(k) = \mathbb{E}(k + X)^n, \quad T_n(\lambda) = M_n(0). \quad (4.1)$$

Their exponential generating functions are

$$\sum_{n \geq 0} M_n(k) \frac{z^n}{n!} = \exp(kz + \lambda(e^z - 1)), \quad \sum_{n \geq 0} T_n(\lambda) \frac{z^n}{n!} = \exp(\lambda(e^z - 1)). \quad (4.2)$$

Expanding the last exponential proves (1.5); at  $\lambda = 1$  these polynomials give the Bell numbers. Standard Bell identities are recorded in [3]; the identities needed below also follow directly from (4.2).

For  $n \geq 1$  set

$$D_n(k) = M_n(k) - M_n(k - 1), \quad S_n(k) = \sum_{i=0}^{n-1} M_i(k) M_{n-1-i}(k + 1). \quad (4.3)$$

The central estimate is the following.

**Lemma 4.1** (Shifted Poisson convolution inequality). *For every  $n \geq 1$ ,  $k \geq 1$ , and  $\lambda > 0$ ,*

$$S_n(k) \leq \left(1 + \frac{2}{\lambda}\right) D_n(k). \quad (4.4)$$

*In fact, after multiplication by  $\lambda$ , the difference between the right and left sides is a polynomial with nonnegative coefficients in  $\lambda$  and  $k - 1$ .*

### 4.1 Reduction to the shift $k = 1$

Write  $u = k - 1$ . The binomial theorem gives

$$D_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} D_r(1). \quad (4.5)$$

There is a matching identity for the ordinary convolution:

$$S_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} S_r(1). \quad (4.6)$$

To verify it, expand both shifted moments in the definition of  $S_n(k)$ . For fixed nonnegative integers  $p, q$  with  $p + q = r - 1$ , the coefficient sum over the convolution index is

$$\sum_i \binom{i}{p} \binom{n-1-i}{q} = \binom{n}{p+q+1} = \binom{n}{r}.$$

This identity counts a subset of  $r$  elements of an  $n$ -element ordered set by the location of its  $(p + 1)$ st element. Thus it holds including all boundary cases. Equations (4.5) and (4.6) reduce the lemma to  $k = 1$ .

Differentiating (4.2) with respect to  $\lambda$  shows

$$D_n(1) = T'_n(\lambda). \quad (4.7)$$

Suppress the argument  $\lambda$  temporarily, write  $S_n = S_n(1)$ , and set

$$H_n = (\lambda + 2)T'_n - \lambda S_n. \quad (4.8)$$

We shall prove that  $H_n$  has nonnegative coefficients for every  $n \geq 1$ .

## 4.2 A recurrence for the difference polynomial

The moment identities imply

$$T_{n+1} = \lambda(T_n + T'_n), \quad (4.9)$$

$$M_{n+1}(k) = (k + \lambda)M_n(k) + \lambda M'_n(k). \quad (4.10)$$

Here and below a prime means differentiation in  $\lambda$ , not in  $k$ . Applying (4.10) to both factors in  $S_n$  gives

$$S_{n+1} = \frac{\lambda}{2}S'_n + \left(\lambda + \frac{3}{2}\right)S_n + \frac{1}{2}(M_n(1) + M_n(2)). \quad (4.11)$$

Indeed, the two shifted convolution sums add up to  $2S_{n+1} - M_n(1) - M_n(2)$ . Furthermore,

$$M_n(1) = T_n + T'_n, \quad M_n(2) = T_n + 2T'_n + T''_n.$$

Substitute these formulas and (4.9) into (4.8). A direct simplification gives the exact recurrence

$$H_{n+1} = \frac{\lambda}{2}H'_n + (\lambda + 1)H_n + K_n, \quad (4.12)$$

where

$$K_n = \frac{\lambda(\lambda + 1)}{2}T''_n - 2\lambda T'_n + 2T_n. \quad (4.13)$$

The initial difference polynomials are

$$\begin{aligned} H_1 &= 2, \\ H_2 &= 2 + 2\lambda, \\ H_3 &= 2 + 6\lambda + \lambda^2, \\ H_4 &= 2 + 14\lambda + 8\lambda^2. \end{aligned}$$

They all have nonnegative coefficients.

## 4.3 Why the forcing polynomial is nonnegative

From (1.5), the coefficient of  $\lambda^j$  in  $K_n$  is

$$[\lambda^j]K_n = \frac{(j-1)(j-4)}{2} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} + \frac{j(j+1)}{2} \left\{ \begin{matrix} n \\ j+1 \end{matrix} \right\}. \quad (4.14)$$

For  $n \geq 4$ , this is visibly nonnegative except possibly when  $j = 2$  or  $j = 3$ . Those cases require only

$$\left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} \leq 3 \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \quad (n \geq 3), \quad \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \leq 6 \left\{ \begin{matrix} n \\ 4 \end{matrix} \right\} \quad (n \geq 4). \quad (4.15)$$

Here is a common elementary proof. If a partition of  $n$  elements into  $r$  blocks has  $n > r$ , it has a nonsingleton block. Choose such a block by any fixed deterministic rule and split off a specified element of it. Remember the unordered pair of blocks created. Merging that pair recovers the original partition. This is an injection into partitions with  $r + 1$  blocks together with a choice of two of those blocks. Therefore

$$\left\{ \begin{matrix} n \\ r \end{matrix} \right\} \leq \binom{r+1}{2} \left\{ \begin{matrix} n \\ r+1 \end{matrix} \right\} \quad (n > r). \quad (4.16)$$

Taking  $r = 2, 3$  proves (4.15). Thus  $K_n$  has nonnegative coefficients for all  $n \geq 4$ .

Starting from  $H_4$ , recurrence (4.12) preserves coefficientwise nonnegativity: differentiation, multiplication by  $\lambda/2$  or by  $\lambda + 1$ , and addition of  $K_n$  all preserve it. Together with the three earlier base cases, this proves  $H_n \geq 0$  coefficientwise for all  $n \geq 1$ . By (4.8) and (4.7),

$$\lambda S_n(1) \leq (\lambda + 2)D_n(1).$$

Finally, (4.5)–(4.6) extend the result to all  $k \geq 1$ , proving Lemma 4.1.

## 5 The global upper bound and completion of the proof

### 5.1 Comparison with the triangular recurrence

Fix any  $\lambda > 0$  and write

$$\gamma = 1 + \frac{2}{\lambda}.$$

We claim that, simultaneously for all  $n \geq 0$  and  $k \geq 0$ ,

$$c_{n,k} \leq \gamma^n M_n(k). \quad (5.1)$$

For  $n = 0$ , both sides equal 1. Assume the bound for smaller first indices. Equations (2.8) and (4.4) imply, for  $n \geq 1$ ,

$$\begin{aligned} c_{n,k} &\leq \gamma^{n-1} \sum_{j=1}^k S_n(j) \\ &\leq \gamma^n \sum_{j=1}^k (M_n(j) - M_n(j-1)) \\ &= \gamma^n (M_n(k) - M_n(0)) \leq \gamma^n M_n(k). \end{aligned} \quad (5.2)$$

The boundary  $k = 0$  is immediate. This proves the induction.

Importantly, the penultimate expression is sharper than the induction claim. At  $k = 1$ , (4.7) yields

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (5.3)$$

Because  $\lambda$  was arbitrary, this proves the upper half of Theorem 1.2.

Differentiation of (4.2) gives

$$\sum_{n \geq 0} T'_n(\lambda) \frac{r^n}{n!} = (e^r - 1) \exp(\lambda(e^r - 1)).$$

All coefficients are nonnegative. Any one term is bounded by the whole sum when  $r > 0$ , and (1.7) follows. This argument is just a positive-coefficient bound; it does not require any analyticity of  $A(x)$ .

## 5.2 An explicit parameter choice

Let  $L = \log n$  and  $\ell = \log L$  as before. For all sufficiently large  $n$ , take

$$\lambda = L, \quad r = L - 2\ell > 0. \quad (5.4)$$

Then  $e^r = n/L^2$ . Taking logarithms in (1.7), and using  $\log(e^r - 1) \leq r$ , gives

$$\begin{aligned} \log a_n &\leq \log n! - n \log r + n \log(1 + 2/L) + L(e^r - 1) + r \\ &= \log n! - n \log(L - 2\ell) + n \log(1 + 2/L) + \frac{n}{L} - 2\ell. \end{aligned} \quad (5.5)$$

The elementary estimates

$$\begin{aligned} \log n! &= nL - n + O(L), \\ \log(L - 2\ell) &= \ell - \frac{2\ell}{L} + O\left(\frac{\ell^2}{L^2}\right), \\ \log(1 + 2/L) &= \frac{2}{L} + O(L^{-2}) \end{aligned}$$

now show

$$\log a_n \leq n(L - \ell - 1) + \frac{2n\ell + 3n}{L} + O\left(\frac{n\ell^2}{L^2} + L\right). \quad (5.6)$$

For the first estimate, even Stirling's full formula is unnecessary: bounding the sum  $\sum_{j=1}^n \log j$  between its neighboring integrals already gives  $\log n! = n \log n - n + O(\log n)$ .

The lower estimate (3.5) and upper estimate (5.6) together give (1.2). Dividing by  $n$  and exponentiating proves (1.3) and (1.4). This completes the proof of Theorem 1.1.

*Remark 5.1* (Why a tunable comparison is necessary). A bound of the form  $a_n \leq C^n n!$  cannot identify the factor  $(\log n)^{-n}$ . Even a comparison with  $C^n$  times a fixed Bell-type sequence leaves an unwanted constant  $C$  in the  $n$ th root. Here the factor  $1 + 2/\lambda$  can approach 1, and the valid choice  $\lambda = \log n$  makes its total logarithmic cost only  $O(n/\log n)$ . The dependence of the Poisson mean on  $n$  is fully allowed because (5.3) holds for every positive  $\lambda$  at every index.

## 6 Consequences and comparison with Bell numbers

### 6.1 Superexponential but subfactorial growth

The main theorem immediately implies

$$\frac{a_n}{C^n} \longrightarrow \infty \quad \text{for every fixed } C > 0. \quad (6.1)$$

On the other hand,

$$\left(\frac{a_n}{n!}\right)^{1/n} = \frac{1}{\log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right) \longrightarrow 0. \quad (6.2)$$

Thus  $a_n/(C^n n!) \rightarrow 0$  for every fixed  $C > 0$ . These two assertions locate the growth more precisely than the phrase “almost factorial”: there is a factor of approximately  $(\log n)^n$  between  $n!$  and  $a_n$  on the logarithmic scale.

For every fixed  $s < 1$  and  $C > 0$ ,

$$\frac{a_n}{C^n (n!)^s} \longrightarrow \infty,$$

because the logarithm of this quotient has dominant term  $(1-s)n \log n$ . In the convention that Gevrey order  $s$  means a bound by  $CK^n(n!)^s$ , the series has exact Gevrey order 1, with zero factorial exponential type. This statement specifies the convention, as Gevrey indexing is not uniform across the literature.

## 6.2 The two generating functions behave differently

By the Cauchy–Hadamard formula,

$$\text{rad } A = \frac{1}{\limsup_{n \geq 0} a_n^{1/n}} = 0.$$

In particular, there is no holomorphic germ at zero whose Taylor series has these coefficients and satisfies (1.1). The formal series itself remains perfectly well-defined and unique.

In contrast, the exponential generating function

$$\mathcal{A}(t) = \sum_{n \geq 0} a_n \frac{t^n}{n!} \quad (6.3)$$

is entire by (6.2). Summing the finite-index upper bound also gives, for every  $t \geq 0$  and  $\lambda > 0$ ,

$$\mathcal{A}(t) \leq 1 + (e^{\gamma t} - 1) \exp(\lambda(e^{\gamma t} - 1)), \quad \gamma = 1 + 2/\lambda. \quad (6.4)$$

**Corollary 6.1** (Double-exponential scale). *Along the positive real axis,*

$$\lim_{t \rightarrow \infty} \frac{\log \log \mathcal{A}(t)}{t} = 1. \quad (6.5)$$

*Proof.* For a fixed  $\lambda$ , (6.4) bounds the limit superior by  $1 + 2/\lambda$ . Letting  $\lambda \rightarrow \infty$  gives the upper bound 1. For the lower bound fix  $0 < \varepsilon < 1$  and take  $n = \lfloor e^{(1-\varepsilon)t} \rfloor$ . The single term  $a_n t^n / n!$  and Theorem 1.1 give

$$\log \left( \frac{a_n t^n}{n!} \right) = n(\log t - \log \log n + o(1)) = n(-\log(1 - \varepsilon) + o(1)).$$

The constant inside the last parentheses is positive. Taking a further logarithm shows that the limit inferior in (6.5) is at least  $1 - \varepsilon$ . Let  $\varepsilon \downarrow 0$ .  $\square$

Since the coefficients of  $\mathcal{A}$  are nonnegative, its maximum modulus on  $|z| = t$  equals  $\mathcal{A}(t)$ . Consequently its usual entire function order is infinite. Equation (6.5), rather than just “infinite order,” states the useful growth scale.

## 6.3 Bell-number scale is not Bell-number equivalence

Let  $B_n = T_n(1)$  denote the Bell numbers. A standard asymptotic, with  $w = W(n)$ , is [3]

$$B_n \sim \frac{\exp(n(w - 1 + 1/w) - 1)}{\sqrt{1 + w}}. \quad (6.6)$$

In particular  $B_n^{1/n} \sim n/(e \log n)$ . Therefore our theorem implies

$$\left( \frac{a_n}{B_n} \right)^{1/n} \longrightarrow 1, \quad a_n = B_n \exp(o(n)). \quad (6.7)$$

These are the senses in which the sequence has Bell-number-scale growth. They do not assert that  $a_n/B_n$  is bounded or converges. Indeed, the finite computations in the next section show a substantial and growing discrepancy. The formula (6.6) is a comparison formula for Bell numbers, not a formula silently transferred to A088714.

## 7 Exact computation and numerical diagnostics

### 7.1 A triangular dynamic program

To obtain  $a_0, \dots, a_N$ , store  $c_{0,k} = 1$  for  $0 \leq k \leq N + 1$ . For each  $n = 1, \dots, N$ , compute

$$c_{n,0} = 0, \quad c_{n,k} = c_{n,k-1} + \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1} \quad (1 \leq k \leq N - n + 1).$$

Then output  $a_n = c_{n,1}$ . The triangular cutoff is sufficient: every requested entry on the right was computed at an earlier first index. The number of products is exactly

$$\sum_{n=1}^N n(N - n + 1) = \frac{N(N + 1)(N + 2)}{6}.$$

Thus the algorithm uses  $O(N^3)$  arbitrary-precision arithmetic operations and  $O(N^2)$  integer cells. These are arithmetic counts, not unit-cost bit-complexity claims; the integers grow with  $N$ .

The archive contains both a dependency-free Python implementation and a faster C++ implementation using GMP. The supplied data were computed with exact integer arithmetic through  $n = 1200$ . The Python and C++ outputs agree through  $n = 400$ . Independent truncated-polynomial checks verify (1.1) and (2.3); the supplied verification script also checks the auxiliary polynomial recurrences and finite bounds on explicitly reported finite ranges.

### 7.2 What the diagnostics measure

Define

$$Q_n = \frac{e \log n}{n} a_n^{1/n}, \quad (7.1)$$

$$R_n = \frac{a_n}{a_{n-1}}, \quad (7.2)$$

$$D_n^{\text{num}} = \log \left( \frac{a_n}{B_n} \right) - W(n)^2 - 3W(n). \quad (7.3)$$

The notation  $D_n^{\text{num}}$  is distinct from the Poisson moment difference  $D_n(k)$  in Section 4. The main theorem proves  $Q_n \rightarrow 1$ . The quantities  $R_n$  and  $D_n^{\text{num}}$  examine finer behavior not established by that theorem. (2026-10-01: the first-order law  $R_n \sim n / \log n$  is now proved in Part II, Theorem 12.3;  $D_n^{\text{num}}$  and Conjecture 8.1 remain open.) [Batch 86, 4 October 2026.] Part V proves Conjecture 8.1 (Theorem 46.1), so that  $D_n^{\text{num}} \rightarrow \log C_*$ ; that proof has not yet been independently reviewed.

The relatively slow approach of  $Q_n$  to 1 is not surprising:  $\log \log n / \log n$  is still appreciable at these indices. Substituting only the first term  $(n / (e \log n))^n$  is therefore not a numerically accurate approximation to  $a_n$ , despite its correct  $n$ th-root scale. The optimized finite lower and upper bounds offer certified alternatives when actual error control is required.

The normalization in (7.3) is motivated next. Its numerical behavior is evidence for a conjecture, not evidence that upgrades Theorem 1.1 to a multiplicative equivalent. In particular no rigorous enclosure for a limiting constant is extracted from the table.

| $n$  | $\log_{10} a_n$ | $Q_n$    | $R_n - n/W(n)$ | $D_n^{\text{num}}$ |
|------|-----------------|----------|----------------|--------------------|
| 20   | 17.180354       | 2.942966 | 2.650843       | -3.494768          |
| 50   | 53.134060       | 2.457011 | 2.312570       | -3.262262          |
| 100  | 123.688846      | 2.159875 | 2.205968       | -3.172516          |
| 200  | 286.269488      | 1.944240 | 2.146342       | -3.115335          |
| 400  | 657.374892      | 1.791450 | 2.110948       | -3.078375          |
| 800  | 1495.731414     | 1.682461 | 2.089104       | -3.054662          |
| 1200 | 2408.704723     | 1.633120 | 2.080197       | -3.045054          |

Table 1: Diagnostics from independently generated exact coefficients. Only the limit  $Q_n \rightarrow 1$  is proved here. The final column tests the conjecture of Section 8.

## 8 A finer asymptotic: formal motivation and its missing step

### 8.1 The leading endpoint balance

Write  $R_n = a_n/a_{n-1}$  and consider (2.1). Near its outer endpoint put  $k = n - 1 - m$ . Then the corresponding summand is

$$a_{n-1-m}[x^m]A(x)^{n+1-m}.$$

Since  $a_1 = 1$ , the first approximation to the coefficient is  $n^m/m!$ . If neighboring ratios can be treated as locally constant, the endpoint sum divided by  $a_n$  is approximately

$$\frac{1}{R_n} \sum_{m \geq 0} \frac{(n/R_n)^m}{m!} = \frac{e^{n/R_n}}{R_n}.$$

Setting this expression equal to 1 leads to

$$R_n \log R_n \approx n, \quad R_n \approx \frac{n}{W(n)}.$$

This explains the natural Lambert  $W$  scale of the ratios. It is not a proof: local ratio regularity and uniform control of the other summands have been assumed in deriving the approximation.

### 8.2 The first correction and a candidate normalization

There is more information in the same calculation. From the first few coefficients,

$$\log A(x) = x + \frac{5}{2}x^2 + O(x^3).$$

Expanding the near-endpoint coefficient for  $m$  of logarithmic size gives the formal local approximation

$$[x^m]A(x)^{n+1-m} = \frac{n^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} + \dots \right). \quad (8.1)$$

The factor  $3/2$  combines the quadratic term of  $\log A$  with the replacement of  $n + 1 - m$  by  $n$ . [Batch 76, 2 October 2026.] This expansion is now proved as Lemma 28.1 of Part III, uniformly for  $0 \leq m \leq C \log n$  with the error  $O((1+m)^6/n^2)$  in the bracket, and without any hypothesis on the ratios.

Suppose, for this calculation, that the ratio has a smooth expansion near  $R_n = n/w$ , where  $w = W(n)$ . Put

$$t = n/R_n, \quad d = \frac{w}{1+w}.$$

The local variation of the ratios contributes  $m(m+1)d/(2n)$  to the bracket multiplying the Poisson kernel. Averaging  $m(m-1)$  and  $m(m+1)$  against that kernel replaces them by  $t^2$  and  $t^2 + 2t$ , respectively. The opposite endpoint, where one inner factor has almost all the degree, begins with  $2a_{n-1}$  from  $A(x)^2$ . At this formal order the equation becomes

$$1 \approx \frac{e^t}{R_n} \left( 1 + \frac{(3+d)t^2 + 2dt}{2n} \right) + \frac{2}{R_n}. \quad (8.2)$$

Writing  $R_n = n/w + \delta$  and expanding gives

$$\delta(1+w) \approx 2 + \frac{(3+d)w + 2d}{2},$$

and hence the candidate ratio approximation

$$R_n \approx \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2}. \quad (8.3)$$

This is consistent with the exact data, but the approximation signs in this subsection are intentional.

A normalization whose successive quotient has these same first terms is

$$\mathcal{N}_n = \frac{\exp(n(w-1+1/w) - 1 + w^2 + 3w)}{\sqrt{1+w}}, \quad w = W(n). \quad (8.4)$$

Indeed, using  $w' = w/(n(1+w))$  and Taylor expansion in the real variable  $n$  shows

$$\frac{\mathcal{N}_n}{\mathcal{N}_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^3}{n}\right). \quad (8.5)$$

One way to check the differentiation is to note that

$$\frac{d}{dn} [n(w-1+1/w)] = w.$$

The additional derivative from  $w^2 + 3w - \frac{1}{2} \log(1+w)$  supplies the constant and rational correction in (8.5).

**Conjecture 8.1** (Finer Bell comparison). *There exists a constant  $C > 0$  such that, with  $w = W(n)$ ,*

$$a_n \sim C B_n \exp(w^2 + 3w) \sim C \mathcal{N}_n. \quad (8.6)$$

The existence or numerical value of  $C$  is not established in this article. The data and the endpoint calculation motivate the displayed form, rather than merely an arbitrary fit to an exponential correction.

[Batch 86, 4 October 2026.] Part V proves this conjecture: Theorem 46.1 gives  $a_n = C_* \mathcal{N}_n (1 + O(W(n)^5/n))$  with  $0 < C_* < \infty$ , hence (8.6) with  $C = C_*$ . The proof comes from manuscript 02 of batch 86 and has not yet been independently reviewed; it does not identify  $C_*$ , and it does not prove the ratio estimate (8.7). Manuscript 01 of the same batch, printed as Part IV and written without knowledge of manuscript 02, still calls the conjecture open.

### 8.3 A precise sufficient condition for the conjecture

The unresolved analytic step can be stated cleanly.

**Proposition 8.2** (Conditional implication). *Suppose that for some fixed  $p \geq 0$  the actual ratios satisfy*

$$\frac{a_n}{a_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^p}{n}\right). \quad (8.7)$$

*Then Conjecture 8.1 follows with some finite, strictly positive constant  $C$ .*

*Proof.* Compare (8.7) with (8.5). Since both ratios are asymptotic to  $n/W(n)$ , their quotient is

$$1 + O\left(\frac{W(n)^{\max(p,3)+1}}{n^2}\right).$$

The error is summable because  $W(n) = O(\log n)$ . Therefore the successive differences of  $\log(a_n/\mathcal{N}_n)$  are absolutely summable. This logarithm converges to a finite real number, so  $a_n/\mathcal{N}_n$  tends to a finite positive constant. Formula (6.6) then gives the Bell-number version.  $\square$

The main theorem does not supply (8.7). In particular, it does not justify replacing all nearby coefficient ratios by a smooth function, nor does it control every non-endpoint contribution at a summable error level. A proof of such uniform endpoint estimates would be a genuine strengthening of the present result. This identifies the remaining task without presenting a formal asymptotic calculation as a completed proof.

[Batch 76, 2 October 2026.] Part III reduces the conjecture to the single estimate  $\mathcal{E}_n = O(W(n)^q/n^2)$  for the scalar defect (30.8) (Theorem 30.1), and proves the one-sided correction  $\liminf_{n \rightarrow \infty} (R_n - n/W(n)) \geq 3/2$  (Theorem 29.1), whereas (8.7) needs the correction  $2 + W(n)/(2(1+W(n))^2) \rightarrow 2$ . Estimate (8.7) and Conjecture 8.1 remain open.

[Batch 86, 4 October 2026.] Part V proves Conjecture 8.1 by a route that bypasses (8.7): it proves the complementary estimate (30.7) (Theorem 44.1) and the unconditional bound  $R_n = n/W(n) + O(1)$  (Lemma 46.2), and then sums a linearized Poisson equation (Section 46). That proof has not yet been independently reviewed. Estimate (8.7) remains open (Research question 49.2).

## 9 Reproducibility and verification scope

The archive is organized so that the mathematical theorem does not depend on a numerical black box. The principal files are `article.tex`, `article.pdf`, the exact data file `data/coefficients.txt`, and the scripts under `code/`.

The dependency-free command

```
python3 code/verify.py
```

checks the initial OEIS terms; verifies the defining composition and inverse identities by independent truncated-polynomial arithmetic; checks the Poisson difference-polynomial identity and coefficient signs on a finite range; and tests both sides of the finite-index bound using exact integer arithmetic. Its actual output is supplied in `data/verification.txt`. These are useful implementation audits; the all-index positivity proof is the argument in Section 4, not an extrapolation from those finite checks.

The command

```
python3 code/coefficients.py -n 400 -output coefficients.txt
```

recomputes the first 401 terms without third-party Python packages. For a larger table, `code/compute_gmp.cpp` provides the same recurrence in C++ with GMP. Instructions in `README.md` describe compilation and analysis. The diagnostic calculations use exact integers as input and arbitrary-precision floating-point logarithms only at the presentation stage; no computed coefficient is rounded.

No computer-assisted proof of the finer conjecture is claimed. The article's rigorous conclusion is Theorem 1.1, supported by the explicit all-index inequalities of Theorem 1.2.

## 10 Conclusion

OEIS A088714 has the precise  $n$ th-root growth

$$a_n^{1/n} \sim \frac{n}{e \log n}.$$

The accompanying logarithmic error estimate is

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right).$$

The mechanism is visible in the proof. Composition iterates turn the functional equation into a positive triangular recurrence. Moving along its increasing-index direction gives roughly  $h^{n-h}$ , whose optimum occurs at  $h$  of order  $n/\log n$ . Shifted Poisson moments furnish a compatible upper comparison and recover the same leading scale, including the factor  $1/e$ .

The result determines the growth at logarithmic precision  $o(n)$  and settles the convergence classification of both generating functions. The finer proposed factor  $\exp(W(n)^2 + 3W(n))$  remains explicitly separate from the proved theorem.

## Part II

# Golden-ratio moment laws for A088714 and A088713

## 11 Source, scope and notation of Part II

**Summary.** Let  $A(z) = \sum a_n z^n$  be the formal solution of  $A(z) = 1 + zA(z)^2 A(zA(z))$ , and let  $C(z) = (1 - zA(z))^{-1} = \sum c_n z^n$ . These are OEIS A088714 and A088713. Part II constructs unique probability measures  $\mu, \nu$  on  $[0, \infty)$  with moments  $a_n, c_n$ . The construction factors a nonlinear series transformation into two positivity-preserving operator constructions. It proves strict total positivity of both Hankel matrices, and strict log-convexity after any finite number of iterations of the log-convexity operator. Their Stieltjes transforms  $F(s)$  and  $G(s)$  satisfy exact identities  $F(s) = 1 - sF(s)^2 F(sF(s))$  and  $G(s) = (1 + sF(s))^{-1}$  for  $s > 0$ . Writing  $\alpha = (3 - \sqrt{5})/2$  and  $\beta = (\sqrt{5} - 1)/2$ , Part II proves  $F(s) \sim s^{-\alpha}$  and  $G(s) \sim s^{-\beta}$ , including the leading constants. Consequently the distribution functions at zero are asymptotic to  $\sin(\pi\alpha)t^\alpha/(\pi\alpha)$  and  $\sin(\pi\beta)t^\beta/(\pi\beta)$ , respectively. Compact algebraic approximants bracket  $F$  monotonically; their exponents are ratios of Fibonacci numbers. Combining the moment structure with the Bell-scale growth bound of Part I (Theorem 1.1) yields  $a_{n+1}/a_n \sim c_{n+1}/c_n \sim n/\log n$  and  $\log \mu([x, \infty)) \sim \log \nu([x, \infty)) \sim -x \log x$ . The package includes independent exact checks, outward-rounded orbit enclosures, and a research agenda. These are written mathematical proofs, not Lean-checked theorems; historical priority is not asserted as certified.

**Provenance.** Part II is manuscript 35 of batch 73 of the repository’s incoming-reports intake (cluster O2): *Golden-ratio moment laws for OEIS A088714 and A088713. Positive operators, strict Hankel positivity, ratio asymptotics, and two endpoint laws*, by OpenAI ChatGPT, prepared for Vladimir Reshetnikov and dated October 1, 2026 (a 1,524-line source and a 23-page PDF). It arrived as `golden_moment_laws.zip` in commit `f8c3a392a` and was placed in commit `6e193dd4f`. The manuscript pins the repository at commit `1f1981f682b2878bde51a6ad40c22777f362fc05` and identifies Part I by the Git blob `ccb50ef335ac4de27bd98a5a152fb48c978c7a96` of its source; that blob was still the source of this report at the placement commit, so the manuscript read the text now printed as Part I. Its code, data and source audit are shipped with the prefix `02-golden-moments-`; its own L<sup>A</sup>T<sub>E</sub>X source, README and PDF are not shipped and survive in the archive of the arrival commit. Part I was written independently, on September 20, 2026; the two Parts share no text.

**What is not reprinted.** Part I proves the Bell-scale growth theorem (Theorem 1.1, with the finite bounds of Theorem 1.2); the golden-moment manuscript re-proves it in an appendix that is not reprinted here. That appendix runs through Part I’s Sections 2–5 step for step: the triangular recurrence (2.8), the lower bound of Lemma 3.1, the Poisson comparison of Lemma 4.1 with the same polynomials  $H_n, K_n$  and the same initial values, the merge inequality (4.16), and the parameter choice (5.4). It states nothing that Part I does not, and the manuscript itself calls it “not claimed as new work”. The manuscript included it so that its deductions would not depend on accepting the earlier report; inside one report that purpose lapses, and every use of the growth bound below refers to Part I. Everything else in the manuscript is printed: all sections, all proofs, remarks and research questions, and its dependency and novelty audit (Appendix A).

**Editorial changes.** Section  $k$  of the manuscript is Section  $k + 11$  here, and its Appendix B is Appendix A; every label carries the prefix `gml:`. References to “the earlier ProveIt article” now point to Part I, and the manuscript’s link to a GitHub copy of Part I at the pin is replaced by an internal reference. The table fragments and file names in Section 22 are the shipped `02-golden-moments-`files. The manuscript’s bibliography entry for A088714 is merged with Part I’s. Nothing else in the mathematics was changed. Two dated notes in Part I (Section 8 and the paragraph after Theorem 1.1) record what Part II settles; Conjecture 8.1 is not settled by it (Section 23).

**Status.** Part II, like Part I, is AI-assisted and unrefereed, and it is not formalized: no Lean or Rocq development checks any statement of this report, and placement in this collection confers no formal status. Its novelty audit is bounded (Section 12.3 and Appendix A).

**Related repository material.** The positive inverse  $U^{-1}(t) = t(1 + U(t))$  of Theorem 12.1 and the determinacy argument through an entire exponential generating function belong to the circle of ideas of the transseries package *Moment Determinacy and Nonlinear Transseries* in the directory

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Analysis/Transseries/docs/series-and-transseries/
Moment_Determinacy_Nonlinear_Transseries/
```

which treats general positive Stieltjes inverses  $T(y) = yS(T(y))$ . That package does not mention A088714 or A088713, and Part II does not depend on it.

**Notation.** The two Parts were written independently and reuse letters. Every symbol of Part II is local to Part II; Table 2 lists each letter that also occurs in Part I with both meanings. No symbol of the manuscript was renamed.

## 12 The sequences, the contribution, and its scope

### 12.1 Formal definitions and the earlier result

Throughout,  $a_0 = c_0 = 1$ , logarithms are natural, and a power series without an explicitly specified analytic domain is a *formal* power series. The defining equations are

$$A(z) = 1 + zA(z)^2A(zA(z)), \quad (12.1)$$

$$C(z) = \frac{1}{1 - zA(z)}, \quad A(z) = C(zA(z)). \quad (12.2)$$

The second identity in (12.2) follows from (12.1); it is also recorded in the OEIS entries [1, 5]. Coefficients are determined successively because the right side of (12.1) at degree  $n \geq 1$  uses only coefficients of  $A$  of degrees below  $n$ . Thus there is exactly one solution with constant term 1; its coefficients are positive integers. The initial values are

| $n$   | 0 | 1 | 2 | 3  | 4  | 5   | 6    | 7     |
|-------|---|---|---|----|----|-----|------|-------|
| $a_n$ | 1 | 1 | 3 | 13 | 69 | 419 | 2809 | 20353 |
| $c_n$ | 1 | 1 | 2 | 6  | 24 | 118 | 674  | 4308. |

The notation  $C$  is used here for the generating function called  $A$  in OEIS A088713, whose defining identity is  $C(z/C(z)) = (1 - z)^{-1}$ . For verification, put  $B = zA$ . Equation (12.1) gives  $B^{(-1)}(z) = z - z^2A(z) = z/C(z)$ . Composing  $A = C \circ B$  with this inverse and substituting in (12.2) proves the identity for  $C$ .

Table 2: Letters used in more than one Part (the lower block was added with Part III; Parts IV and V list their letters in Tables 5 and 6).

| letter                                                                                       | Part I                                                                                                          | Part II                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $c, C$                                                                                       | $c_{n,k} = [x^n]F_k(x)$ , coefficients of the composition iterates (2.4); $C$ is the constant of Conjecture 8.1 | $c_n$ is A088713 and $C(z) = \sum c_n z^n = (1 - zA(z))^{-1}$                                                                                                                                                                                   |
| $F, G$                                                                                       | $F_k(x) = B^{\circ k}(x)/x$ , formal series; $F_1 = A$                                                          | $F(s), G(s), F_r(s), G_r(s)$ are Stieltjes transforms, functions of $s > 0$ ; $F(s)$ has the asymptotic expansion $\sum (-1)^j a_j s^j$ as $s \downarrow 0$ . Part II's $F_1(s) = 2/(1 + \sqrt{1 + 4s})$ is <i>not</i> Part I's $F_1(x) = A(x)$ |
| $B$                                                                                          | $B(x) = xA(x)$                                                                                                  | $B(z)$ is a moment series (Lemma 13.1); the iterates $B_r$ of (14.1) approximate $A$ , not $xA$                                                                                                                                                 |
| $T$                                                                                          | $T_n(\lambda)$ , Touchard polynomials                                                                           | $\mathcal{T}$ is the nonlinear transform (13.6); $T$ is a multiplication operator inside proofs                                                                                                                                                 |
| $D$                                                                                          | $D_n(k)$ , Poisson differences (4.3); $D_n^{\text{num}}$ (7.3)                                                  | $D_n^{(h)}(m)$ , Hankel determinants (Proposition 15.4)                                                                                                                                                                                         |
| $H, K, S, M$                                                                                 | polynomials $H_n, K_n$ in $\lambda$ ; shifted Poisson moments $M_n(k)$ and convolutions $S_n(k)$                | $H(s)$ and $K$ in Lemma 19.1; kernels $H_M, S_M$ ; moment series $M(z)$                                                                                                                                                                         |
| $R, r$                                                                                       | $R_n = a_n/a_{n-1}$ (7.2)                                                                                       | $R(z)$ is the renewal series (13.1); $r_n = m_{n+1}/m_n$ is a <i>forward</i> ratio, so Part I's $R_n$ is $r_{n-1}$ for $m = a$                                                                                                                  |
| $Q, X$                                                                                       | $Q_n$ (7.1); $X$ a Poisson variable                                                                             | $Q$ and $X$ are positive operators (Lemmas 13.1 and 13.2)                                                                                                                                                                                       |
| $\gamma, L, r$                                                                               | $\gamma = 1 + 2/\lambda$ ; $L = \log n$ and $r = L - 2\ell$ (5.4)                                               | $\gamma$ is a Tauberian exponent; $\mathcal{L}$ is the log-convexity operator; $r$ indexes the iterates $B_r, F_r$                                                                                                                              |
| <i>Added with Part III (Section 25): middle column Parts I and II, right column Part III</i> |                                                                                                                 |                                                                                                                                                                                                                                                 |
| $r, R$                                                                                       | Part I: $R_n = a_n/a_{n-1}$ . Part II: $r_n = a_{n+1}/a_n$ (forward), $R(z)$ the renewal series                 | $r_n = a_n/a_{n-1}$ , Part I's $R_n$ and Part II's $r_{n-1}$ ; $R(n) = n/w + 2 + w/(2(1+w)^2)$ is the candidate (8.3), <i>not</i> a ratio                                                                                                       |
| $w, t, d$                                                                                    | Part I, Section 8: $w = W(n)$ , $t = n/R_n$ , $d = w/(1+w)$ . Part II: $t$ is the variable of $\mu, \nu$        | as in Part I's Section 8: $w = W(n)$ , $t = n/r_n$ , $d = w/(1+w)$ ; $d\mu$ is still the measure differential                                                                                                                                   |
| $L$                                                                                          | $L = \log n$ (5.4)                                                                                              | $L = \lceil 16w \rceil$ , the endpoint window                                                                                                                                                                                                   |
| $c, C$                                                                                       | as above                                                                                                        | $C > 0$ bounds the window in Lemma 28.1; $C_1$ a constant; $C_{n,m} = m! b_{n,m}/n^m$ (30.3); $c(w) = 3/2 + 1/(2(1+w))$                                                                                                                         |
| $D$                                                                                          | as above; Part I's $D_n^{\text{num}}$ (7.3)                                                                     | $D_{n,m} = \prod_{j \leq m} r_n/r_{n-j}$ (30.2); $D_n^B$ is Part I's $D_n^{\text{num}}$ ; $D_n^N = \log(a_n/\mathcal{N}_n)$                                                                                                                     |
| $H, K, M$                                                                                    | as above                                                                                                        | $H_n(r)$ , a decreasing function of $r > 0$ (29.2); $K$ with $a_j \leq K^{j-1}j!$ ; $M = n + 1 - m$ , or the power in $A(z)^M$                                                                                                                  |
| $B, b, \mathcal{N}$                                                                          | Part I: $B(x) = xA(x)$ , Bell numbers $\mathbf{B}_n$ , normalizer $\mathcal{N}_n$ (8.4). Part II: $B(z), B_r$   | $b_{n,m} = [z^m]A(z)^{n+1-m}$ ; Bell numbers $\mathbf{B}_n$ (printed $B_n$ in the table fragments); $\mathcal{N}_n$ as in Part I (the manuscript's $N_n$ )                                                                                      |
| $X, Z, \mu_{n-1}$                                                                            | Part I: $X$ a Poisson variable. Part II: $X$ an operator; $\sigma_n$ the size-biased law                        | $Z \sim \text{Poisson}(t)$ ; $X$ the variable of the tilt $\mu_{n-1}$ , which is Part II's $\sigma_{n-1}$ for $\sigma = \mu$                                                                                                                    |
| $p, q, Q, s$                                                                                 | Part I: $p$ in (8.7), $Q_n$ (7.1). Part II: $Q$ an operator, $s > 0$ the transform variable                     | $p, q$ fixed exponents; $p_t(m)$ Poisson weights (30.1); $Q = \max(q, 6)$ ; $s$ is a degree count, an index, or $s_n = n/w + c(w)$ , and $F(s)$ as in Part II                                                                                   |
| $J, \mathcal{E}, \Delta, \eta$                                                               | not used                                                                                                        | $J_n$ (30.4); $\mathcal{E}_n$ (30.8); $\Delta_n = r_n - R(n)$ ; $\eta$ an excess degree (the manuscript's $e$ )                                                                                                                                 |

Part I of this report, *Bell-Scale Growth of OEIS A088714*, proves (Theorem 1.1)

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (12.3)$$

Part I explicitly separates this result from consecutive-coefficient asymptotics and from a finer multiplicative Bell-number conjecture. The golden-moment manuscript re-proves this bound, with the finite bounds of Theorem 1.2, in an appendix that is not reprinted here (Section 11). Part I's proof uses no numerical evidence, so the deductions below rest on written proofs only.

## 12.2 Main results

A normalized Stieltjes moment sequence is a sequence  $m_n = \int_{[0,\infty)} t^n d\sigma(t)$  for a probability measure  $\sigma$  with all moments finite. It is *determinate* when this measure is unique. Define

$$\alpha = \frac{3 - \sqrt{5}}{2}, \quad \beta = \frac{\sqrt{5} - 1}{2}, \quad \alpha + \beta = 1, \quad \beta^2 + \beta = 1. \quad (12.4)$$

**Theorem 12.1** (Moment and analytic structure). *There are unique probability measures  $\mu, \nu$  on  $[0, \infty)$  such that*

$$a_n = \int t^n d\mu(t), \quad c_n = \int t^n d\nu(t) \quad (n \geq 0). \quad (12.5)$$

*Both have all positive exponential moments finite. Their supports are unbounded, and all minors of their infinite Hankel matrices with strictly increasing row and column index lists are strictly positive. For  $(\mathcal{L}m)_n = m_n m_{n+2} - m_{n+1}^2$ , every  $\mathcal{L}^k a$  and  $\mathcal{L}^k c$ ,  $k \geq 1$ , is a Stieltjes moment sequence with strictly positive entries.*

For  $s > 0$ , set

$$F(s) = \int \frac{d\mu(t)}{1 + st}, \quad G(s) = \int \frac{d\nu(t)}{1 + st}. \quad (12.6)$$

These transforms satisfy

$$F(s) = 1 - sF(s)^2 F(sF(s)), \quad G(s) = \frac{1}{1 + sF(s)}. \quad (12.7)$$

Moreover,  $U(s) = sF(s)$  is an increasing homeomorphism of  $(0, \infty)$  onto itself and has inverse

$$U^{-1}(t) = t(1 + U(t)). \quad (12.8)$$

**Theorem 12.2** (Golden-ratio endpoint laws). *As  $s \rightarrow \infty$ ,*

$$F(s) \sim s^{-\alpha}, \quad G(s) \sim s^{-\beta}. \quad (12.9)$$

As  $t \downarrow 0$ ,

$$\mu([0, t]) \sim \frac{\sin(\pi\alpha)}{\pi\alpha} t^\alpha, \quad \nu([0, t]) \sim \frac{\sin(\pi\beta)}{\pi\beta} t^\beta. \quad (12.10)$$

*Neither measure has an atom at zero. For  $q > 0$ , its negative  $q$ th moment is finite precisely when  $q < \alpha$  for  $\mu$ , or  $q < \beta$  for  $\nu$ .*

**Theorem 12.3** (Consecutive ratios and the opposite tail). *The two sequences satisfy*

$$\frac{a_{n+1}}{a_n} \sim \frac{n}{\log n}, \quad \frac{c_{n+1}}{c_n} \sim \frac{n}{\log n}. \quad (12.11)$$

*Their representing measures satisfy*

$$\lim_{x \rightarrow \infty} \frac{\log \mu([x, \infty))}{x \log x} = \lim_{x \rightarrow \infty} \frac{\log \nu([x, \infty))}{x \log x} = -1. \quad (12.12)$$

These statements concern entire infinite families of inequalities and asymptotic regimes, rather than additional finite entries in an OEIS table. The two ends of the same measures are quite different: an irrational power law at zero and a logarithmically Poisson-scale tail at infinity.

### 12.3 Attribution and limitations

The sequence definitions and reversion identities are established OEIS material [1, 5]. Bound (12.3) is the theorem of Part I, not a new theorem of Part II. Fock-space models for noncrossing moments are standard [6]; so are the general implications from Stieltjes moments to infinite log-convexity [7] and the relevant Tauberian principle [8]. We give the particular proofs needed, including positivity of the operators and the Tauberian constant.

The contributions developed here are the application of these constructions to this nonlinear fixed point, its determinate positive-measure realization, the exact negative-axis equations and certified approximants, the constant-one golden-ratio laws, and the ratio and tail deductions. The selected repository report (Part I) and targeted searches did not reveal these specific conclusions. That is a bounded novelty audit, not an exhaustive priority determination. The new statements are unrefereed written proofs; no formal verification is claimed. We do not prove absolute continuity of either measure, and do not settle the earlier finer Bell-number conjecture.

## 13 Two operator constructions that preserve positivity

This section supplies the central existence mechanism. All operators here are bounded: input measures have compact support. Unbounded support is introduced only by taking weak limits in Section 14.

### 13.1 The renewal transformation

**Lemma 13.1** (A positive block operator). *Let  $\xi$  be a compactly supported probability measure on  $[0, \infty)$ , and write  $B(z) = \int (1 - zt)^{-1} d\xi(t)$  near zero. Then*

$$R(z) = \frac{1}{1 - zB(z)} \quad (13.1)$$

*is the moment series of a compactly supported probability measure on  $[0, \infty)$ .*

*Proof.* On  $K = \mathbb{C} \oplus L^2(\xi)$  let  $T$  be multiplication by  $t$  and put

$$Q = \begin{pmatrix} 1 & \langle \sqrt{t}, \cdot \rangle \\ \sqrt{t} & T \end{pmatrix}.$$

For  $(u, v) \in K$ , its quadratic form is

$$\langle (u, v), Q(u, v) \rangle = \int |u + \sqrt{t} v(t)|^2 d\xi(t) \geq 0.$$

The mass-one assumption on  $\xi$  is used in the coefficient of  $|u|^2$ . Thus  $Q$  is bounded, self-adjoint and positive. Its spectral measure in the unit vector  $e_0 = (1, 0)$  is a compactly supported probability measure on  $[0, \infty)$ . A Schur complement, initially for sufficiently small  $z$ , gives its moment series as

$$\langle e_0, (I - zQ)^{-1} e_0 \rangle = \frac{1}{1 - z - z^2 \int t(1 - zt)^{-1} d\xi(t)}$$

$$= \frac{1}{1 - zB(z)}.$$

The identity  $B(z) = 1 + z \int t(1 - zt)^{-1} d\xi(t)$  proves the last step.  $\square$

In coefficients, if  $B = \sum b_n z^n$  and  $R = \sum d_n z^n$ , then

$$d_0 = 1, \quad d_n = \sum_{j=0}^{n-1} b_j d_{n-1-j}. \quad (13.2)$$

Positivity of these coefficients alone would not prove the moment property; the positive operator is the stronger ingredient.

### 13.2 From a positive moment sequence to noncrossing moments

Let  $\text{NC}(n)$  denote the partitions of  $\{1, \dots, n\}$  with no two distinct blocks containing  $i < j < k < \ell$  with  $i, k$  in one block and  $j, \ell$  in the other. These are the noncrossing partitions. The empty partition has weight 1.

**Lemma 13.2** (A positive full-Fock operator). *Suppose  $d_n = \int t^n d\rho(t)$ , where  $\rho$  is a compactly supported probability measure on  $[0, \infty)$ . Put*

$$m_0 = 1, \quad m_n = \sum_{\pi \in \text{NC}(n)} \prod_{V \in \pi} d_{|V|}. \quad (13.3)$$

*Then  $(m_n)$  is also a compactly supported Stieltjes moment sequence. Writing  $R(z) = \sum d_n z^n$ , its moment series is the unique formal solution*

$$M(z) = R(zM(z)), \quad M(0) = 1. \quad (13.4)$$

*Proof.* Set  $H = L^2(\rho)$  and let  $T$  be multiplication by  $t$ . The full Fock space is

$$\mathcal{F}(H) = \mathbb{C}\Omega \oplus \bigoplus_{k \geq 1} H^{\otimes k}.$$

The vector  $\Omega$  is the vacuum. Define maps to  $H \otimes \mathcal{F}(H)$  by

$$D\Omega = 0, \quad D(h_1 \otimes \dots \otimes h_k) = h_1 \otimes (h_2 \otimes \dots \otimes h_k), \quad J\psi = 1 \otimes \psi.$$

The empty tail means  $\Omega$ . Both maps have norm 1, and

$$X = (D + J)^*(T \otimes I)(D + J) \geq 0. \quad (13.5)$$

It is bounded. Expanding gives four operations: multiplication by  $T$  on the first tensor factor of a nonempty word; creation of a first factor  $t$ ; annihilation by the linear functional  $\langle t, \cdot \rangle$ ; and  $d_1 I$ . The gauge operation vanishes on  $\Omega$ .

We verify the vacuum moments, not merely positivity. Expand  $X^n$  into words in these four operations and apply them from right to left. A term contributing to the vacuum never attempts to remove a factor from an empty word and ends at tensor length zero. Match each created tensor factor with its eventual annihilation, using the last-in, first-out order of first tensor factors. Gauges acting on that factor, together with its creation and annihilation, form a block. Scalar operations form singleton blocks. Operations acting on factors above it form nested blocks. This correspondence

is a bijection with noncrossing partitions; reading such a partition from one end reconstructs the operator choices and the stack of active factors.

A block of size  $r \geq 2$  has weight  $\langle t, T^{r-2}t \rangle = \int t^r d\rho(t) = d_r$ ; a singleton has weight  $d_1$ . Nested closed pieces leave the lower tensor factors untouched and contribute their own scalar weights. Thus  $\langle \Omega, X^n \Omega \rangle$  is exactly (13.3). The positive spectral measure of  $X$  at  $\Omega$  proves the assertion.

Finally, consider the block containing the first point of a noncrossing partition. If its size is  $r$ , its weight is  $d_r$ , and there are  $r$  independent intervening or final intervals, each carrying a noncrossing partition. Consequently

$$M(z) = 1 + \sum_{r \geq 1} d_r z^r M(z)^r = R(zM(z)).$$

The degree- $n$  equation determines  $m_n$  from earlier  $m_j$ , proving uniqueness.  $\square$

For example,  $m_1 = d_1$ ,  $m_2 = d_1^2 + d_2$ , and  $m_3 = d_1^3 + 3d_1d_2 + d_3$ . In free-probability terminology, this is the rate-one free compound Poisson construction with jump law  $\rho$ ; the operator proof above is sufficient without adopting that terminology.

### 13.3 The nonlinear transform

Combining the two lemmas gives the following general result.

**Theorem 13.3** (Moment-preserving nonlinear transform). *For a compact Stieltjes moment series  $B$  with  $B(0) = 1$ , the unique formal solution  $M = \mathcal{T}(B)$  of*

$$M(z) = 1 + zM(z)^2 B(zM(z)), \quad M(0) = 1, \quad (13.6)$$

*is again a compact Stieltjes moment series.*

*Proof.* Apply Lemma 13.1 to obtain  $R = (1 - zB)^{-1}$ , and then Lemma 13.2 to obtain  $M = R(zM)$ . The resulting identity is  $M = (1 - zMB(zM))^{-1}$ , which is equivalent to (13.6).  $\square$

**Corollary 13.4** (Removal of compact support). *The same transformation preserves normalized Stieltjes moment sequences with all moments finite, without a compact-support assumption.*

*Proof.* Replace an input random variable  $X \geq 0$  by  $\min(X, R)$ . Its moments converge to the input moments. Every output moment in the two lemmas is a fixed polynomial in finitely many input moments, so the output moments converge as well. The output first moment is 1, and bounds on each next moment give tightness and uniform integrability. A weak subsequential limit therefore represents all output moments. No assertion of determinacy is needed for this preservation statement.  $\square$

This is a preservation statement on an entire class of series. The special feature of A088714 is that it is the coefficientwise attracting fixed point of this transformation.

## 14 The fixed point as a determinate moment law

### 14.1 Exact coefficient stabilization

Start with  $B_0(z) = 1$ , the moment series of  $\delta_0$ , and define

$$B_{r+1} = \mathcal{T}(B_r), \quad C_r = (1 - zB_r)^{-1}. \quad (14.1)$$

Let  $\mu_r, \nu_r$  be the compact probability measures supplied above for  $B_r, C_r$ , respectively. All have nonnegative support.

**Lemma 14.1** (One additional correct coefficient per step). *If  $B - \tilde{B}$  is divisible by  $z^N$ ,  $N \geq 1$ , then  $\mathcal{T}(B) - \mathcal{T}(\tilde{B})$  is divisible by  $z^{N+1}$ . In particular,*

$$B_r \equiv A \pmod{z^{r+1}}, \quad C_r \equiv C \pmod{z^{r+2}}. \quad (14.2)$$

*Proof.* For  $M = \mathcal{T}(B)$ , the coefficient  $m_n$ ,  $n \geq 1$ , depends only on  $b_0, \dots, b_{n-1}$  and  $m_0, \dots, m_{n-1}$ . Induction on  $n \leq N$  therefore proves agreement. Since  $B_0$  and  $A$  agree at degree zero, iteration proves the first congruence. Equation (13.2) proves the second.  $\square$

## 14.2 Existence and uniqueness of the limiting measures

The first moment of  $\mu_r$  is 1 for  $r \geq 1$ . Hence  $\mu_r([R, \infty)) \leq 1/R$ ; the measures are tight. Every subsequential weak limit has all moments  $a_n$ . Indeed, for fixed  $n$ , the moments of orders  $n$  and  $n+1$  eventually stabilize, and

$$\int_{[R, \infty)} t^n d\mu_r(t) \leq \frac{a_{n+1}}{R}$$

for sufficiently large  $r$ . Bounded continuous cutoffs, followed by  $R \rightarrow \infty$ , justify passage of the  $n$ th moment to the limit. Denote one such limit by  $\mu$ .

Bound (12.3) and  $\log(n!) = n \log n - n + O(\log n)$  imply

$$\left(\frac{a_n}{n!}\right)^{1/n} \rightarrow 0. \quad (14.3)$$

Thus for every  $v > 0$ , monotone convergence gives

$$\int e^{vt} d\mu(t) = \sum_{n \geq 0} \frac{a_n v^n}{n!} < \infty.$$

The same argument applies to any measure with these moments. Its characteristic function is then the absolutely justified moment expansion  $\sum a_n (iu)^n / n!$ . Equality of characteristic functions determines the probability measure. Thus  $\mu$  is unique, and the whole sequence  $\mu_r$  converges weakly to it.

For  $\nu_r$ , the first moment is again 1, and its moments converge coefficientwise to  $c_n$ . Tightness and uniform integrability give a representing measure  $\nu$ . To prove uniqueness, observe first that the moments of a nonnegative random variable satisfy

$$a_i a_j \leq a_{i+j}. \quad (14.4)$$

For example, take independent copies  $X, Y$  with law  $\mu$  and use  $\mathbb{E}[(X^i - Y^i)(X^j - Y^j)] \geq 0$ . Also  $a_j \leq a_{j+1}$ : log-convexity of moments follows from Cauchy-Schwarz, and the initial ratio is  $a_1/a_0 = 1$ . Expanding the geometric series in (12.2), for  $n \geq 1$  we obtain

$$c_n = \sum_{k=1}^n \sum_{\substack{s_1 + \dots + s_k = n \\ s_i \geq 1}} \prod_{i=1}^k a_{s_i-1}. \quad (14.5)$$

Each product is at most  $a_{n-k} \leq a_{n-1}$ , and there are  $2^{n-1}$  compositions of  $n$ . Therefore

$$a_{n-1} \leq c_n \leq 2^{n-1} a_{n-1}. \quad (14.6)$$

It follows that  $\sum c_n v^n / n!$  is entire, proving determinacy and all positive exponential moments for  $\nu$  as well. Consequently  $\nu_r \Rightarrow \nu$ .

Finally, (12.3) gives  $a_n^{1/n} \rightarrow \infty$ . The moments of a probability measure supported in  $[0, R]$  are at most  $R^n$ . Thus  $\mu$  has unbounded support. Inequality (14.6) proves the same for  $\nu$ .

## 15 Strict Hankel positivity and iterated log-convexity

We provide the strict statements explicitly, since existence of a representing measure alone only guarantees nonnegative minors in general.

**Lemma 15.1** (Generalized Vandermonde positivity). *If  $0 \leq e_1 < \dots < e_k$  are integers and  $0 < t_1 < \dots < t_k$ , then  $\det[t_j^{e_i}]_{i,j=1}^k > 0$ .*

*Proof.* A nonzero linear combination of  $k$  distinct nonnegative powers has at most  $k - 1$  distinct positive zeros. This follows by induction: divide by the smallest power, differentiate, and use Rolle's theorem. Consequently the determinant cannot vanish in the connected chamber  $0 < t_1 < \dots < t_k$ , so its sign is constant. Letting the nodes approach a common positive point identifies its sign with that of the Wronskian

$$t^{\sum e_i - k(k-1)/2} \prod_{i < j} (e_j - e_i) > 0$$

times a positive ordinary Vandermonde factor. This proves the sign.  $\square$

**Theorem 15.2** (Strict total positivity). *Let  $m_n$  be the moments of a probability measure  $\sigma$  on  $[0, \infty)$  with all moments finite and infinitely many positive support points. For any strictly increasing nonnegative index lists  $i_1 < \dots < i_k$  and  $j_1 < \dots < j_k$ ,*

$$\det[m_{i_p+j_q}]_{p,q=1}^k > 0. \quad (15.1)$$

*In particular this holds for both  $(a_n)$  and  $(c_n)$ .*

*Proof.* Expanding two determinants and integrating termwise gives

$$\det[m_{i_p+j_q}] = \frac{1}{k!} \int \dots \int \det[t_\ell^{i_p}]_{p,\ell} \det[t_\ell^{j_q}]_{q,\ell} \prod_{\ell=1}^k d\sigma(t_\ell). \quad (15.2)$$

All terms are integrable since all moments are finite. The product is unchanged by a permutation of the  $t_\ell$ , and is nonnegative everywhere: on distinct positive points this follows from Lemma 15.1; on repeated points it vanishes, and at zero it follows by continuity. Choose  $k$  separated positive support points and disjoint small ordered neighborhoods. Each neighborhood has positive mass, and the integrand is strictly positive on their product. Thus the integral is strictly positive. Both measures constructed above have unbounded support, so the hypothesis holds for every  $k$ .  $\square$

**Theorem 15.3** (Strict infinite log-convexity). *For  $m = a$  or  $m = c$ , every  $\mathcal{L}^k m$ ,  $k \geq 1$ , is a Stieltjes moment sequence with an unbounded positive support representation. In particular,  $(\mathcal{L}^k m)_n > 0$  for every  $n \geq 0$ .*

*Proof.* If  $m_n = \int t^n d\sigma(t)$ , then

$$m_n m_{n+2} - m_{n+1}^2 = \frac{1}{2} \iint (xy)^n (x - y)^2 d\sigma(x) d\sigma(y). \quad (15.3)$$

Thus  $\mathcal{L}m$  is the moment sequence of the pushforward under  $(x, y) \mapsto xy$  of the finite positive measure  $\frac{1}{2}(x - y)^2 d\sigma(x) d\sigma(y)$ . All moments remain finite. If  $\sigma$  has unbounded support and a positive support point  $y_0$ , choose support points  $x_j \rightarrow \infty$ . Small product neighborhoods of  $(x_j, y_0)$  have positive weighted mass and yield arbitrarily large positive product support points. Therefore the new measure also has unbounded support and positive total mass. Repeat this argument. The representing measures after the first iteration need not have mass one, but identity (15.3) does not require normalization.  $\square$

The general preservation phenomenon is known [7]; its use here is justified by the newly constructed representing measures. Ordinary strict log-convexity,  $a_n a_{n+2} > a_{n+1}^2$ , is only the first layer.

### 15.1 An exact determinant bridge between the two sequences

For a sequence  $m$ , write  $D_n^{(h)}(m) = \det[m_{i+j+h}]_{i,j=0}^{n-1}$ , with  $D_0^{(h)} = 1$ . The following identities explain the repeated determinants in the numerical tables.

**Proposition 15.4** (Renewal Hankel identities). *For every  $n \geq 1$ ,*

$$D_n^{(1)}(c) = D_n^{(0)}(a), \quad D_n^{(0)}(c) = D_{n-1}^{(1)}(a). \quad (15.4)$$

*These are algebraic consequences of  $C = (1 - zA)^{-1}$ , independent of moment positivity.*

*Proof.* For a series  $M = \sum m_n z^n$ , its two bivariate Hankel kernels are

$$H_M(x, y) = \frac{xM(x) - yM(y)}{x - y}, \quad S_M(x, y) = \frac{M(x) - M(y)}{x - y}.$$

They have coefficients  $m_{i+j}$  and  $m_{i+j+1}$ , respectively. A direct substitution gives

$$S_C(x, y) = C(x)C(y)H_A(x, y), \quad H_C(x, y) = C(x)C(y)(1 + xyS_A(x, y)).$$

Multiplication of a bivariate kernel by  $C(x)C(y)$  is congruence by a lower-triangular Toeplitz matrix with diagonal entries  $c_0 = 1$ . Its finite leading determinants are unchanged. The first kernel identity proves the first formula; in the second identity the inside leading matrix is the block diagonal matrix with blocks 1 and the leading  $(n-1) \times (n-1)$  shifted Hankel matrix of  $a$ . This proves the second.  $\square$

## 16 Exact functions behind the divergent series

### 16.1 Passing the nonlinear identity to the limit

Let

$$F_r(s) = \int \frac{d\mu_r(t)}{1 + st}, \quad G_r(s) = \int \frac{d\nu_r(t)}{1 + st}.$$

For the compact measures, the formal identities become analytic identities near  $s = 0$ :

$$F_{r+1}(s) = 1 - sF_{r+1}(s)^2 F_r(sF_{r+1}(s)), \quad G_r(s) = \frac{1}{1 + sF_r(s)}. \quad (16.1)$$

They hold for every  $s > 0$ . Indeed, both sides are real-analytic on this connected interval, their composed arguments remain positive, and they agree on an initial interval. This continuation argument concerns compact approximants, not the divergent limiting power series.

Weak convergence gives  $F_r(s) \rightarrow F(s)$  and  $G_r(s) \rightarrow G(s)$  for each  $s > 0$ . It also permits the variable-argument passage in (16.1). For example, on any interval  $s \geq \delta > 0$ ,

$$|F'_r(s)| = \int \frac{t}{(1 + st)^2} d\mu_r(t) \leq \frac{1}{4\delta}.$$

Since  $sF_{r+1}(s) \rightarrow sF(s) > 0$ , this equicontinuity bound and pointwise convergence imply  $F_r(sF_{r+1}(s)) \rightarrow F(sF(s))$ . Thus (12.7) follows.

## 16.2 The inverse equation and absence of an atom

For every  $s > 0$ ,  $0 < F(s) < 1$  because  $\mu$  has positive mass away from zero. Moreover

$$U'(s) = \int \frac{d\mu(t)}{(1+st)^2} > 0, \quad U(s) < s. \quad (16.2)$$

The first identity in (12.7) is equivalent to

$$s = U(s)(1 + U(U(s))). \quad (16.3)$$

If  $U$  were bounded, its right side would be bounded, which is impossible. As  $U(0+) = 0$ , strict monotonicity proves it is a homeomorphism onto  $(0, \infty)$ , and (12.8) follows.

Also  $F(s) \downarrow \mu(\{0\})$ . A positive limit  $p$  would give  $sF(s)^2F(sF(s)) \geq sp^3$ , contradicting (12.7). Hence  $F(s) \rightarrow 0$ . Since  $U(s) \rightarrow \infty$ ,  $G(s) = (1 + U(s))^{-1} \rightarrow 0$  as well. This proves the absence of atoms at zero for both measures.

## 16.3 Asymptotic series with explicit remainders

The identity for a finite geometric sum gives, for every integer  $N \geq 0$  and every  $s > 0$ ,

$$F(s) = \sum_{j=0}^N (-1)^j a_j s^j + (-1)^{N+1} s^{N+1} \int \frac{t^{N+1}}{1+st} d\mu(t). \quad (16.4)$$

In particular, the remainder has its displayed sign and absolute value at most  $a_{N+1}s^{N+1}$ . The analogous statement holds for  $G$  with  $c_j$ . Thus the alternating formal series is a genuine Poincaré asymptotic expansion as  $s \downarrow 0$ , with finite-order upper and lower bounds, even though its infinite sum has radius zero.

The entire functions

$$E_a(z) = \sum_{n \geq 0} \frac{a_n z^n}{n!}, \quad E_c(z) = \sum_{n \geq 0} \frac{c_n z^n}{n!}$$

are the exponential moment transforms of  $\mu, \nu$ . Consequently

$$F(s) = \int_0^\infty e^{-v} E_a(-sv) dv, \quad G(s) = \int_0^\infty e^{-v} E_c(-sv) dv \quad (s > 0). \quad (16.5)$$

To justify this, first use exponential integrability to identify  $E_a(-u) = \int e^{-ut} d\mu(t)$ , and then use Tonelli's theorem in (16.5). The integrand  $E_a(-sv)$  lies between 0 and 1. This is a Borel–Laplace representation on the negative real axis. It means integrating the *whole entire function*; exchanging its infinite series with this integral is not justified and would recover the divergent ordinary series.

These constructions select canonical transforms from the moment problem. They do not assert uniqueness among every possible smooth, non-Stieltjes solution of the nonlinear equation.

## 17 Certified algebraic approximants and Fibonacci exponents

### 17.1 Alternating pointwise bounds

**Proposition 17.1** (Order reversal). *Given the Stieltjes transform  $H$  of a probability measure on  $[0, \infty)$ , the equation*

$$y + sy^2 H(sy) = 1 \quad (17.1)$$

*has exactly one solution in  $(0, 1)$  for each  $s > 0$ . Increasing  $H$  pointwise decreases this solution pointwise.*

*Proof.* The left side is continuous, is 0 at  $y = 0$ , exceeds 1 at  $y = 1$ , and is strictly increasing. For the last assertion, each function  $y^2/(1 + sty)$  has derivative  $y(2 + sty)/(1 + sty)^2 > 0$  on  $y > 0$ , so its average is increasing. If  $H \geq \tilde{H}$ , the left side for  $H$  is at least that for  $\tilde{H}$  at every  $y$ , and its unique crossing occurs no later.  $\square$

Since  $F_0 = 1 \geq F$ , repeated order reversal in (16.1) gives

$$F_1 \leq F_3 \leq F_5 \leq \cdots \leq F \leq \cdots \leq F_4 \leq F_2 \leq F_0 \quad \text{on } (0, \infty). \quad (17.2)$$

For nesting, start with  $F_2 \leq F_0$  and apply the same reversal successively. Section 14 proves that both bounding sequences converge to  $F$ . The convergence is uniform on compact subintervals of  $(0, \infty)$  by monotonicity and continuity.

Every  $B_r$  is algebraic. Indeed, if  $C_r$  satisfies a nonzero polynomial relation  $P(z, C_r(z)) = 0$ , then  $B_{r+1} = C_r(zB_{r+1})$  satisfies  $P(zB_{r+1}, B_{r+1}) = 0$ . The substitution  $z^i y^j \mapsto z^i y^{i+j}$  is injective on polynomials, so this relation is nonzero. Start with  $B_0 = 1$  and use  $C_r = (1 - zB_r)^{-1}$ .

The first two bounds are especially simple:

$$\frac{2}{1 + \sqrt{1 + 4s}} = F_1(s) \leq F(s) \leq F_2(s), \quad F_2(s) + sF_2(s)^3 = 1. \quad (17.3)$$

The upper bound is the unique positive root of the cubic. To derive it, write  $v = F_1(sy)$ , so that  $v + syv^2 = 1$  and  $1 - y = sy^2v$ ; eliminating  $v$  gives  $1 - y = sy^3$ . Therefore (17.2) is also an effective route to certified bounds using algebraic root isolation or nested interval bisection, without truncating a divergent ordinary series at large  $s$ .

## 17.2 The Fibonacci pattern

Let  $f_0 = 0$ ,  $f_1 = 1$ , and  $f_{r+2} = f_{r+1} + f_r$  be the Fibonacci numbers.

**Theorem 17.2** (Endpoint exponents of every approximant). *For each fixed  $r \geq 0$ ,*

$$F_r(s) \sim s^{-\alpha_r}, \quad \alpha_r = \frac{f_r}{f_{r+2}}, \quad s \rightarrow \infty. \quad (17.4)$$

*Every leading constant is 1. The exponents alternate around  $\alpha$  and converge to it:*

$$0, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{2}{5}, \quad \frac{3}{8}, \quad \frac{5}{13}, \quad \frac{8}{21}, \dots \rightarrow \alpha.$$

*Proof.* Put  $U_r(s) = sF_r(s)$ . The finite identity yields

$$U_{r+1}^{-1}(t) = t(1 + U_r(t)), \quad U_0(t) = t.$$

This inverse relation also proves inductively that every  $U_r$  is an increasing homeomorphism onto  $(0, \infty)$ . If  $U_r(t) \sim t^{b_r}$ , then its successor's inverse is asymptotic to  $t^{1+b_r}$ , so  $U_{r+1}(s) \sim s^{1/(1+b_r)}$ . With  $b_0 = 1$ , this gives  $b_r = f_{r+1}/f_{r+2}$ , including the constant 1. Since  $\alpha_r = 1 - b_r$ , the formula follows. The elementary recurrence  $b \mapsto 1/(1 + b)$  has fixed point  $\beta$  and reverses the order around it, giving the stated alternating convergence.  $\square$

The brackets and this theorem alone determine  $\log F(s)/\log s \rightarrow -\alpha$ . They do *not* justify the full constant-one equivalent by interchanging  $r \rightarrow \infty$  with  $s \rightarrow \infty$ . The next section supplies a separate rigidity argument that closes precisely this gap.

## 18 Why the golden-ratio law has leading constant one

The derivation below avoids assuming regular variation or differentiable asymptotics. Its conclusion is stronger than simply guessing a power in the nonlinear equation.

**Lemma 18.1** (An asymptotic iteration rigidity principle). *Suppose  $g : \mathbb{R} \rightarrow \mathbb{R}$  is increasing and continuous,  $g(x) < x$ ,  $g(x) \rightarrow \infty$  as  $x \rightarrow \infty$ , and*

$$g(g(x)) + g(x) = x - \varepsilon(x), \quad \varepsilon(x) \rightarrow 0. \quad (18.1)$$

*Then  $g(x) - \beta x \rightarrow 0$ , where  $\beta^2 + \beta = 1$  and  $\beta > 0$ .*

*Proof.* Put  $d(x) = g(x) - \beta x$ . Substitution in (18.1) gives

$$d(x) = -\beta d(g(x)) - \beta \varepsilon(x). \quad (18.2)$$

Fix a sufficiently large real  $X$ . For  $x > X$ , iterate  $g$  until it first reaches a point at most  $X$ , and let  $N = N(x)$  be that number of steps. It is finite: otherwise the decreasing orbit has a limit at least  $X$ , and continuity makes that limit a fixed point, contrary to  $g(y) < y$ . By monotonicity, the exit point lies in  $[g(X), X]$ . Also  $N(x) \rightarrow \infty$  as  $x \rightarrow \infty$ , because every fixed finite iterate of  $g$  tends to infinity.

Let  $M_X = \sup_{[g(X), X]} |d|$  and  $E_X = \sup_{y \geq X} |\varepsilon(y)|$ . Iterating (18.2) gives

$$|d(x)| \leq \beta^{N(x)} M_X + \frac{\beta}{1 - \beta} E_X.$$

Take the limit superior as  $x \rightarrow \infty$ , then let  $X \rightarrow \infty$ . The result is zero, proving the lemma.  $\square$

Apply this lemma to

$$g(x) = \log U(e^x). \quad (18.3)$$

The properties of  $U$  prove all its monotonicity, continuity and range hypotheses. From (16.3),

$$U(U(s)) = \frac{s}{U(s)} - 1 = \frac{s}{U(s)} (1 - F(s)).$$

Taking logarithms at  $s = e^x$  yields

$$g(g(x)) + g(x) = x + \log(1 - F(e^x)).$$

Thus  $\varepsilon(x) = -\log(1 - F(e^x)) \rightarrow 0$ , since  $F(e^x) \rightarrow 0$ . Lemma 18.1 now gives

$$\log \frac{U(s)}{s^\beta} \rightarrow 0.$$

Exponentiating proves  $U(s) \sim s^\beta$ , hence  $F(s) = U(s)/s \sim s^{-\alpha}$ . Finally  $G(s) = (1 + U(s))^{-1} \sim s^{-\beta}$ . This proves (12.9), with no unspecified multiplicative constant and no exchange of the two limits in the preceding section.

## 19 Recovering the measures near zero

For completeness we prove the precise special case of the Stieltjes Tauberian theorem used here. General versions and their history are available in [8]. The normalization of the transform matters.

**Lemma 19.1** (Power-law Stieltjes Tauberian lemma). *Let  $\sigma$  be a probability measure on  $[0, \infty)$  and  $H(s) = \int (1 + st)^{-1} d\sigma(t)$ . If*

$$H(s) \sim Ks^{-\gamma} \quad (s \rightarrow \infty), \quad K > 0, \quad 0 < \gamma < 1,$$

then

$$\sigma([0, t]) \sim K \frac{\sin(\pi\gamma)}{\pi\gamma} t^\gamma \quad (t \downarrow 0). \quad (19.1)$$

*Proof.* Let  $\eta_s$  be the probability measure of the rescaled variable  $y = st$ , weighted by  $(1 + st)^{-1}/H(s)$ . Regard it as a measure on the compact interval  $[0, 1]$  using the coordinate  $w = y/(1 + y)$ , with  $y = \infty$  corresponding to  $w = 1$ . For  $u > 0$ ,

$$\int \frac{1 + y}{1 + uy} d\eta_s(y) = \frac{H(us)}{H(s)} \rightarrow u^{-\gamma}. \quad (19.2)$$

The integrand is the bounded continuous function  $[1 + (u - 1)w]^{-1}$  on  $[0, 1]$ .

Consider the probability measure on  $(0, \infty)$  with density

$$\frac{\sin(\pi\gamma)}{\pi} \frac{y^{\gamma-1}}{1 + y}. \quad (19.3)$$

Euler's beta integral  $\int_0^\infty y^{\gamma-1}(1 + y)^{-1} dy = \pi/\sin(\pi\gamma)$  normalizes it. Substitution  $v = uy$  shows that its integral in (19.2) is  $u^{-\gamma}$ . This probability measure is the unique one on the compact  $w$ -interval with those integrals. Indeed, for  $|u - 1| < 1$  expand  $[1 + (u - 1)w]^{-1}$  uniformly as a power series. The integrals determine all moments of  $w$ ; polynomials are dense in continuous functions on  $[0, 1]$ , so these moments determine the measure.

Every subsequential weak limit of  $\eta_s$  therefore equals (19.3). Compactness proves convergence of the whole family. In particular, the limit has no mass at  $y = 1$  or at either compactification endpoint. The bounded function  $(1 + y)\mathbf{1}_{\{y \leq 1\}}$  is continuous except at  $y = 1$ , so weak convergence gives

$$\frac{\sigma([0, 1/s])}{H(s)} = \int_{y \leq 1} (1 + y) d\eta_s(y) \rightarrow \frac{\sin(\pi\gamma)}{\pi} \int_0^1 y^{\gamma-1} dy = \frac{\sin(\pi\gamma)}{\pi\gamma}.$$

Using the assumed equivalent for  $H(s)$  proves (19.1).  $\square$

Apply the lemma to  $F$  with  $(K, \gamma) = (1, \alpha)$  and to  $G$  with  $(K, \gamma) = (1, \beta)$ . This proves (12.10). In particular, arbitrarily small positive intervals adjacent to zero have positive mass. The result is about *distribution functions*: it does not justify differentiating the equivalent or asserting a density.

**Corollary 19.2** (Sharp negative-moment thresholds). *For  $q > 0$ ,*

$$\int t^{-q} d\mu(t) < \infty \iff q < \alpha, \quad \int t^{-q} d\nu(t) < \infty \iff q < \beta.$$

*At the critical exponent  $\gamma$  for either law  $\sigma$ ,*

$$\int_{[\varepsilon, 1]} t^{-\gamma} d\sigma(t) \sim \frac{\sin(\pi\gamma)}{\pi} \log \frac{1}{\varepsilon}. \quad (19.4)$$

*Proof.* For  $0 < x \leq 1$ ,  $x^{-q} = 1 + q \int_x^1 t^{-q-1} dt$ . Tonelli's theorem reduces finiteness near zero to that of  $\int_0^1 t^{-q-1} \sigma([0, t]) dt$ . The distribution-function equivalent makes this the elementary integral of  $t^{\gamma-q-1}$ . The part on  $[1, \infty)$  is bounded by the total mass. At  $q = \gamma$ , integration by parts on  $[\varepsilon, 1]$  gives a bounded endpoint term plus  $\gamma \int_\varepsilon^1 t^{-\gamma-1} \sigma([0, t]) dt$ . Its equivalent is the asserted logarithm. Possible endpoint atoms do not alter it: the distribution-function equivalent makes their scaled contribution bounded, hence negligible relative to the logarithm.  $\square$

## 20 From global growth to consecutive-coefficient asymptotics

The earlier root equivalent by itself cannot control neighboring ratios. The moment construction supplies the missing convexity.

**Lemma 20.1** (A convex-slope principle). *Suppose  $m_n > 0$  is log-convex and*

$$\log m_n = n \log n - n \log \log n - n + o(n). \quad (20.1)$$

*Then  $m_{n+1}/m_n \sim n/\log n$ .*

*Proof.* Let  $h(n) = \log m_n$ , and interpolate linearly between integers. Log-convexity makes this interpolation convex. Fix  $0 < \delta < 1$ . The forward slope  $h(n+1) - h(n)$  is bounded below by the secant from  $\lfloor(1-\delta)n\rfloor$  to  $n$ , and above by the secant from  $n$  to  $\lceil(1+\delta)n\rceil$ . Inserting (20.1), accounting for rounding by  $o(1)$  in the quotients, gives

$$\begin{aligned} \liminf_{n \rightarrow \infty} (h(n+1) - h(n) - \log n + \log \log n) &\geq -\frac{1-\delta}{\delta} \log(1-\delta) - 1, \\ \limsup_{n \rightarrow \infty} (h(n+1) - h(n) - \log n + \log \log n) &\leq \frac{1+\delta}{\delta} \log(1+\delta) - 1. \end{aligned}$$

Both right sides tend to zero as  $\delta \downarrow 0$ . Hence the expression inside parentheses tends to zero. Exponentiation proves the result.  $\square$

Strict log-convexity from Section 15 and (12.3) immediately prove the first equivalent in (12.11). It remains to transfer the sharp logarithmic growth to  $c_n$ ; the factor  $2^{n-1}$  in (14.6) is too large for this purpose.

**Proposition 20.2** (Sharper comparison for the companion sequence). *As  $n \rightarrow \infty$ ,*

$$\log c_n = \log a_{n-1} + O(\log n) = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (20.2)$$

*Consequently  $c_{n+1}/c_n \sim n/\log n$ .*

*Proof.* Keeping track of the number of parts in (14.5), and using (14.4), gives

$$a_{n-1} \leq c_n \leq \sum_{j=0}^{n-1} \binom{n-1}{j} a_j. \quad (20.3)$$

Write  $r_j = a_{j+1}/a_j$ , which is increasing and satisfies  $r_j \sim j/\log j$ . Put  $m = \lfloor(n-1)/2\rfloor$ . The part  $j \leq m$  of the upper sum is at most  $2^{n-1} a_m = o(a_{n-1})$  by (12.3). For  $j > m$ , monotonicity of  $r_j$  gives

$$a_j \leq a_{n-1} r_m^{-(n-1-j)}.$$

Thus the rest of the sum is bounded by

$$a_{n-1}(1 + 1/r_m)^{n-1} \leq a_{n-1} \exp((n-1)/r_m) = a_{n-1} \exp(O(\log n)).$$

Together with the lower bound this proves the first equality in (20.2). The second follows from (12.3) and  $\log r_{n-1} = O(\log n)$ . Finally apply Lemma 20.1 to  $c$ , whose log-convexity has already been proved.  $\square$

**Corollary 20.3** (A quantitative, though slow, ratio estimate). *For  $m = a$  or  $m = c$ ,*

$$\frac{m_{n+1}}{m_n} = \frac{n}{\log n} \left( 1 + O \left( \sqrt{\frac{\log \log n}{\log n}} \right) \right).$$

*Proof.* Let  $h_0(t) = t \log t - t \log \log t - t$  and  $e_n = \log \log n / \log n$ . On  $[n/2, 2n]$ , the available logarithmic estimate has error  $O(ne_n)$ , uniformly. For secants over length  $\delta n$ , this contributes  $O(e_n/\delta)$ . Since  $h_0''(t) = O(1/n)$  there, the secant differs from  $h_0'(n)$  by  $O(\delta)$ ; also  $h_0'(n) = \log n - \log \log n + O(1/\log n)$ . Take  $\delta = \sqrt{e_n}$  in the convex-slope bounds. Rounding endpoints contributes only  $O(\log n/(\delta n))$ . Exponentiation gives the result.  $\square$

This is not a proof that  $c_n \sim a_{n-1}$ . The polynomial-factor comparison is sufficient for the asserted logarithmic and ratio laws, and no stronger equivalent is used below.

## 21 The far tails from size-biased concentration

We prove a general lemma and then apply it to both measures. This keeps the lower-tail-bound argument separate from an insufficient Markov upper bound.

**Lemma 21.1** (A moment-growth tail principle). *Let  $\sigma$  be a probability measure on  $[0, \infty)$  with positive moments  $m_n$ , unbounded support, and*

$$\log m_n = n \log n - n \log \log n - n + o(n), \quad r_n := m_{n+1}/m_n \sim n/\log n.$$

*Then  $\log \sigma([x, \infty)) \sim -x \log x$ .*

*Proof.* For the upper bound, Markov's inequality gives  $\sigma([x, \infty)) \leq m_n/x^n$ . Choose  $n = \lfloor x \log x \rfloor$ . Then  $\log n - \log \log n = \log x + o(1)$ , and consequently

$$\log \sigma([x, \infty)) \leq -(1 + o(1))x \log x. \quad (21.1)$$

For the reverse inequality define the probability measure

$$d\sigma_n(t) = \frac{t^n}{m_n} d\sigma(t).$$

Its mean is  $r_n$  and its relative variance is

$$\frac{\text{Var}_{\sigma_n}(t)}{r_n^2} = \frac{r_{n+1}}{r_n} - 1 \longrightarrow 0.$$

The last limit follows directly from the ratio equivalent at  $n$  and  $n+1$ . Thus for each fixed  $0 < \varepsilon < 1$ ,

$$\sigma_n([(1-\varepsilon)r_n, (1+\varepsilon)r_n]) \longrightarrow 1. \quad (21.2)$$

Fix  $\delta > 0$  and choose  $\varepsilon > 0$  so small that  $(1 - \varepsilon)(1 + \delta) > 1$ . Now take  $n = \lfloor (1 + \delta)x \log x \rfloor$ . Then  $r_n \sim (1 + \delta)x$ , so the interval in (21.2) lies in  $[x, \infty)$  for large  $x$ . On it,  $t^n \leq ((1 + \varepsilon)r_n)^n$ , which implies

$$\sigma([x, \infty)) \geq \frac{m_n}{((1 + \varepsilon)r_n)^n} \sigma_n([(1 - \varepsilon)r_n, (1 + \varepsilon)r_n]).$$

Taking logarithms and using both hypotheses gives

$$\liminf_{x \rightarrow \infty} \frac{\log \sigma([x, \infty))}{x \log x} \geq -(1 + \delta)(1 + \log(1 + \varepsilon)).$$

Let  $\delta \downarrow 0$  with admissible  $\varepsilon \downarrow 0$ , and combine with (21.1). This proves the limit.  $\square$

Apply the lemma to  $\mu$  using (12.3) and to  $\nu$  using (20.2), with their already proved ratio laws. This proves Theorem 12.3. Equivalently, for every fixed  $\varepsilon > 0$ , for sufficiently large  $x$  each tail lies between  $\exp(-(1 + \varepsilon)x \log x)$  and  $\exp(-(1 - \varepsilon)x \log x)$ . Nothing here supplies a multiplicative tail equivalent or a density.

## 22 Computational audits and certified orbit enclosures

### 22.1 Exact calculations actually performed

The computations are checks of implementations and finite instances, not substitutes for the all-index proofs. The file `code/02-golden-moments-verify.py` uses only the Python standard library and performs the following tasks. It independently generates  $a_0, \dots, a_{401}$  by the triangular recurrence (2.8) of Part I (whose  $c_{n,k}$  are not the  $c_n$  of A088713), obtains  $c_0, \dots, c_{401}$  by (13.2), and checks all 21 displayed A088714 terms and all 22 displayed A088713 terms retrieved from the OEIS entries.

Two different formal calculations check the defining functional equation and the noncrossing relation  $A = C(zA)$  through degree 30. The compact transform iteration is separately checked through stage 16, including its exact stabilization property. A direct rational-arithmetic implementation of the full-Fock operator for  $\rho = \frac{1}{2}(\delta_1 + \delta_3)$  agrees with the noncrossing formula through moment 10. In that calculation the tensor factors are functions on  $\{1, 3\}$ , with their weighted inner product; no square-root or floating-point approximation enters.

For both sequences, every minor of order 1 through 4 whose row and column indices lie in  $\{0, \dots, 7\}$  was checked: 17,768 minors in total, all strictly positive. An exact fraction-free determinant algorithm also checks all 40 leading and once-shifted determinants of orders 1 through 10 and all 20 determinant identities in Proposition 15.4 for those orders. Six iterations of  $\mathcal{L}$  on the first 101 moments of each sequence give 1,128 strictly positive values. Complete outputs are in the files

`data/02-golden-moments-verification.json`  
`data/02-golden-moments-verification.txt`

Some exact determinants are displayed below. Large integers have been computed directly, not inferred from the determinant identities.

| $n$ | $D_n^{(0)}(a)$ | $D_n^{(1)}(a)$    | $D_n^{(0)}(c)$ | $D_n^{(1)}(c)$ |
|-----|----------------|-------------------|----------------|----------------|
| 1   | 1              | 1                 | 1              | 1              |
| 2   | 2              | 4                 | 1              | 2              |
| 3   | 20             | 100               | 4              | 20             |
| 4   | 1,440          | 20,864            | 100            | 1,440          |
| 5   | 970,624        | 45,897,152        | 20,864         | 970,624        |
| 6   | 7,618,255,104  | 1,300,185,348,608 | 45,897,152     | 7,618,255,104  |

The repetition is explained exactly by (15.4).

The next table shows the scaled consecutive ratios. These are rounded floating-point diagnostics obtained only after the integer coefficients were computed exactly. The theorem gives the limit 1; the table makes clear that the convergence is still slow at the displayed indices.

| $n$ | $\frac{a_{n+1}}{a_n}$ | $\frac{\log n}{n}$ | $\frac{c_{n+1}}{c_n}$ | $\frac{\log n}{n}$ |
|-----|-----------------------|--------------------|-----------------------|--------------------|
|     |                       |                    |                       |                    |
| 20  | 1.798430              |                    | 1.743890              |                    |
| 50  | 1.568256              |                    | 1.545876              |                    |
| 100 | 1.472240              |                    | 1.461060              |                    |
| 200 | 1.410484              |                    | 1.404905              |                    |
| 400 | 1.368842              |                    | 1.366053              |                    |

No monotonicity of these *scaled* ratios is asserted from this table.

## 22.2 Enclosures far beyond the range of a power-series evaluation

Equation (12.8) has a useful exact numerical consequence. Choose  $s_0 > 0$ , set  $s_{-1} = U(s_0)$ , and define

$$s_{k+1} = s_k(1 + s_{k-1}) \quad (k \geq 0). \quad (22.1)$$

Then  $U(s_k) = s_{k-1}$  for every  $k \geq 0$ . Indeed,  $U^{-1}(s_k) = s_k(1 + U(s_k)) = s_k(1 + s_{k-1})$ . Thus values of the true transform can be transported along its inverse orbit without any approximation to the functional equation.

For a rigorous seed, take  $s_0 = 1/100$ . Equation (16.4) gives

$$\sum_{j=0}^{31} (-1)^j a_j s_0^j \leq F(s_0) \leq \sum_{j=0}^{30} (-1)^j a_j s_0^j. \quad (22.2)$$

Both endpoints are rational and computed exactly. Positive additions and multiplications preserve lower and upper enclosures under (22.1). The script `code/02-golden-moments-orbit_bounds.py` propagates these enclosures with 100-digit Decimal arithmetic, rounding down for lower endpoints and up for upper endpoints. Correctly rounded logarithms, square roots and exponentials are enlarged by one adjacent representable value on each side before they are used in an enclosure [12]. Subtracted products are rounded in the opposite direction before subtraction.

For each row in the following table there is a true inverse-orbit point  $s_k$  in the full logarithmic interval recorded in `data/02-golden-moments-orbit_bounds.json`. The middle column is only a rounded location; the last column is a widened outward enclosure of

$$s_k^\alpha F(s_k) = \frac{s_{k-1}}{s_k^\beta}.$$

The enclosure concerns that true point, not every point in the interval.

| $k$ | approximate $\log s_k$ | enclosure of $s_k^\alpha F(s_k)$ |
|-----|------------------------|----------------------------------|
| 110 | 2.310822               | [0.871023967670, 0.871023967671] |
| 120 | 293.675802             | [0.999996764030, 0.999996764031] |
| 130 | 36119.735741           | [0.999999973689, 0.999999973690] |
| 140 | 4442433.820372         | [0.999999999786, 0.999999999787] |
| 150 | 546383240.170012       | [0.999999999998, 0.999999999999] |

These are finite computational enclosures conditional only on the exact identities proved above and the correctly rounded arithmetic operations. They illustrate the constant-one law at extremely large arguments. They are not its proof: Lemma 18.1 proves that law at all sufficiently large arguments, not merely along the selected orbit.

### 22.3 Reproduction and complexity

In the delivered package layout, where the three scripts and their data files carry no prefix, the checks and the delivered three-pass build were

```
python3 code/verify.py -n 401
python3 code/orbit_bounds.py
python3 code/make_tables.py
sh build.sh
```

The scripts write their outputs under those unprefixed names, not over the shipped files with the prefix `02-golden-moments-`; the report's README gives the commands for rerunning them on a copy. The PDF of this report is built by its `Makefile`. The coefficient generator uses  $N(N+1)(N+2)/6$  integer products and  $O(N^2)$  integer storage cells. These are arithmetic-operation counts, not unit-cost bit-complexity claims: the coefficients grow rapidly. The small formal-series, Fock, and determinant checks use independent formulas to reduce the chance of an indexing or recurrence error. The PDF build requires a normal pdfLaTeX installation and the packages listed in the preamble; the mathematical checks do not require a computer algebra system.

## 23 Further research questions and concrete next steps

The moment representation changes the available methods: it allows orthogonal polynomials, resolvent bounds, probability tilting, and Tauberian arguments to be used alongside formal reversion. The following questions are not treated as established consequences of finite data. They are follow-up tasks for this construction, not claims that every item is a previously recognized open problem; existing general theorems may resolve some of them once their hypotheses are checked.

**Research question 23.1** (Regularity and the support of the two laws). Are  $\mu$  and  $\nu$  absolutely continuous on  $(0, \infty)$ ? Do their supports fill all of  $[0, \infty)$ , or can gaps or positive atoms occur?

The present proof gives unbounded support, no atom at zero, and precise mass near zero, but none of these excludes gaps or a singular component. A concrete first step is to check applicable free-convolution regularity theorems, and then continue the coupled Cauchy-transform identities into the upper half-plane and analyze boundary values. Any proposed density asymptotic must include a regularity argument; differentiating (12.10) is not sufficient. [Batch 86, 4 October 2026.] Answered in Part IV: both laws are absolutely continuous, with strictly positive real-analytic densities on  $(0, \infty)$  and supports  $[0, \infty)$ , so there are no gaps and no atoms; the densities have explicit equivalents at zero (Theorem 37.1), as do their derivatives of every fixed order (Corollary 37.8). The proof continues the transform identities into the upper half-plane and does not differentiate (12.10).

**Research question 23.2** (A full asymptotic equivalent for the coefficients). What is the multiplicative asymptotic equivalent of  $a_n$ ? In particular, does the finer Bell-number normalization proposed in Part I (Conjecture 8.1) converge to a finite positive constant?

Part I proposes a correction involving  $\exp(W(n)^2 + 3W(n))$  relative to Bell numbers. The ratio law proved here is weaker: an error tending to zero in a single ratio can still accumulate

into an unbounded logarithmic error. A practical next step is to derive a uniform expansion of the tilted-law mean beyond  $n/\log n$ , with a summable remainder after the proposed normalization. [Batch 76, 2 October 2026.] Part III makes partial progress and does not answer the question. It proves  $\liminf_{n \rightarrow \infty} (a_n/a_{n-1} - n/W(n)) \geq 3/2$ , where the proposed normalization needs the correction 2 (Theorem 29.1), reduces convergence of the normalization to the single scalar estimate (30.8) (Theorem 30.1), and shows that a quantitative relative-variance estimate for the tilted law suffices for its curvature half (Proposition 31.1). [Batch 86, 4 October 2026.] Proved in Part V (batch 86, manuscript 02), not yet independently reviewed: the finer normalization converges,  $a_n = C_* \mathcal{N}_n(1 + O(W(n)^5/n))$  and  $a_n \sim C_* \mathbf{B}_n \exp(W(n)^2 + 3W(n))$  with  $0 < C_* < \infty$  (Theorem 46.1). The value of  $C_*$  and the ratio expansion (8.7) remain open (Research questions 49.1 and 49.2). Part IV, from manuscript 01 of the same batch, which was written without knowledge of manuscript 02, still calls the question open.

**Research question 23.3** (Second-order golden-ratio asymptotics). Can  $F(s) = s^{-\alpha}(1 + o(1))$  be sharpened to an explicit second-order expansion, and does its correction exhibit oscillation under the inverse iteration  $s_{k+1} = s_k(1 + s_{k-1})$ ?

The exact equation  $d(x) = -\beta d(g(x)) - \beta \varepsilon(x)$  is a suitable starting point. Its negative contraction coefficient makes orbit-dependent alternating corrections plausible, but the present proof does not establish a global oscillatory profile or an optimal error rate. [Batch 86, 4 October 2026.] Answered in Part IV: Theorem 38.1 expands  $\log(s^\alpha F(s))$  in powers of  $1/\log s$ , with coefficients built from a nonzero real-analytic profile  $P$  of  $\log_\varphi \log s$  satisfying  $P(t+1) = -P(t)$ , and Corollary 38.3 shows that the correction oscillates:  $(\log s)(s^\alpha F(s) - 1)$  has upper and lower limits  $\pm M$  with  $M > 0.00095$ , and alternates in sign along the inverse orbit. The equation for  $d(x)$  above is the starting point there, in the form (38.20).

**Research question 23.4** (Effective convergence of the certified approximants). How quickly do  $F_{2r}(s) - F_{2r+1}(s)$  tend to zero, uniformly when  $s$  varies over a growing interval? Can one obtain practical a priori stopping bounds for numerical evaluation?

The proof gives monotone convergence on fixed compact intervals, but not a useful global complexity estimate. The known finite-stage Fibonacci exponents indicate why the limits  $r \rightarrow \infty$  and  $s \rightarrow \infty$  require care. Interval evaluation with adaptive algebraic root isolation is a concrete computational approach.

**Research question 23.5** (Orthogonal polynomials and continued fractions). What are the asymptotics of the Jacobi or Stieltjes continued-fraction coefficients associated with  $\mu$  and  $\nu$ ?

Strict Hankel positivity guarantees nondegeneracy of every finite orthogonalization step, while (15.4) ties the two systems together. Exact determinant tables can generate candidate recurrence coefficients, but the main challenge is to prove their large-degree behavior and connect it to the two endpoint laws.

**Research question 23.6** (Sharper comparison of the companion sequence). Does  $c_n/a_{n-1} \rightarrow 1$ , and what is the first correction if it does?

Proposition 20.2 supplies only a polynomial-factor bound. The renewal recurrence suggests that endpoint terms should dominate its convolutions. Turning that observation into a uniform bound on the entire convolution, rather than on a fixed number of terms, would resolve the question without assuming the full asymptotics of  $a_n$ . [Batch 86, 4 October 2026.] Answered in Part IV, exactly along these lines:  $c_n = \sum_{j=0}^K \gamma_j a_{n-1-j} + O_K(a_{n-K-2})$  for every fixed  $K$ , with  $\gamma_j = [z^j]C(z)^2$ , so

$c_n/a_{n-1} \rightarrow 1$  with first correction  $2 \log n/n$  (Theorem 39.1 and (39.11)). Part V proves the same expansion independently (Theorem 47.1) and adds  $c_n \sim C_*(W(n)/n)\mathcal{N}_n$  with the constant of the Bell normalization, which is not yet independently reviewed.

**Research question 23.7** (Other reversion fixed points and positivity-preserving maps). Which variants of  $M = 1 + zM^pB(zM^q)$  preserve normalized Stieltjes moment sequences, and which OEIS reversion sequences have fixed points of such maps?

The factorization in Section 13 proves the case  $(p, q) = (2, 1)$ . It is not a proof for arbitrary  $p, q$ . Positive block operators and noncrossing transforms offer testable sufficient mechanisms; negative Hankel minors provide exact obstructions in unsuccessful cases.

**Research question 23.8** (A modular formalization in ProveIt). Can the proof be formalized with reusable lemmas for formal transforms, positive kernels, compact moment limits, and iteration rigidity?

A feasible first stage is the formal coefficient stabilization and the bivariate determinant identities, followed by the finite-remainder formula and the convex-slope lemma. The operator-to-measure limit is a separate analytic component. The scalar rigidity lemma is comparatively isolated and could be formalized independently. No code included here is presented as a completed Lean formalization.

## 24 Conclusion

A088714 and A088713 are not merely two rapidly growing coefficient lists. They are the determinate moment sequences of a coupled pair of positive measures. This representation proves every strict Hankel inequality, provides monotone algebraic bounds for the associated transforms, and makes their divergent ordinary generating functions meaningful on the negative axis through explicit Stieltjes and Borel–Laplace formulas.

The nonlinear inverse relation  $U^{-1}(t) = t(1 + U(t))$  forces complementary golden-ratio exponents, including their leading constants. Part I’s Bell-scale coefficient bound, together with the new convexity, determines consecutive ratios and the far-tail logarithmic law. The resulting picture is rigorous at both endpoints while leaving the finer coefficient normalization, measure regularity, and second-order behavior clearly open. [Batch 86, 4 October 2026.] Measure regularity and the second-order behavior at the golden-ratio endpoint are settled in Part IV, and the finer normalization in Part V, whose proof has not yet been independently reviewed.

## Part III

# Local estimates for the finer Bell comparison

## 25 Source, scope and notation of Part III

**Summary.** Part II proves positive-measure realizations and consecutive-ratio equivalents for A088714, but does not prove Part I’s finer Bell normalization (Conjecture 8.1). Part III reconciles the two arguments and isolates the extra precision needed. A uniform small-degree coefficient expansion is proved without assuming any smoothness of the ratios (Lemma 28.1); it is Part I’s formal near-endpoint expansion (8.1), now a theorem. Combined with the log-convexity of Part II, it yields an unconditional additive lower correction tending to  $3/2$  (Theorem 29.1). The conjectured correction tends instead to 2; the missing half is generated by curvature of ratios across a logarithmic window. An exact Poisson-weighted endpoint identity reduces the conjecture to two quantitative estimates, averaged local curvature and the mass outside that endpoint (Theorem 30.1), or to the single scalar estimate  $\mathcal{E}_n = O(W(n)^q/n^2)$  of (30.8). A sufficient tilted-variance estimate is stated explicitly (Proposition 31.1), and every subsequent implication is proved. Exact-integer computations test both the proposed normalization and the individual missing estimates. The full ratio expansion and the existence or value of the normalization constant remain unproved: **Conjecture 8.1 is still open after Part III.**

**Provenance.** Part III is manuscript 01 of batch 76 of the repository’s incoming-reports intake: *The missing local estimate in the finer Bell comparison. An endpoint reduction, a rigorous one-sided correction, and numerical stress tests for OEIS A088714*, “Research continuation prepared for Vladimir Reshetnikov”, dated October 2, 2026 (UTC) (a 372-line source and a 10-page PDF). Its author line names no person and no model. It arrived as `A088714_Bell_Local_Estimates.zip` in commit `6914ccca6` and was placed in commit `2a04b60f2`. The manuscript pins the repository at commit `82afb955918e5af9ccd2a7740424b2a66976f339` (October 1, 2026), a descendant of the commit `0a5908ec0` that wrote Part II. The source of this report is the same at the pin and at the placement commit, so the manuscript read Parts I and II as printed here, together with Part II’s source audit `02-golden-moments-SOURCE_AUDIT.md`; it also consulted library copies of the two delivered articles. Its code and data are shipped with the prefix `03-local-estimates-`. Not shipped, and surviving in the archive of the arrival commit: its own  $\text{\LaTeX}$  source, README and PDF; its copy of the repository’s root LICENSE; its copies of Part I’s `code/coefficients.py`, `code/compute_gmp.cpp` and `data/diagnostics.csv` (the last delivered as `data/prior_diagnostics.csv`), which equal the shipped files byte for byte; and its 2.49 MB table `data/coefficients_snapshot.txt` of  $a_0, \dots, a_{1600}$ , which is regenerable (see the README) and agrees with `data/02-golden-moments-coefficients.csv` on the overlap  $n \leq 401$ .

**What is not reprinted.** The manuscript’s Section 2 re-derives Proposition 8.2 without saying that Part I proves it: it assumes the same ratio estimate (8.7), uses the same normalizer (8.4) and the same quotient expansion (8.5), obtained from the same derivative identity, and reaches the same summability conclusion. That argument is replaced by a pointer in Section 27, and the manuscript’s displays of the defining equation and of the normalizer are replaced by (1.1) and (8.4). The inherited results (26.1)–(26.3), which the manuscript itself calls “not new assertions”, are kept as a summary, with pointers to the theorems of Parts I and II that prove them. Everything else in the manuscript is printed: all other sections, proofs and remarks, its status table, and its list of next investigations.

**Editorial changes.** Section  $k$  of the manuscript is Section  $k + 25$  here, and every printed label carries the prefix `ble:`. Three symbols were renamed, to match Parts I and II or to avoid a clash: the manuscript’s normalizer  $N_n$  is Part I’s  $\mathcal{N}_n$  (the same function), its Bell number  $B_n$  is Part I’s  $\mathcal{B}_n$  in the text (the shipped table fragments still print  $B_n$ ), and the excess degree called  $e$  in the proof of Lemma 28.1 is  $\eta$ , since  $e$  is also Euler’s number. No normalization changed. One sentence of Section 31 read “This proves (30.6)” although the deduction uses the hypothesis (31.1); it now reads “Under (31.1), this proves (30.6).” The manuscript defined a theorem environment “Unproved estimate” that it never used; it is not carried over. File names in Section 33 are the shipped ones. In the bibliography, the manuscript’s entry for the repository is replaced by internal references (its caveat is kept below), its OEIS entry is merged with Part I’s, and its two remaining entries are added [9, 10]. The manuscript’s statement that it makes no repository modifications described the delivery. At placement, Part I gained dated notes at (8.1) and after Proposition 8.2, Part II gained one after its question on a full asymptotic equivalent (Section 23), and Table 2 gained Part III’s letters. Nothing else in the mathematics was changed.

**Status.** Part III, like Parts I and II, is AI-assisted and unrefereed, and it is not formalized: no Lean or Rocq development checks any statement of this report, and placement in this collection confers no formal status. The manuscript treats Parts I and II as “written research reports, not assumed to be peer reviewed or formally checked”. It makes no priority claim; “new” in it means new relative to Parts I and II. Its numerical tests are finite (all  $n \leq 1600$ ) and are not used in any proof.

**Related repository material.** The Bell asymptotic behind  $\mathcal{N}_n$  is (6.6) [3, 9]. An all-orders form of the Bell saddle expansion, with  $W(n)$  as the saddle  $we^w = n$ , is Theorem `q2:thm:bell`, with Remark `q2:rem:bell-scale`, of the transseries volume in

Analysis/Transseries/docs/series-and-transseries/  
Transseries\_And\_Inversion/

Part III uses  $W(n)$  only as that saddle. The tilt  $\mu_{n-1}$  of Section 31 is Part II’s size-biased law  $\sigma_{n-1}$  for  $\sigma = \mu$  (proof of Lemma 21.1), and identity (31.3) is the relative-variance identity of that proof with the ratio index shifted by one.

**Notation.** Part III uses the letters of Part I’s Section 8,  $w = W(n)$ ,  $t = n/r_n$  and  $d = w/(1 + w)$ , but writes the backward ratio  $a_n/a_{n-1}$  as  $r_n$ . Every symbol of Part III is local to Part III; the lower block of Table 2 lists each letter it shares with Part I or Part II, with all meanings. The tempting false readings:  $R(n)$  below is the candidate right side of (8.3), *not* a ratio, while Part I’s ratio  $R_n$  is Part III’s  $r_n$ ; and Part II’s  $r_n = a_{n+1}/a_n$  is a forward ratio, Part III’s  $r_{n+1}$ .

## 26 Sources, notation, and current proof status

Let  $A(z) = \sum_{n \geq 0} a_n z^n$  be the formal solution of (1.1) (Part I writes  $x$  for the variable), so that  $a_0 = 1$ ,  $a_1 = 1$ ,  $a_2 = 3$ ,  $a_3 = 13$ . Throughout Part III,

$$r_n = \frac{a_n}{a_{n-1}}, \quad w = W(n), \quad R(n) = \frac{n}{w} + 2 + \frac{w}{2(1+w)^2}, \quad d = \frac{w}{1+w}.$$

Part II’s source audit, dated October 1, says explicitly that the earlier finer Bell normalization is not established. The manuscript inspected Part I (the September 20 Bell-scale article), Part II (the October 1 golden-moment work) and that audit at the pin recorded in Section 25.

The usable conclusions of that work are

$$\log a_n = n \log n - n \log \log n - n + O(n \log \log n / \log n), \quad (26.1)$$

$$a_n = \int_0^\infty x^n d\mu(x), \quad (26.2)$$

$$r_n \sim n / \log n \sim n / W(n), \quad r_{n+1} \geq r_n. \quad (26.3)$$

They are Theorem 1.1, Theorem 12.1, and Theorem 12.3 with the monotonicity of the ratios given by the strict log-convexity of Theorem 15.3. The probability measure  $\mu$  has all positive exponential moments finite. Its normalized Stieltjes transform satisfies

$$F(s) = \int \frac{d\mu(x)}{1 + sx}, \quad F(s) = 1 - sF(s)^2 F(sF(s)),$$

and, with  $\alpha = (3 - \sqrt{5})/2$ ,  $F(s) \sim s^{-\alpha}$  as  $s \rightarrow \infty$  (Theorem 12.2). These are inherited results, not new assertions of Part III.

| Item                     | What is proved                                                             | What remains needed                                                   |
|--------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| New consecutive ratios   | $r_n \sim n / \log n$                                                      | Additive corrections at constant scale                                |
| Quantitative ratio bound | Relative error $O(\sqrt{\log \log n / \log n})$                            | Additive error $O(w^p/n)$ after $R(n)$                                |
| Moment representation    | Relative tilted variance tends to zero                                     | Relative variance $d/n$ with error $O(w^q/n^2)$ is a sufficient input |
| Golden transform law     | Mass near zero and its leading constant                                    | Quantitative information about high-moment tilts in the far tail      |
| This continuation        | Uniform coefficient expansion; one-sided correction; conditional reduction | Curvature and complementary-endpoint estimates                        |

## 27 Why the new ratio theorem does not settle normalization

The convex-slope proof (Lemma 20.1 and the corollary after Proposition 20.2) uses secants over intervals of length  $\delta n$ . The global error in (26.1) contributes  $O(\log \log n / (\delta \log n))$  to such a secant, and curvature of the comparison function contributes  $O(\delta)$ . Optimizing gives a relative ratio error of order  $\sqrt{\log \log n / \log n}$ . This is much too large even to distinguish a constant correction in  $r_n$ .

In particular, a relative  $o(1)$  error in one ratio can accumulate over infinitely many ratios. Neither strict Hankel positivity nor strict iterated log-convexity provides the particular numerical curvature coefficient required below.

The  $s \rightarrow \infty$  transform law concerns  $x \downarrow 0$ . High moments tilt toward  $x$  of order  $n/W(n) \rightarrow \infty$ . Improving the former endpoint law does not by itself improve the latter regime. The exact transform equation still contains the whole sequence, but using it for high moments requires a new uniform asymptotic argument; differentiating a cumulative-mass law or treating a divergent ordinary series as convergent would not supply one.

Part I's normalizer  $\mathcal{N}_n$  of (8.4) satisfies  $B_n \exp(w^2 + 3w) \sim \mathcal{N}_n$  by the classical Bell asymptotics (6.6) [9]. The exact summability threshold is Proposition 8.2: if

$$r_n = R(n) + O(w^p/n)$$

for some fixed  $p$ , which is (8.7), then  $a_n/\mathcal{N}_n$  tends to a finite positive constant. (The manuscript re-derives this:  $\mathcal{N}_n/\mathcal{N}_{n-1} = R(n) + O(w^3/n)$  is (8.5), and the successive differences of  $\log(a_n/\mathcal{N}_n)$

are  $O(w^{\max(p,3)+1}/n^2)$ , as in the proof of Proposition 8.2.) This argument proves convergence, but does not determine the constant from its asymptotic local input.

More generally, absolute summability of  $|r_n - \mathcal{N}_n/\mathcal{N}_{n-1}|/r_n$  suffices. The fixed-power condition (8.7) is convenient rather than necessary.

## 28 A coefficient estimate that can be proved now

Direct coefficient extraction in (1.1) gives the exact endpoint recurrence

$$a_n = \sum_{k=0}^{n-1} a_k [z^{n-1-k}] A(z)^{k+2} = \sum_{m=0}^{n-1} a_{n-1-m} b_{n,m}, \quad b_{n,m} = [z^m] A(z)^{n+1-m}. \quad (28.1)$$

(The first sum is Part I's (2.1); the second puts  $m = n - 1 - k$ .) All terms are nonnegative. Only coefficients of finite degree are used.

**Lemma 28.1** (Uniform logarithmic-window coefficient expansion). *For every fixed  $C > 0$ , uniformly for integers  $0 \leq m \leq C \log n$ ,*

$$b_{n,m} = \frac{n^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} + O\left(\frac{(1+m)^6}{n^2}\right) \right). \quad (28.2)$$

*No local ratio smoothness hypothesis is used.*

*Proof.* Equation (26.1) implies  $(a_j/j!)^{1/j} \rightarrow 0$ . Consequently there is a finite  $K \geq 1$  with  $a_j \leq K^{j-1}j!$  for every  $j \geq 2$ . Put  $M = n + 1 - m$ . The coefficient in question is obtained by choosing nonconstant terms from  $M$  factors. The contribution using only  $m$  singletons is  $(M)_m/m!$ . Its normalized value is

$$\frac{(M)_m}{n^m} = \prod_{h=0}^{m-1} \left( 1 + \frac{1-m-h}{n} \right) = 1 - \frac{3m(m-1)}{2n} + O((1+m)^4/n^2).$$

The contribution with one degree-two term and  $m-2$  singletons is  $3(M)_{m-1}/(m-2)!$ , whose normalized value is

$$\frac{3m(m-1)}{n} \{1 + O((1+m)^2/n)\}.$$

The sum of these two contributions supplies the displayed first correction.

For completeness, all higher patterns admit a uniform bound. Let  $l_j$  count the chosen degree- $j$  terms for  $j \geq 2$ . Put  $\eta = \sum_{j \geq 2} (j-1)l_j$  and  $s = \sum_{j \geq 2} j l_j$ . The number of chosen nonconstant factors is  $m - \eta$ ; there are  $m - s$  singletons. After normalization by  $n^m/m!$ , the contribution is at most

$$n^{-\eta} \frac{m!}{(m-s)!} \prod_{j \geq 2} \frac{a_j^{l_j}}{l_j!} \leq \left( \frac{K m^3}{n} \right)^\eta.$$

Here  $s \leq 2\eta$ ,  $j! \leq m^{j-1}$  for  $2 \leq j \leq m$ , and  $(M)_{m-\eta} \leq n^{m-\eta}$ . For fixed  $\eta$ , the possible lists  $(l_j)$  are partitions of  $\eta$ , at most  $2^\eta$  in number. Summing over  $\eta \geq 2$  bounds their total by a geometric series of order  $m^6/n^2$ , since  $m = O(\log n)$ . The cases  $m = 0, 1$  are exact. This proves (28.2).  $\square$

For two small degrees, exact certificates are

$$C_{n,2} = 1 + \frac{3}{n} - \frac{4}{n^2}, \quad C_{n,3} = 1 + \frac{9}{n} + \frac{14}{n^2} - \frac{72}{n^3},$$

where  $C_{n,m} = m! b_{n,m}/n^m$  as in (30.3) below. They agree with the coefficient  $3m(m-1)/2$  above and are independently checked in the supplied code (`code/03-local-estimates-diagnose.py` asserts both as exact integer identities at  $n = 10, 100, 1000$  and  $10000$ ).

This disposes of one previously formal ingredient: the coefficient expansion itself, Part I's (8.1), is valid uniformly in the relevant growing window. The unresolved input is the coefficient ratio multiplying it, and the contribution outside the window.

## 29 An unconditional one-sided additive correction

**Theorem 29.1** (Lower correction from moment positivity). *Assuming the inherited results (26.1)–(26.3),*

$$r_n \geq \frac{n}{W(n)} + \frac{3}{2} + \frac{1}{2(1+W(n))} - O\left(\frac{W(n)^4}{n}\right). \quad (29.1)$$

In particular,

$$\liminf_{n \rightarrow \infty} \left( r_n - \frac{n}{W(n)} \right) \geq \frac{3}{2}.$$

*Proof.* Let  $L = \lceil 16w \rceil$  and  $t = n/r_n$ . For sufficiently large  $n$ ,  $L < n - 1$ . Monotonicity of the ratios gives

$$\frac{a_{n-1-m}}{a_n} = \prod_{j=0}^m r_{n-j}^{-1} \geq r_n^{-m-1}.$$

The  $k = 0$  term of (28.1), disjoint from  $0 \leq m \leq L$ , is  $[z^{n-1}]A^2 \geq 2a_{n-1}$ . Thus Lemma 28.1 implies

$$1 \geq \frac{1}{r_n} \sum_{m=0}^L \frac{t^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} - O(w^6/n^2) \right) + \frac{2}{r_n}.$$

By (26.3),  $t/w \rightarrow 1$ . The standard Chernoff estimate, obtained by applying Markov's inequality to  $e^{uZ}$  for  $Z \sim \text{Poisson}(t)$ , shows that the probability of  $Z > L$  and its fixed polynomial moments are  $O(n^{-10})$ . Indeed, for large  $n$ ,  $t \leq 2w$  and  $L \geq 16w$ ; the exponent at  $L/t \geq 8$  is at most  $-(16 \log 8 - 16 + 2)w$ , much smaller than  $-10 \log n$  asymptotically. First the preceding lower sum shows  $e^t/r_n$  is bounded. Completing the Poisson sums is consequently justified with an error smaller than  $O(w^6/n^2)$ , giving

$$1 \geq H_n(r_n) - O(w^6/n^2), \quad H_n(r) = \frac{e^{n/r}}{r} \left( 1 + \frac{3n}{2r^2} \right) + \frac{2}{r}. \quad (29.2)$$

This function is strictly decreasing for  $r > 0$ . Put

$$s_n = \frac{n}{w} + c(w), \quad c(w) = \frac{2 + 3w/2}{1 + w} = \frac{3}{2} + \frac{1}{2(1+w)}.$$

Since  $e^w = n/w$ , expansion yields

$$H_n(s_n) = 1 + O(w^4/n^2), \quad -H'_n(r) \asymp w^2/n$$

uniformly in a fixed bounded neighborhood of  $s_n$ . For a sufficiently large constant  $C_1$ , evaluation at  $s_n - C_1 w^4/n$  contradicts (29.2) if  $r_n$  lies below that point. Monotonicity handles points further below. This gives the stated error  $O(w^4/n)$ .  $\square$

The gap between this lower correction and the conjectured correction is

$$\left(2 + \frac{w}{2(1+w)^2}\right) - \left(\frac{3}{2} + \frac{1}{2(1+w)}\right) = \frac{1}{2} - \frac{1}{2(1+w)^2}.$$

It tends to  $1/2$ . This is precisely the extra contribution supplied by the local curvature calculation in the next section. The lower bound does not prove the conjectured upper bound.

### 30 An exact Poisson reduction and a sharp sufficient lemma

Fix  $L = \lceil 16w \rceil < n - 1$ , and define

$$p_t(m) = e^{-t} \frac{t^m}{m!}, \quad t = \frac{n}{r_n}, \quad (30.1)$$

$$D_{n,m} = r_n^{m+1} \frac{a_{n-1-m}}{a_n} = \prod_{j=0}^m \frac{r_n}{r_{n-j}}, \quad (30.2)$$

$$C_{n,m} = \frac{m! b_{n,m}}{n^m}, \quad (30.3)$$

$$J_n = \frac{1}{a_n} \sum_{m=L+1}^{n-1} a_{n-1-m} b_{n,m}. \quad (30.4)$$

Then (28.1) is exactly

$$1 = \frac{e^t}{r_n} \sum_{m=0}^L p_t(m) D_{n,m} C_{n,m} + J_n. \quad (30.5)$$

There is no infinite-series substitution into  $A$  here.

**Theorem 30.1** (Averaged endpoint criterion). *Suppose that for some fixed  $q \geq 0$  the actual sequence satisfies*

$$\sum_{m=0}^L p_t(m) D_{n,m} C_{n,m} = 1 + \frac{(3+d)t^2 + 2dt}{2n} + O(w^q/n^2), \quad (30.6)$$

$$J_n = \frac{2}{r_n} + O(w^q/n^2). \quad (30.7)$$

Then (8.7) holds for some fixed  $p$ , and the finer Bell normalization converges to a finite strictly positive constant.

*Proof.* By (26.3),  $t \asymp w$  and  $r_n \asymp n/w$ . Equations (30.5)–(30.7) first show  $e^t/r_n = 1 + O(w^2/n) + O(w^q/n^2)$ . Hence

$$t - \log r_n = O(w^2/n) + O(w^q/n^2).$$

Since  $r \mapsto n/r - \log r$  has derivative  $-(t+1)/r \asymp -w^2/n$  near its zero  $n/w$ , the mean value theorem implies  $r_n = n/w + O(1)$  (fixed powers of  $w$  divided by  $n$  tend to zero). Write  $r_n = n/w + \delta$ ,  $\delta = O(1)$ . Then

$$\frac{e^t}{r_n} = 1 - \frac{\delta w(1+w)}{n} + O(w^4/n^2), \quad t = w + O(w^2/n).$$

Insert these in (30.5). With  $Q = \max(q, 6)$ ,

$$\delta w(1+w) = \frac{(3+d)w^2 + 2dw}{2} + 2w + O(w^Q/n).$$

Therefore

$$\delta = \frac{2 + (3 + d)w/2 + d}{1 + w} + O(w^Q/n) = 2 + \frac{w}{2(1 + w)^2} + O(w^Q/n).$$

This is (8.7); Proposition 8.2 completes the proof.  $\square$

The two hypotheses are the missing estimates, not proved conclusions. The averaged criterion is weaker than a uniform local expansion: rare values of  $m$  may behave differently if their weighted contribution is controlled. It avoids asking for a density of  $\mu$  or a complete tail transseries. (In Part I's terms, under (30.6) and (30.7) the exact identity (30.5) is the formal balance (8.2) with an error  $O(w^q/n^2)$ , and the proof above is Part I's formal derivation of (8.3) made rigorous under those hypotheses.)

[Batch 86, 4 October 2026.] Part V proves (30.7) with  $q = 2$ , unconditionally (Theorem 44.1):  $J_n = 2/r_n + 5/(r_n r_{n-1}) + O(w^3/n^3)$ . Estimate (30.6) remains unproved; Part V reaches the convergence of the finer normalization without it (Theorem 46.1), in a proof that has not yet been independently reviewed.

There is also a logically minimal scalar defect. Define

$$\mathcal{E}_n = \frac{e^t}{r_n} \left\{ \sum_{m \leq L} p_t(m) D_{n,m} C_{n,m} - 1 - \frac{(3 + d)t^2 + 2dt}{2n} \right\} + J_n - \frac{2}{r_n}. \quad (30.8)$$

The exact identity says

$$1 = \frac{e^t}{r_n} \left( 1 + \frac{(3 + d)t^2 + 2dt}{2n} \right) + \frac{2}{r_n} + \mathcal{E}_n.$$

Thus the single estimate  $\mathcal{E}_n = O(w^q/n^2)$  suffices. It allows cancellation between the two errors, whereas (30.6)–(30.7) controls them separately. By substituting (8.7) into this identity one sees that some such fixed-power bound on  $\mathcal{E}_n$  is also necessary for (8.7). This scalar reformulation is sharp in precision, but the separated estimates explain what must be analyzed.

## 31 Local curvature: the moment representation's precise missing input

The following uniform condition is sufficient for the averaged estimate:

$$\log r_{n-j} - \log r_n = -\frac{dj}{n} + O\left(\frac{w^q}{n^2}\right), \quad 0 \leq j \leq L, \quad (31.1)$$

for a fixed  $q$ ; polynomial factors in  $j \leq 16w + 1$  can be absorbed into a larger  $q$ . Summing this relation in (30.2) gives, uniformly,

$$D_{n,m} = 1 + \frac{d m(m+1)}{2n} + O(w^{q'}/n^2). \quad (31.2)$$

Multiply by Lemma 28.1 and use

$$\mathbb{E}[Z(Z-1)] = t^2, \quad \mathbb{E}[Z(Z+1)] = t^2 + 2t, \quad Z \sim \text{Poisson}(t),$$

with the same Chernoff truncation as before. Under (31.1), this proves (30.6). It is now clear exactly where the coefficient  $d$  comes from. Monotonicity alone gives  $D_{n,m} \geq 1$  and loses this contribution.

The moment formulation is especially transparent. Define the probability tilt

$$d\mu_{n-1}(x) = \frac{x^{n-1}}{a_{n-1}} d\mu(x).$$

Then the following identities are exact:

$$\mathbb{E}_{\mu_{n-1}} X = r_n, \quad \frac{\text{Var}_{\mu_{n-1}} X}{r_n^2} = \frac{r_{n+1}}{r_n} - 1. \quad (31.3)$$

**Proposition 31.1** (Tilted variance sufficient for local curvature). *A sufficient input for (31.1) is the all-large-index estimate*

$$\frac{\text{Var}_{\mu_{i-1}} X}{r_i^2} = \frac{W(i)}{i(1+W(i))} + O\left(\frac{W(i)^q}{i^2}\right). \quad (31.4)$$

*Proof.* Take the logarithm of 1+ the right side of (31.3):

$$\log r_{i+1} - \log r_i = \frac{W(i)}{i(1+W(i))} + O(W(i)^{\max(q,0)}/i^2).$$

On  $[n-L, n]$ , the derivative of  $W(x)/(x(1+W(x)))$  is  $O(n^{-2})$ . Telescoping over  $j \leq L = O(w)$  therefore gives (31.1), after enlarging the fixed exponent to absorb  $j$  and  $j^2$ .  $\square$

The golden-moment ratio law gives only  $\text{Var}_{\mu_{i-1}} X/r_i^2 \rightarrow 0$ . Estimate (31.4) requires its leading value at scale  $1/i$ , and an error at scale  $\text{polylog}(i)/i^2$ . Even a first-order equivalent at scale  $1/i$  is insufficient for the displayed summable-error target unless its error is quantified much more sharply. The variance condition is a sufficient route, not asserted to be necessary: the averaged condition (30.6) could hold without it.

A conventional saddle analysis would aim at mean near  $n/w$ , variance near  $n/[w(1+w)]$ , and higher central moments with uniform errors. But Part II has not proved a density or a local tail law that permits such an analysis. Assuming such a density would move the gap rather than close it.

## 32 The complementary endpoint is an independent missing estimate

The  $k = 0$  term in (28.1) contains  $2a_{n-1}$ . For (30.7) one must bound everything else outside  $m \leq L$  by

$$\sum_{m=L+1}^{n-1} a_{n-1-m} b_{n,m} - 2a_{n-1} = O\left(a_n \frac{w^q}{n^2}\right). \quad (32.1)$$

Its left side is nonnegative for sufficiently large  $n$ , since  $k = 0$  lies in the complementary range. Thus this is a positive upper-bound problem, and it cannot rely on cancellation inside that range.

For orientation, degree concentration in the first two outer terms supplies a lower-order contribution. The  $k = 0$  term has  $2a_{n-1} + 2a_1 a_{n-2} + \dots$ ; the  $k = 1$  term has  $3a_1 a_{n-2} + \dots$ . Hence, for large enough  $n$  that both terms lie outside the endpoint window,

$$J_n - \frac{2}{r_n} \geq \frac{5a_{n-2}}{a_n} = \frac{5}{r_n r_{n-1}} \sim \frac{5w^2}{n^2}.$$

This explains why it would be wrong to discard the complementary error as smaller than  $w^2/n^2$ . A bound of that order is compatible with the desired normalization.

Similarly, the main Poisson endpoint has second-order contributions from two degree-two choices, degree-three choices, and the square of the local curvature. Its natural next error is of polynomial-logarithmic order divided by  $n^2$ , which is precisely sufficient after solving for the ratio. The current proof establishes the small-degree coefficient part, but does not establish a uniform upper bound for all configurations containing a large inner coefficient. The inherited  $r_n \sim n/\log n$  controls fixed neighboring ratios but does not automatically justify summing a growing number of such configurations.

[Batch 86, 4 October 2026.] Part V supplies that uniform upper bound by decomposing according to the position of the unique large inner index: Theorem 44.1 gives (32.1) with  $q = 2$ , and in fact the complementary sum to every fixed order, beginning  $2a_{n-1} + 5a_{n-2} + 24a_{n-3}$ .

### 33 Numerical stress tests and their limits

The accompanying computation generates  $a_0, \dots, a_{1600}$  as exact integers by the inherited triangular recurrence of Section 7 (the delivered run used the C adapter `code/03-local-estimates-compute_runtime.c`; its output table is not shipped). The script `code/03-local-estimates-diagnose.py` checks a prefix independently with Python, compares against Part I's stored high-precision ratios (`data/diagnostics.csv`), and checks (1.1) through degree 35 using independent finite polynomial multiplication. Exact Bell numbers are generated with the Bell triangle. Decimal logarithms, exponentials and a Newton solution of  $we^w = n$  provide the diagnostic arithmetic. Runs at 80 and 120 decimal digits agree: all reported diagnostics pass the comparison thresholds, with maximum observed absolute difference below  $2 \times 10^{-75}$  (`data/03-local-estimates-precision-check.json`). This checks numerical stability. No limiting claim is derived from those checks.

For each index we record

$$\Delta_n = r_n - R(n), \quad D_n^B = \log(a_n/B_n) - w^2 - 3w, \quad D_n^N = \log(a_n/\mathcal{N}_n).$$

The first normalization uses exact Bell numbers; the second uses the smooth leading model. Confusing them at finite  $n$  would contaminate a remainder fit. ( $D_n^B$  is Part I's  $D_n^{\text{num}}$  of (7.3); Table 3, whose  $B_n$  is  $\mathcal{B}_n$ , agrees with Table 1 at  $n \leq 1200$ , and its column  $r_n - R(n)$  is Part I's  $R_n - n/W(n)$  minus  $2 + w/(2(1+w)^2)$ .)

| $n$  | $r_n - R(n)$ | $n\Delta_n/w^2$ | $D_n^B$     | $a_n/(B_n e^{w^2+3w})$ |
|------|--------------|-----------------|-------------|------------------------|
| 100  | 0.11795489   | 1.029051        | -3.17251620 | 0.04189804             |
| 200  | 0.06549107   | 0.848172        | -3.11533505 | 0.04436364             |
| 400  | 0.03645943   | 0.723500        | -3.07837536 | 0.04603398             |
| 800  | 0.02023522   | 0.631585        | -3.05466187 | 0.04713866             |
| 1200 | 0.01430547   | 0.588026        | -3.04505401 | 0.04759374             |
| 1600 | 0.01117179   | 0.560602        | -3.03966580 | 0.04785088             |

Table 3: Exact integer inputs; high-precision decimal diagnostics. No limit is inferred from this finite table.

The observed scaled ratio residual  $n\Delta_n/w^2$  is a diagnostic for a candidate  $O(w^2/n)$  remainder, considerably stronger than an unscaled plot. Its bounded appearance is evidence only. A finite range cannot exclude a later term such as  $1/w^k$ , which would be much larger than any fixed  $w^p/n$  and could destroy normalization convergence.

For selected indices, the endpoint sum is computed exactly before Decimal conversion. Coefficients of  $A(z)^M$  are generated by the independent recurrence

$$b_0 = 1, \quad sb_s = \sum_{j=1}^s ((M+1)j - s)a_j b_{s-j},$$

which follows from  $A(A^M)' = MA'A^M$ . Every division is checked to have zero integer remainder. This directly measures  $J_n$  as 1– the endpoint sum, without assuming endpoint dominance.

| $n$  | $L$ | $n^2(J_n - 2/r_n)/w^2$ | $n^2\mathcal{E}_n/w^4$ |
|------|-----|------------------------|------------------------|
| 100  | 55  | 5.20367730             | 3.50364995             |
| 200  | 63  | 5.10012491             | 3.53256190             |
| 400  | 72  | 5.05204992             | 3.51066312             |
| 800  | 82  | 5.02778152             | 3.45070163             |
| 1200 | 87  | 5.01935819             | 3.40469189             |
| 1600 | 91  | 5.01500173             | 3.36919924             |

Table 4: The complementary mass and the complete balance defect are tested separately. The 5 suggested by the third column has a rigorous lower-bound explanation, but no matching upper-bound proof is supplied.

The complementary residual’s approach to 5 in its natural scaling (Table 4) agrees with the two explicit  $a_{n-2}$  contributions above. This does not prove a matching upper bound. We also test the curvature average, the local-window log-ratio error, and the complete scalar balance defect. Their full values and the cutoffs are in the CSV and JSON outputs:

- all indices: `data/03-local-estimates-diagnostics.csv`;
- the selected indices: `data/03-local-estimates-endpoint_diagnostics.json`;
- the window sweep: `data/03-local-estimates-window_sweep.json`.

Changing the window coefficient from 16 to 8, 12 and 24 checks sensitivity to truncation. Windows with the larger constants can require substantially larger  $n$  to be asymptotic; the diagnostic does not claim uniformity beyond the stated finite values.

The normalization seems compatible with a finite positive limit and with the summable ratio remainder. No certified value of that limit is supplied. In particular, proximity to  $e^{-3}$  in a finite range would not prove an exact constant. Initial coefficients affect the normalization constant, so a local all-orders expansion alone would not automatically identify it.

### 34 What has been resolved and the next precise task

The new proof and the old sufficient criterion are consistent. The new ratio law is an essential enabling result because it gives monotone ratios and a legitimate moment tilt; its error estimate remains far from the normalization precision.

This continuation proves the uniform coefficient expansion and the additive lower correction (29.1). It also proves every implication from the precise missing estimates to the desired ratio and Bell normalization. It does not prove those missing estimates for A088714, and therefore does not claim a proof of the full finer comparison.

The sharpest single remaining lemma in this formulation is

$$\mathcal{E}_n = O(W(n)^q/n^2) \quad \text{for some fixed } q.$$

The more informative two-part route is (30.6) plus (32.1). From the moment representation, (31.4) supplies a concrete sufficient curvature estimate. An attack through the endpoint recurrence should prove the complementary positive bound and bootstrap the local curvature without presupposing the target ratio expansion. Using the target expansion to justify the window estimate would be circular.

Useful next investigations are: (i) a positive combinatorial upper bound for configurations with a large inner coefficient; (ii) a stability theorem for the endpoint balance that controls discrete curvature from the exact recurrence; (iii) quantitative high-moment tilt estimates from the coupled resolvents, without assuming absolute continuity; and (iv) a higher-order calculation retaining the  $5a_{n-2}$  term, followed by a proved error bound. Only after those steps should one attempt a certified interval for the normalization constant or all-orders transseries inversion.

[Batch 86, 4 October 2026.] Items (i) and (iv) are carried out in Part V (Theorem 44.1), which then proves the finer comparison without the curvature estimate (Section 46); that proof has not yet been independently reviewed. A certified interval for the constant remains open (Research question 49.1).

## Part IV

# Densities, the correction profile, and the companion sequence

## 35 Source, scope and notation of Part IV

**Summary.** Let  $A(z) = 1 + zA(z)^2A(zA(z))$  and  $C(z) = (1 - zA(z))^{-1}$ , and let  $\mu, \nu$  be the probability measures of Part II with moments  $a_n$  and  $c_n$ . Part IV answers three of Part II's Research questions (Section 23).

- *Question 23.1, regularity and support.* Both measures have strictly positive real-analytic densities throughout  $(0, \infty)$  and support exactly  $[0, \infty)$ ; with  $\beta = (\sqrt{5} - 1)/2$  and  $\alpha = 1 - \beta$ ,  $p(x) \sim \pi^{-1} \sin(\pi\alpha) x^{-\beta}$  and  $q(x) \sim \pi^{-1} \sin(\pi\beta) x^{-\alpha}$  as  $x \downarrow 0$ , with every fixed derivative (Theorem 37.1, Corollary 37.8). A coupled inverse relation rules out every hypothetical positive density zero; a complex inversion argument gives the endpoint equivalents, without differentiating the distribution-function laws.
- *Question 23.3, second-order golden-ratio asymptotics.* There is a nonzero real-analytic function  $P$  with  $P(t+1) = -P(t)$  such that  $\log(s^\alpha F(s)) = P(\log_\varphi \log s)/\log s + O((\log s)^{-3})$ ,  $\varphi = (1 + \sqrt{5})/2$ , together with an explicit expansion to all inverse-logarithmic orders that converges for large  $\log s$ , with an exponentially smaller, separately bounded discrepancy (Theorem 38.1). An interval certificate proves  $P \not\equiv 0$ , so the correction oscillates and the  $O(1/\log s)$  rate is optimal (Corollary 38.3).
- *Question 23.6, the companion sequence.* For every fixed  $K$ ,  $c_n = \sum_{j=0}^K \gamma_j a_{n-1-j} + O_K(a_{n-K-2})$  with  $\gamma_j = [z^j]C(z)^2$ ; in particular  $c_n/a_{n-1} \rightarrow 1$  with first relative correction  $2 \log n/n$  (Theorem 39.1). The theorem is proved for every log-convex sequence with  $a_n/a_{n-1} \sim n/\log n$ , by a weighted-composition argument that assumes no fine expansion of the ratios.

Part IV does not bear on Part I's finer Bell normalization (Conjecture 8.1); Part V, from another manuscript of the same batch, proves it.

**Provenance.** Part IV is the part on A088714 and A088713 of manuscript 01 of batch 86 of the repository's incoming-reports intake: *Three advances on OEIS conjectures and asymptotics. Bridgeless toroidal maps, golden-ratio moment laws, and unitary-divisor partitions*, "Prepared with ChatGPT, building on Vladimir Reshetnikov's ProveIt research repository", dated 4 October 2026 (UTC) (a 2,894-line source and a 40-page PDF). It arrived as `oeis_advances.zip` in commit `ae9baa422` and was placed in commit `0f084afa9`. The manuscript pins the repository at commit `eaf08931cbfd0132dd36ab82a7743ccb02506dc7` (4 October 2026, 00:59:12 UTC); the source of this report is the same at the pin and at the placement commit, so the manuscript read Parts I–III as printed here. Its Part II (Sections 3–6) is printed here as Sections 36–39, and the corresponding passages of its Sections 1, 8 and 9 as this section and Sections 40 and 41. Its other two parts are printed elsewhere in the collection: its Section 2, a proof of the generating function that Peter Bala conjectured for A343093, is the report

`enumerative-combinatorics/a343093-bridgeless-toroidal-maps/`

and its Section 7, two-sided oscillation for A301981 and A301982, is Part II of the report

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a301981-unitary-divisor-partitions/`

(both paths relative to the collection’s root, `SetTheory/Cardinals/docs/reports/`). [Batch 86 reciprocal notes, 4 October 2026.] The A301981 Part is Sections 12–13 of that report (label prefix `udp:tw:`), and its Theorem 12.5, `udp:tw:thm:main`, is the two-sided oscillation. The A343093 report prints the manuscript’s introduction (Sections 1.1 and 1.4) and the map parts of its Sections 8 and 9, and keeps the manuscript-wide non-claims quoted in the Status paragraph below (internal reviews are not refereeing; a bounded search is not a priority claim; no OEIS update or submission was made); its Section 3 points to the sections and theorems of this Part. Part IV’s two programs and their outputs are shipped with the prefix `04-densities-profile-`. The manuscript’s  $\text{\LaTeX}$  source and delivery README were staged as the base of the A343093 report; they and the delivered PDF survive in the archive of the arrival commit.

**What is not reprinted.** Proposition 36.1 restates results of Parts I and II (the manuscript calls it a package of “established statements”) and is kept as a summary, with pointers to the theorems that prove it. Nothing else of the manuscript’s Sections 3–6 is omitted. Its introduction is absorbed into the summary above, and the passages of its Sections 8 and 9 that concern its other parts (the map and partition programs, the map diagnostics, the directions on maps and partitions) are printed with those parts.

**Editorial changes.** Every label carries the prefix `gdy:`; the manuscript’s labels are unchanged after it. References to “the pinned report” or “the repository manuscript” now point to Parts I–III, its question numbers are Part II’s Research questions 23.1, 23.3 and 23.6, and file names are the shipped ones. In the bibliography, the manuscript’s entries for A088714, A088713 and Python’s `decimal` module are merged with the existing ones, Huang and Wang’s article is added [11], and its entry for this report is replaced by internal references. Five symbols are renamed (see the notation paragraph); no normalization changed. A dated note in Section 38.6 and one in Section 41 record what Part V proves. Nothing else in the mathematics was changed.

**Status.** Part IV, like Parts I–III, is AI-assisted and unrefereed, and it is not formalized: no Lean or Rocq development checks any statement of this report, and placement in this collection confers no formal status. The manuscript reports “independent internal mathematical reviews”; it states that they are not external refereeing or machine-checked proofs. Its novelty claim is bounded: “The search did not identify earlier proofs of these particular new conclusions. It does not establish exhaustive historical priority”; its delivery README adds that a bounded literature search is not an exhaustive priority search. The global inversion theorem of Huang and Wang [11, Propositions 3.1–3.2] is imported, not re-proved (Theorem 37.3). Finite prefix agreement and numerical diagnostics check implementations and constants and prove no asymptotic statement; the interval certificate of Section 38.5 is part of the proof that  $P \neq 0$ . The manuscript’s statement that no OEIS update, repository change or external submission was made described the delivery.

**Relation to Part V.** Manuscript 02 of the same batch, printed as Part V, pins commit `b7e4f25b6`, an ancestor of manuscript 01’s pin made six minutes earlier; neither manuscript was in the repository at either pin, and neither cites the other. Both prove the fixed-order expansion of the companion sequence: Theorem 39.1 here, more general, is the statement printed as the theorem, and Part V’s proof of its Theorem 47.1 is printed there as a second route. Manuscript 01 calls the finer Bell normalization open; Part V proves it (Theorem 46.1), in a proof that has not yet been independently reviewed.

**Notation.** Section 36 fixes  $A, C, a_n, c_n, \mu, \nu, F, G, U, \alpha, \beta, \varphi$  and  $r_n = a_n/a_{n-1}$  (Part I's  $R_n$ , Part III's  $r_n$ , Part II's  $r_{n-1}$ ). Every other symbol of Part IV is local to the section that defines it, as in the manuscript, except that Sections 40 and 41, which discuss Sections 37–39, reuse their symbols with the same meanings; Table 5 lists the letters that also occur in Parts I–III or V. Five symbols are renamed:

1.  $\gamma_j = [z^j]C(z)^2$  in Section 39 is the manuscript's  $d_j$ , since  $d$  is the stable coordinate  $d(x)$  of Section 38 and Part I's  $w/(1+w)$ ;  $\gamma_j$  is Part V's  $\gamma_h$ ;
2.  $\omega_n = a_{n-1}$  in Section 39 is the manuscript's  $w_n$ , since  $w$  is  $W(n)$  in Parts I, III and V;
3.  $\theta$ , the exponent in Lemma 37.6, is the manuscript's  $\gamma$ ;
4.  $\varrho(y) = \log(1 + e^{-y})$  in Section 38 is the manuscript's  $r(y)$ , since  $r_n$  is the ratio;
5.  $\chi_n = (1 + e^{-x_{n-1}})^{-1}$  in the proof of Proposition 38.2 is the manuscript's  $c_n$ , since  $c_n$  is A088713.

Sections 36–40 do not use the Lambert function; Section 41 writes  $W(n)$  for it, as Part I does. (In manuscript 01's Section 7, printed in the A301981 report,  $W(n)$  denotes  $n^{\sigma^*/3}(\log n)^{m-1}$ ; that meaning does not occur in this report, where  $W(n)$  is always the solution of  $we^w = n$ .) The tempting false readings: Section 38's  $P$  is a function, the oscillation profile, not Part V's fixed truncation order  $P$ , and its  $Q_n(t)$  are coefficient functions, not Part I's  $Q_n$ ; its  $D(x)$  is the alternating invariant coordinate, not Part V's endpoint series  $D(z)$ ;  $H, K$  and  $h$  in Section 37 are Cauchy-transform maps, not Part I's polynomials  $H_n, K_n$ ; and  $w$  in Section 37 is a complex variable, never  $W(n)$ .

Table 5: Letters of Part IV that also occur in Parts I–III or V.

| letter                   | elsewhere in the report                                                               | Part IV                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $r_n, q_n$               | Part I $R_n$ ; Part II forward $r_n$ ; Part III $r_n$                                 | $r_n = a_n/a_{n-1}$ ; in Section 39 $q_n = \omega_{n+1}/\omega_n = r_n$ ; in Section 38 $p_n, q_n$ are orbit derivatives                             |
| $d, D$                   | Part I/III $d = w/(1+w)$ , $D_n(k)$ , $D_n^{\text{num}}$ , $D_{n,m}$ ; Part V $D(z)$  | $d(x) = g(x) - \beta x$ as in Part II's Research question 23.3, $d_n = d(x_n)$ ; $D(x)$ the alternating coordinate; $\mathcal{D} = \lambda^{-1}d/dt$ |
| $\gamma$                 | Part I $1 + 2/\lambda$ ; Part II a Tauberian exponent; Part V $\gamma_h = [z^h]C^2$   | $\gamma_j = [z^j]C^2$ (as Part V); the Abelian exponent is $\theta$                                                                                  |
| $P, Q$                   | Part I $Q_n$ ; Part V fixed order $P$ , $Q_n = R_n/R_{N+2}$                           | $P$ the profile; $Q_n(t)$ its coefficient functions                                                                                                  |
| $H, K, h$                | Part I $H_n, K_n$ ; Part II $H(s)$ ; Part III $H_n(r)$                                | $H(w) = w^2\mathcal{G}_\nu(w)$ , $K = 1/\mathcal{G}_\mu$ , $h$ the boundary map (Section 37); $h = g^{-1}$ (Section 38)                              |
| $w, \omega$              | $w = W(n)$ in Parts I, III, V                                                         | $w$ a complex variable (Section 37); $\omega_n = a_{n-1}$ (Section 39)                                                                               |
| $E, S, T$                | Part III $\mathcal{E}_n$ ; Part V $E_n, E_{n,m}, S_n, T_n(L)$ ; Part I $T_n(\lambda)$ | $E(x)$ a series (Section 38); $E_{K,n}$ a diagnostic; $S_n, T_{n,j}$ convolution quantities (Section 39)                                             |
| $\eta, \lambda, \varrho$ | Part III $\eta$ an excess degree; Part I $\lambda$ a Poisson mean                     | $\eta = 1/(\sqrt{5}x^2)$ (Section 38) or a positive constant (Section 39); $\lambda = \log \varphi$ ; $\varrho(y) = \log(1 + e^{-y})$                |
| $p, q$                   | Part I $p$ in (8.7); Part III $p_t(m)$ , $q$                                          | the densities of $\mu$ and $\nu$                                                                                                                     |

## 36 The nonlinear reversion pair and the inherited foundations

The ordinary generating functions in this Part are formal series

$$A(z) = \sum_{n \geq 0} a_n z^n = 1 + zA(z)^2 A(zA(z)), \quad C(z) = \sum_{n \geq 0} c_n z^n = \frac{1}{1 - zA(z)}. \quad (36.1)$$

Here  $a_n$  is A088714 and  $c_n$  is A088713. Their initial values are

$$(a_0, \dots, a_6) = (1, 1, 3, 13, 69, 419, 2809), \quad (c_0, \dots, c_6) = (1, 1, 2, 6, 24, 118, 674).$$

The definitions, reversion relation, and identity  $A(z) = C(zA(z))$  are recorded in the OEIS entries [1, 5]. The factor  $z$  makes the first equation triangular in degree and gives a unique formal solution. The series have radius of convergence zero; analytic statements below concern their Stieltjes transforms, not evaluations of these divergent ordinary series at nonzero arguments.

Parts I–III of this report are the starting point. Section 23 of Part II explicitly asks about regularity and support (Research question 23.1), second-order golden-ratio asymptotics (Research question 23.3), and the sharper companion comparison (Research question 23.6). We resolve these questions below. They are questions formulated in this research report; the live OEIS entries themselves do not state all three as conjectures.

**Proposition 36.1** (Foundation imported from Parts I and II). *There are unique probability measures  $\mu, \nu$  on  $[0, \infty)$  with*

$$a_n = \int_0^\infty t^n d\mu(t), \quad c_n = \int_0^\infty t^n d\nu(t).$$

*They have every positive exponential moment, unbounded support, and no atom at zero. Define, for  $s > 0$ ,*

$$F(s) = \int \frac{d\mu(t)}{1+st}, \quad G(s) = \int \frac{d\nu(t)}{1+st}, \quad U(s) = sF(s). \quad (36.2)$$

*Then*

$$F(s) = 1 - sF(s)^2 F(sF(s)), \quad G(s) = \frac{1}{1+U(s)}, \quad U^{-1}(s) = s(1+U(s)). \quad (36.3)$$

*Here  $U$  is an increasing homeomorphism of  $(0, \infty)$  onto itself. With*

$$\beta = \frac{\sqrt{5}-1}{2}, \quad \alpha = 1 - \beta = \beta^2, \quad \varphi = \beta^{-1},$$

*the endpoint equivalents are*

$$F(s) \sim s^{-\alpha}, \quad G(s) \sim s^{-\beta} \quad (s \rightarrow \infty), \quad (36.4)$$

$$\mu([0, t]) \sim \frac{\sin(\pi\alpha)}{\pi\alpha} t^\alpha, \quad \nu([0, t]) \sim \frac{\sin(\pi\beta)}{\pi\beta} t^\beta \quad (t \downarrow 0). \quad (36.5)$$

*The coefficients are log-convex, and*

$$r_n := \frac{a_n}{a_{n-1}} \sim \frac{n}{\log n}. \quad (36.6)$$

*For every integer  $N \geq 0$ , the finite expansion of  $F(s)$  has the exact signed remainder*

$$F(s) = \sum_{j=0}^N (-1)^j a_j s^j + (-1)^{N+1} s^{N+1} \int \frac{t^{N+1}}{1+st} d\mu(t). \quad (36.7)$$

**Dependency and proof mechanism.** This proposition packages established statements of Part II (Theorems 12.1, 12.2 and 12.3, the log-convexity of Theorem 15.3, and the remainder formula (16.4)), which rests on Part I's growth theorem, rather than assigning them a new proof or a formal verification status here. Its construction iterates a moment-preserving map: a positive block operator gives  $(1 - zB(z))^{-1}$ , and a positive full-Fock operator turns a moment series  $R$  into the solution  $M = R(zM)$ . Their composition is  $M = 1 + zM^2B(zM)$ . Each iteration stabilizes one more coefficient. Tightness and uniform integrability give the two limiting measures. The proved

coefficient growth makes their exponential generating functions entire and hence their moment problems determinate. Weak limits of compact approximants give (36.3) on the negative real axis.

The leading constant in (36.4) is also part of the prior theorem. For orientation, put  $g(x) = \log U(e^x)$  and  $d(x) = g(x) - \beta x$ . The inverse identity implies  $d(x) = -\beta d(g(x)) - \beta \varepsilon(x)$  with  $\varepsilon(x) = -\log(1 - F(e^x)) \rightarrow 0$ . Iteration to a compact interval, followed by a contraction bound, proves  $d(x) \rightarrow 0$ . The Stieltjes Tauberian argument then gives (36.5). Log-convexity and the coefficient logarithmic growth give (36.6). Formula (36.7) follows directly from the finite geometric identity and will make the new oscillation certificate independent of an infinite numerical-series approximation.

The new proofs separate into three directions. Complex inversion improves the integrated endpoint laws to analytic densities and derivative asymptotics. Real inverse iteration identifies the missing oscillatory correction. A composition estimate sharpens the coefficient comparison without requiring the finer Bell-number conjecture.

### 37 A088714: positive analytic densities on the full half-line

The moment representation of Part II leaves a concrete question: can either of its two canonical measures have a singular component, a positive atom, or a gap in its support? We resolve all three issues. The proof combines a global inversion theorem with a descent argument specific to the coupled transforms. A separate argument then determines the densities at the origin; it does not differentiate the previously known distribution-function equivalents.

Throughout this section,  $\mu$  and  $\nu$  are the canonical probability measures on  $[0, \infty)$  with moment series  $A$  and  $C = (1 - zA)^{-1}$ , respectively. We use their inherited finite moments, absence of atoms at zero, and the negative-axis identity

$$s = U(s)(1 + U(U(s))), \quad U(s) = s \int_{[0, \infty)} \frac{d\mu(t)}{1 + st}. \quad (37.1)$$

For the endpoint calculation we also use the inherited law

$$\nu([0, t]) \sim \frac{\sin(\pi\beta)}{\pi\beta} t^\beta, \quad t \downarrow 0, \quad \beta = \frac{\sqrt{5} - 1}{2}, \quad \alpha = 1 - \beta = \beta^2. \quad (37.2)$$

No assertion about the existence of a density is included in these inputs.

**Theorem 37.1** (Regularity, full support, and density at zero). *Both  $\mu$  and  $\nu$  are absolutely continuous with respect to Lebesgue measure. Their densities  $p$  and  $q$  have strictly positive, real-analytic versions on  $(0, \infty)$ , and*

$$\text{supp } \mu = \text{supp } \nu = [0, \infty). \quad (37.3)$$

*In particular, neither law has atoms, a singular continuous component, or a gap in its support. Moreover,*

$$p(x) \sim \frac{\sin(\pi\alpha)}{\pi} x^{-\beta}, \quad q(x) \sim \frac{\sin(\pi\beta)}{\pi} x^{-\alpha}, \quad x \downarrow 0. \quad (37.4)$$

*The constants in the two density equivalents are equal because  $\alpha + \beta = 1$ .*

#### 37.1 The analytic inverse selected by the moment laws

Write

$$\mathcal{G}_\mu(z) = \int \frac{d\mu(t)}{z - t}, \quad \mathcal{G}_\nu(z) = \int \frac{d\nu(t)}{z - t}, \quad K(z) = \frac{1}{\mathcal{G}_\mu(z)}.$$

These functions are holomorphic on  $\mathbb{C} \setminus [0, \infty)$ . Each Cauchy transform maps the upper half-plane into the lower half-plane, so  $K$  maps the upper half-plane into itself. A real argument  $x < 0$  gives  $K(x) < 0$ .

The renewal identity, initially justified on the negative axis, extends by the identity theorem to give

$$\mathcal{G}_\nu(z) = \frac{1}{z(1 - \mathcal{G}_\mu(z))}. \quad (37.5)$$

Define

$$H(w) = w^2 \mathcal{G}_\nu(w) = \frac{w}{1 - \mathcal{G}_\mu(w)}. \quad (37.6)$$

Since  $\nu$  is a probability measure with finite second moment, polynomial division yields the integral representation

$$H(w) = w + m_1(\nu) + \int \frac{t^2}{w - t} d\nu(t). \quad (37.7)$$

In the present normalization  $m_1(\nu) = 1$ .

**Lemma 37.2** (The exact inverse identity). *For every  $z$  in the upper half-plane,*

$$H(K(z)) = z. \quad (37.8)$$

*Proof.* For  $s > 0$  one has  $\mathcal{G}_\mu(-1/s) = -U(s)$  and  $K(-1/s) = -1/U(s)$ . Consequently, (37.6) and (37.1) give

$$H(K(-1/s)) = -\frac{1}{U(s)(1 + U(U(s)))} = -\frac{1}{s}.$$

The composition  $H \circ K$  is holomorphic on  $\mathbb{C} \setminus [0, \infty)$ :  $K$  maps each half-plane into itself and the negative real axis into itself. The identity theorem now proves (37.8).  $\square$

Thus the required inverse relation follows from identities of actual transforms. It does not depend on evaluating the divergent series  $A$  at a positive argument. In free-probability terminology, (37.7) identifies  $\mu$  as the rate-one free compound Poisson law with jump distribution  $\nu$ . The argument below uses the explicit analytic representation.

### 37.2 The global inversion input and its hypotheses

We use the following special form of the global inversion theorem of Huang and Wang [11, Propositions 3.1–3.2].

**Theorem 37.3** (Global inversion; imported result). *Let  $\tau$  be a nonzero finite positive measure on  $\mathbb{R}$ , let  $b \in \mathbb{R}$ , and put*

$$H_\tau(w) = w + b + \int \frac{d\tau(t)}{w - t}, \quad I_\tau(u, v) = \int \frac{d\tau(t)}{(u - t)^2 + v^2}.$$

Define

$$v_\tau(u) = \inf\{v > 0 : I_\tau(u, v) < 1\}, \quad \Omega_\tau = \{u + iv : v > v_\tau(u)\}.$$

*Then  $v_\tau$  is finite and continuous, and  $v_\tau(u) = 0$  exactly when  $\int (u - t)^{-2} d\tau(t) \leq 1$ . The map  $H_\tau : \Omega_\tau \rightarrow \mathbb{C}^+$  is a holomorphic bijection. It and its inverse extend continuously as mutually inverse maps between their closures. The boundary is  $\partial\Omega_\tau = \{u + iv_\tau(u) : u \in \mathbb{R}\}$ , and*

$$h_\tau(u) = H_\tau(u + iv_\tau(u))$$

*is a strictly increasing homeomorphism of  $\mathbb{R}$  onto itself. For  $v_\tau(u) > 0$  one has  $I_\tau(u, v_\tau(u)) = 1$ .*

Here and below a boundary value of  $H_\tau$  at a real point means its continuous extension from  $\Omega_\tau$ . The theorem is a direct specialization of the cited Nevanlinna integral form: take  $d\sigma(t) = (1+t^2)^{-1}d\tau(t)$  and replace the real constant by  $b - \int t d\sigma(t)$ . All these integrals are finite because  $\tau$  is finite.

For (37.7), set

$$d\tau(t) = t^2 d\nu(t), \quad b = m_1(\nu). \quad (37.9)$$

The second moment of  $\nu$  makes  $\tau$  finite; its first moment is positive, so  $\tau \neq 0$ . Thus every hypothesis of Theorem 37.3 holds. Suppress the subscript  $\tau$  and write  $v$ ,  $\Omega$ , and  $h$  for the resulting objects.

For  $z \in \mathbb{C}^+$ , (37.8) implies  $\operatorname{Im} H(K(z)) > 0$ . The integral formula gives

$$\operatorname{Im} H(u + iy) = y \left( 1 - \int \frac{t^2}{(u-t)^2 + y^2} d\nu(t) \right). \quad (37.10)$$

It follows that  $K(z) \in \Omega$ . Uniqueness of the inverse in Theorem 37.3 therefore identifies  $K$  with its global inverse. In particular,  $K$  has a continuous, injective extension to  $\mathbb{C}^+ \cup \mathbb{R}$ , and

$$K(h(u)) = u + iv(u), \quad \operatorname{Re} K(x) = h^{-1}(x). \quad (37.11)$$

The absence of an atom of  $\nu$  at zero gives

$$\int \frac{t^2}{(0-t)^2} d\nu(t) = 1.$$

Hence  $v(0) = 0$ , while (37.7) gives the boundary value  $H(0) = m_1(\nu) - \int t d\nu(t) = 0$ . The evaluation is absolutely convergent, and it also follows by dominated convergence along the positive imaginary axis. Therefore

$$h(0) = 0, \quad K(0) = 0, \quad K(x) \neq 0 \ (x \neq 0), \quad \operatorname{Re} K(x) > 0 \ (x > 0). \quad (37.12)$$

### 37.3 Absolute continuity and the descent that excludes zeros

On every compact interval in  $(0, \infty)$ , the reciprocal  $\mathcal{G}_\mu = 1/K$  has finite, continuous boundary values. This already proves local absolute continuity. Indeed, the Poisson integral of  $\mu$  is  $-\pi^{-1} \operatorname{Im} \mathcal{G}_\mu(x + iy)$ . On such an interval it converges uniformly to a continuous function, whereas integration against a compactly supported continuous test function converges to integration against  $\mu$ . The two limits identify the density as

$$p(x) = -\frac{1}{\pi} \operatorname{Im} \mathcal{G}_\mu(x) = \frac{\operatorname{Im} K(x)}{\pi |K(x)|^2}, \quad x > 0. \quad (37.13)$$

The inherited absence of an atom at zero, together with support in  $[0, \infty)$ , proves that  $\mu$  is globally absolutely continuous. At this point  $p$  is nonnegative and continuous; positivity still needs proof.

**Lemma 37.4** (Descent of a hypothetical density zero). *Let  $Z = \{x > 0 : p(x) = 0\}$ . If  $x \in Z$ , then*

$$w = K(x) \in Z, \quad 0 < w < x, \quad K(w) > 1. \quad (37.14)$$

*Proof.* By (37.13),  $K(x)$  is real, and (37.12) makes  $w = K(x) > 0$ . Since  $K(x)$  lies on  $\partial\Omega$ , one has  $v(w) = 0$ . Thus the vertical path  $w + iy$ ,  $y > 0$ , lies in  $\Omega$ , and the global boundary inverse gives

$$\lim_{y \downarrow 0} H(w + iy) = H(w) = x. \quad (37.15)$$

The value  $\mathcal{G}_\mu(w) = 1/K(w)$  is finite, since  $w > 0$  and  $K$  vanishes only at zero. If  $1 - \mathcal{G}_\mu(w) = 0$ , the quotient in (37.6) would have unbounded modulus along this vertical path, contradicting (37.15). We may therefore pass to the boundary in that quotient and obtain

$$\mathcal{G}_\mu(w) = 1 - \frac{w}{x}. \quad (37.16)$$

This is real, so (37.13) shows that  $w \in Z$ . Applying (37.12) at  $w$  now gives  $\mathcal{G}_\mu(w) = 1/K(w) > 0$ . Equation (37.16) consequently gives  $w < x$ . It also gives  $\mathcal{G}_\mu(w) < 1$ , because  $w > 0$  and  $x > 0$ . Hence  $K(w) > 1$ .  $\square$

**Proposition 37.5** (Every positive point has positive density). *The set  $Z$  is empty.*

*Proof.* Suppose  $x_0 \in Z$  and define  $x_{j+1} = K(x_j)$  for  $j \geq 0$ . Lemma 37.4 makes this an infinite strictly decreasing sequence of positive elements of  $Z$ . Equation (37.16) gives

$$\frac{1}{x_{j+2}} = 1 - \frac{x_{j+1}}{x_j}, \quad x_j - x_{j+1} = \frac{x_j}{x_{j+2}} > 1.$$

It follows that  $x_j < x_0 - j$  for every  $j \geq 1$ , contradicting positivity once  $j > x_0$ .  $\square$

The feedback identity forces a decrease greater than one at every hypothetical zero. Merely knowing that the measures have mass arbitrarily close to zero and unbounded support would not exclude intermediate gaps.

### 37.4 Analyticity and the second density

For  $x > 0$ , write  $K(x) = u + iv$ ; the preceding proposition gives  $v > 0$ . This point lies in the open upper half-plane and on  $\partial\Omega$ , so

$$\int \frac{t^2}{(u-t)^2 + v^2} d\nu(t) = 1.$$

Differentiating (37.7) at this interior point and using this equality gives

$$\operatorname{Re} H'(u + iv) = 2v^2 \int \frac{t^2}{((u-t)^2 + v^2)^2} d\nu(t) > 0. \quad (37.17)$$

Thus  $H'(K(x)) \neq 0$ . The holomorphic inverse function theorem extends  $K$  across a neighborhood of  $x$ . Its reciprocal extends there as well, and (37.13) proves that  $p$  is real analytic.

In (37.5), the boundary denominator cannot vanish at any  $x > 0$ , because  $\operatorname{Im} \mathcal{G}_\mu(x) = -\pi p(x) < 0$ . Consequently  $\mathcal{G}_\nu$  has a finite holomorphic continuation through every positive point. The same Poisson-inversion argument proves local absolute continuity of  $\nu$  and the formula

$$q(x) = \frac{p(x)}{x|1 - \mathcal{G}_\mu(x)|^2} > 0, \quad x > 0. \quad (37.18)$$

It is real analytic. The absence of an atom at zero completes global absolute continuity. Strict positivity on every positive interval proves (37.3) for both laws.

For completeness, the boundary parametrization gives exact simultaneous density formulas. For  $u > 0$ , the number  $v(u)$  is positive and uniquely determined by

$$\int \frac{t^2}{(u-t)^2 + v(u)^2} d\nu(t) = 1, \quad h(u) = H(u + iv(u)). \quad (37.19)$$

Equations (37.13) and (37.18) become

$$\begin{aligned} p(h(u)) &= \frac{v(u)}{\pi(u^2 + v(u)^2)}, \\ q(h(u)) &= \frac{v(u)}{\pi h(u)((u-1)^2 + v(u)^2)}. \end{aligned} \quad (37.20)$$

These identities are exact. They characterize the density using the same probability law  $\nu$ , so they are not by themselves a closed-form numerical solution.

### 37.5 A complex Abelian lemma at the origin

The transition from a distribution-function asymptotic to a density asymptotic requires more than differentiation. We first obtain a uniform complex-transform limit away from the real support and then invert  $H$  at an interior point of the upper half-plane.

**Lemma 37.6** (Scaled Cauchy transforms). *Let  $\sigma$  be a probability measure on  $[0, \infty)$ , and suppose*

$$N(t) = \sigma([0, t]) \sim ct^\theta \quad (t \downarrow 0), \quad c > 0, \quad 0 < \theta < 1.$$

*Then, locally uniformly for  $\zeta \in \mathbb{C} \setminus [0, \infty)$ ,*

$$r^{1-\theta} \mathcal{G}_\sigma(r\zeta) \longrightarrow -\frac{c\pi\theta}{\sin(\pi\theta)} (-\zeta)^{\theta-1} \quad (r \downarrow 0). \quad (37.21)$$

*The power uses the branch of  $\log(-\zeta)$  that is real for  $\zeta < 0$ .*

*Proof.* The hypothesis implies  $N(0) = 0$  and a global bound  $N(t) \leq C_0 t^\theta$  for a finite constant  $C_0$ . Integration by parts, followed by  $t = ry$ , gives

$$\mathcal{G}_\sigma(z) = -\int_0^\infty \frac{N(t)}{(z-t)^2} dt, \quad r^{1-\theta} \mathcal{G}_\sigma(r\zeta) = -\int_0^\infty \frac{N(ry)/r^\theta}{(\zeta-y)^2} dy.$$

On each compact subset of the slit plane, the absolute value of the integrand is bounded by a constant times  $y^\theta/(1+y)^2$ , which is integrable. Dominated convergence is therefore locally uniform and gives the limit  $-c \int_0^\infty y^\theta (\zeta-y)^{-2} dy$ . For  $\zeta = -a < 0$ , Euler's beta integral evaluates it as

$$-ca^{\theta-1} \Gamma(\theta+1) \Gamma(1-\theta) = -\frac{c\pi\theta}{\sin(\pi\theta)} a^{\theta-1}.$$

Both expressions are holomorphic on the slit plane, so the identity theorem evaluates the limit everywhere in that plane.  $\square$

Apply this lemma to (37.2). Since  $\alpha = 1 - \beta$ , it gives, locally uniformly for  $\zeta \in \mathbb{C}^+$ ,

$$r^\alpha \mathcal{G}_\nu(r\zeta) \longrightarrow -(-\zeta)^{-\alpha} = e^{-i\pi\beta} \zeta^{-\alpha}. \quad (37.22)$$

The last expression uses  $0 < \arg \zeta < \pi$ .

### 37.6 Inversion at a nonreal point gives the density constants

**Proposition 37.7** (Boundary Cauchy transforms at zero). *The continuous boundary values on  $(0, \infty)$  satisfy*

$$K(x) \sim e^{i\pi\alpha}x^\beta, \quad \mathcal{G}_\mu(x) \sim e^{-i\pi\alpha}x^{-\beta}, \quad \mathcal{G}_\nu(x) \sim e^{-i\pi\beta}x^{-\alpha} \quad (x \downarrow 0). \quad (37.23)$$

Here a complex equivalent means that the quotient tends to 1.

*Proof.* Use  $H(w) = w^2\mathcal{G}_\nu(w)$ , take  $r = x^\beta$  in (37.22), and note that  $\beta(2 - \alpha) = \beta(1 + \beta) = 1$ . Thus

$$\frac{H(x^\beta\zeta)}{x} \longrightarrow e^{-i\pi\beta}\zeta^{1+\beta} \quad (37.24)$$

locally uniformly in  $\mathbb{C}^+$ . The equation

$$e^{-i\pi\beta}\zeta^{1+\beta} = 1$$

has a simple root at  $\zeta_0 = e^{i\pi\alpha}$ , because  $\alpha(1 + \beta) = \beta$ .

Choose an arbitrarily small closed disk about  $\zeta_0$  contained strictly in the upper half-plane and containing no other zero of  $e^{-i\pi\beta}\zeta^{1+\beta} - 1$ . Uniform convergence on its boundary and Rouché's theorem show that, for all sufficiently small  $x > 0$ , the equation  $H(x^\beta\zeta) = x$  has a root in this disk. Let  $w = x^\beta\zeta$  be that root. It lies in  $\mathbb{C}^+$  and has  $\text{Im } H(w) = 0$ . By (37.10) and strict monotonicity of the integral there as a function of the positive imaginary part,  $w$  lies on  $\partial\Omega$ . The boundary bijection in Theorem 37.3 now forces  $w = K(x)$ . Since the disk can be arbitrarily small,  $K(x)/x^\beta \rightarrow e^{i\pi\alpha}$ .

Taking reciprocals proves the second equivalent in (37.23). Finally, (37.5) and  $|\mathcal{G}_\mu(x)| \rightarrow \infty$  give

$$\mathcal{G}_\nu(x) \sim -\frac{1}{x\mathcal{G}_\mu(x)} \sim -e^{i\pi\alpha}x^{\beta-1} = e^{-i\pi\beta}x^{-\alpha}.$$

□

Taking negative imaginary parts of (37.23) proves (37.4) and finishes Theorem 37.1. The proof uses a complex disk a fixed positive distance from the real axis after scaling. It does not assert uniformity of an Abelian theorem up to the support, where such uniformity would require an additional argument.

**Corollary 37.8** (Every fixed derivative at the origin). *For each fixed integer  $j \geq 0$ ,*

$$\begin{aligned} p^{(j)}(x) &\sim \frac{\sin(\pi\alpha)}{\pi}(-1)^j(\beta)_j x^{-\beta-j}, \\ q^{(j)}(x) &\sim \frac{\sin(\pi\beta)}{\pi}(-1)^j(\alpha)_j x^{-\alpha-j}, \quad x \downarrow 0, \end{aligned} \quad (37.25)$$

where  $(a)_0 = 1$  and  $(a)_j = a(a+1)\cdots(a+j-1)$  for  $j \geq 1$ . In particular, each density is strictly decreasing and strictly convex on a sufficiently short interval adjacent to zero.

*Proof.* The same Rouché argument may be made uniform when the right-hand side 1 in the scaled equation is replaced by a complex variable  $z$  in a small disk about 1. Choose the disk small enough that  $e^{i\pi\alpha}z^\beta$  stays in a fixed disk compactly contained in  $\mathbb{C}^+$ . The limiting inverse is holomorphic with nonzero derivative. Local uniform convergence of the scaled  $H$  also gives local uniform convergence of its derivatives by Cauchy's formula, so the nearby roots remain noncritical and define holomorphic inverses by the implicit function theorem. Uniqueness identifies these inverses with  $x^{-\beta}K(xz)$

for  $\text{Im } z > 0$ , and then with their continuations through the positive axis. They converge locally uniformly to  $e^{i\pi\alpha} z^\beta$ . Consequently

$$x^\beta \mathcal{G}_\mu(xz) \longrightarrow e^{-i\pi\alpha} z^{-\beta}, \quad x^\alpha \mathcal{G}_\nu(xz) \longrightarrow e^{-i\pi\beta} z^{-\alpha}$$

locally uniformly in that disk. Cauchy's differentiation formula permits any fixed number of derivatives at  $z = 1$ . Taking negative imaginary parts gives (37.25). The cases  $j = 1, 2$  give the final assertion.  $\square$

### 37.7 What this resolves and what remains

Theorem 37.1 resolves Part II's Research question 23.1 about regularity and support of both canonical laws. The additional boundary argument provides the density equivalents that were expressly left unjustified by the earlier cumulative-mass estimates. The general global inversion theorem is an imported tool; the exclusion of every positive density zero comes from the particular feedback identity  $H(w) = w/(1 - \mathcal{G}_\mu(w))$  and Lemma 37.4.

These results leave substantive questions. They do not give a multiplicative equivalent for the far tails or a complete large- $n$  asymptotic for the moments. Nor does local decrease near zero prove global monotonicity of either density. Useful next problems are to determine whether  $p$  or  $q$  is globally decreasing, to obtain quantitative large- $x$  boundary estimates from (37.6), and to turn (37.19) into certified density enclosures with a proved convergence rate. The derivative equivalents are statements for each fixed order  $j$ ; they do not establish complete monotonicity on one common interval for all orders.

## 38 A complete oscillatory expansion for the golden-ratio laws

Part II asks (Research question 23.3) whether the leading golden-ratio asymptotics for A088714 admit an explicit second-order expansion, and whether the correction oscillates along inverse orbits. This section answers both questions. The correction is of order  $1/\log s$ , has a nonzero real-analytic profile periodic in  $\log \log s$ , and admits a convergent expansion in inverse powers of  $\log s$ , with an error smaller than all those powers. The profile changes sign after one golden-ratio dilation of the logarithmic variable. An exact interval certificate proves that this oscillation is present in the particular OEIS solution, rather than merely permitted by the functional equation.

### 38.1 Inherited input and the main statement

We use the following results of Part II: the theorems entitled *Moment and analytic structure* and *Golden-ratio endpoint laws* (Theorems 12.1 and 12.2), together with its Stieltjes remainder formula (16.4). In the notation of Part II,

$$F(s) = \int_{[0,\infty)} \frac{d\mu(t)}{1+st}, \quad U(s) = sF(s), \quad G(s) = \frac{1}{1+U(s)}. \quad (38.1)$$

Here  $\mu$  is the probability measure with moments A088714,  $G$  is the Stieltjes transform for A088713,  $U$  is an increasing homeomorphism of  $(0, \infty)$  onto itself, and

$$U^{-1}(t) = t(1+U(t)), \quad U(s) \sim s^\beta, \quad \beta = \frac{\sqrt{5}-1}{2}. \quad (38.2)$$

We put

$$\varphi = \frac{1 + \sqrt{5}}{2} = \frac{1}{\beta}, \quad \alpha = 1 - \beta = \beta^2, \quad \lambda = \log \varphi, \quad \mathcal{D} = \lambda^{-1} \frac{d}{dt}. \quad (38.3)$$

The letter  $P$  below denotes an analytic function, not a polynomial. Its definition is constructive and canonical; it is not a freely chosen periodic multiplier.

**Theorem 38.1** (Complete golden-ratio expansion). *There is a nonzero real-valued real-analytic function  $P$  on  $\mathbb{R}$  such that*

$$P(t+1) = -P(t), \quad P(t+2) = P(t). \quad (38.4)$$

For every fixed integer  $N \geq 1$ , as  $s \rightarrow \infty$ ,

$$\log(s^\alpha F(s)) = \sum_{n=1}^N \frac{Q_n(\log_\varphi \log s)}{(\log s)^{2n-1}} + O((\log s)^{-2N-1}), \quad (38.5)$$

$$\log(s^\beta G(s)) = -\sum_{n=1}^N \frac{Q_n(\log_\varphi \log s)}{(\log s)^{2n-1}} + O((\log s)^{-2N-1}), \quad (38.6)$$

where an empty product is 1 and

$$Q_n(t) = \frac{5^{(1-n)/2}}{n!} \left[ \prod_{j=n}^{2n-2} (\mathcal{D} - j) \right] P(t)^n. \quad (38.7)$$

The remainders hold uniformly over the oscillation phase. Moreover, for all sufficiently large  $x$ , the series

$$S(x) = \sum_{n=1}^{\infty} \frac{Q_n(\log_\varphi x)}{x^{2n-1}} \quad (38.8)$$

converges absolutely, uniformly over the phase, and

$$\log(s^\alpha F(s)) = S(\log s) + O(s^{-\beta}), \quad (38.9)$$

$$\log(s^\beta G(s)) = -S(\log s) + O(s^{-1}). \quad (38.10)$$

In particular a first correction beyond all inverse-logarithmic orders can be isolated explicitly:

$$s^\alpha F(s) = \exp(S(\log s)) - s^{-\beta} + O(s^{-1}). \quad (38.11)$$

For example,

$$Q_1 = P, \quad Q_2 = \frac{P\mathcal{D}P - P^2}{\sqrt{5}}, \quad Q_3 = \frac{1}{30}(\mathcal{D} - 3)(\mathcal{D} - 4)P^3. \quad (38.12)$$

In general  $Q_n(t+1) = (-1)^n Q_n(t)$ : the odd-indexed coefficient functions are antiperiodic and the even-indexed ones are 1-periodic. Thus, writing  $x = \log s$  and  $p = P(\log_\varphi x)$ ,

$$s^\alpha F(s) = 1 + \frac{p}{x} + \frac{p^2}{2x^2} + O(x^{-3}), \quad (38.13)$$

$$s^\beta G(s) = 1 - \frac{p}{x} + \frac{p^2}{2x^2} + O(x^{-3}). \quad (38.14)$$

There is no constant coefficient of  $1/\log s$ : the leading correction is the nonconstant sign-changing function  $P$  evaluated at  $\log_\varphi \log s$ .

### 38.2 The exact perturbed Fibonacci recurrence

Set

$$g(x) = \log U(e^x), \quad h = g^{-1}, \quad d(x) = g(x) - \beta x, \quad \varrho(y) = \log(1 + e^{-y}). \quad (38.15)$$

The inherited equivalent in (38.2) gives  $d(x) \rightarrow 0$ . The inverse identity becomes

$$h(x) = x + \log(1 + e^{g(x)}) = x + g(x) + \varrho(g(x)). \quad (38.16)$$

Fix  $x$  sufficiently large and write

$$x_{-1} = g(x), \quad x_0 = x, \quad x_n = h^n(x) \quad (n \geq 0). \quad (38.17)$$

Then, exactly,

$$x_{n+1} = x_n + x_{n-1} + \varrho(x_{n-1}) \quad (n \geq 0). \quad (38.18)$$

In particular the forcing is positive and becomes smaller than every geometrically decreasing sequence along a positive orbit. If  $f_n$  are the Fibonacci numbers,  $f_0 = 0$ ,  $f_1 = 1$ , induction gives

$$x_n \geq f_{n+1}x + f_n g(x) > 0 \quad (n \geq 0). \quad (38.19)$$

On every compact interval of initial points with  $g(x) > 0$ , the right side is bounded below by  $c\varphi^n$  with some  $c > 0$ .

Define  $d_n = x_{n-1} - \beta x_n = d(x_n)$ . The characteristic roots of the homogeneous recurrence are  $\varphi$  and  $-\beta$ , and direct substitution gives the stable-coordinate recurrence

$$d_{n+1} = -\beta d_n - \beta \varrho(x_{n-1}). \quad (38.20)$$

The negative sign is the exact source of the alternating correction.

**Proposition 38.2** (Canonical growing and alternating coordinates). *On a sufficiently large ray, the following limits and series exist:*

$$\Psi(x) = \lim_{n \rightarrow \infty} \beta^n h^n(x) = x + \frac{d(x)}{\sqrt{5}} + \frac{1}{\sqrt{5}} \sum_{j=0}^{\infty} \beta^j \varrho(x_{j-1}), \quad (38.21)$$

$$D(x) = \lim_{n \rightarrow \infty} (-\beta)^{-n} d(h^n(x)) = d(x) + \sum_{j=0}^{\infty} (-\varphi)^j \varrho(x_{j-1}). \quad (38.22)$$

Both series converge absolutely and locally uniformly. Moreover,

$$\Psi(h(x)) = \varphi \Psi(x), \quad D(h(x)) = -\beta D(x). \quad (38.23)$$

The function  $\Psi$  is positive, real analytic and strictly increasing, has  $\Psi'(x) > 0$ , and maps any sufficiently large closed ray  $[X, \infty)$  homeomorphically onto  $[\Psi(X), \infty)$ .

*Proof.* Iteration of (38.20) proves the formula for  $D$ . Variation of constants in (38.18) gives

$$x_n = f_{n+1}x + f_n g(x) + \sum_{j=0}^{n-1} f_{n-j} \varrho(x_{j-1}).$$

Insert Binet's formula  $f_n = (\varphi^n - (-\beta)^n)/\sqrt{5}$  and multiply by  $\beta^n$ . The bound (38.19), together with  $0 < \varrho(y) \leq e^{-y}$  for  $y > 0$ , justifies the limit and yields (38.21). Absolute and locally uniform

convergence of both series follows from the same bound. Shifting the limits by one iterate proves (38.23). Positivity of  $\Psi$  follows also from its equivalent formula with initial term  $(\varphi x + g(x))/\sqrt{5}$  and positive summands.

We prove analyticity rather than assuming regularity of the limiting coordinate. The Stieltjes representation extends  $U$  holomorphically to a neighborhood of every positive real point. Since  $U$  is positive there,  $g$  is holomorphic in a neighborhood of every real point. On the real axis, differentiation under the integral, with  $w_t(x) = e^x/(1 + te^x)$ , gives

$$0 < g'(x) = \frac{\int w_t(x)(1 + te^x)^{-1} d\mu(t)}{\int w_t(x) d\mu(t)} < 1. \quad (38.24)$$

The strict upper bound follows because  $\mu$  is not supported at zero. In particular  $h' > 0$ .

Fix a real initial point far enough to the right that it and its  $g$ -image exceed 2. On a sufficiently small complex disk around that point, both  $\operatorname{Re} z$  and  $\operatorname{Re} g(z)$  exceed 2. Set  $z_{-1} = g(z)$ ,  $z_0 = z$ , and define the complex orbit by the recurrence

$$z_{n+1} = z_n + z_{n-1} + \varrho(z_{n-1}).$$

Here  $\varrho(w) = \log(1 + e^{-w})$  is holomorphic throughout  $\operatorname{Re} w > 0$ , by its convergent logarithm series. For  $y = \operatorname{Re} w \geq 2$  it satisfies

$$|\varrho(w)| \leq -\log(1 - e^{-y}) < \frac{y}{2}.$$

Thus, inductively, all real parts stay above 2 and

$$\operatorname{Re} z_{n+1} \geq \operatorname{Re} z_n + \frac{1}{2} \operatorname{Re} z_{n-1}.$$

They grow at least like  $c\kappa^n$ , uniformly on the disk, where  $\kappa = (1 + \sqrt{3})/2 > 1$ . Consequently the complex versions of both invariant series converge locally uniformly and absolutely; even the larger summands are bounded by  $C\varphi^j \exp(-c\kappa^j)$ . They therefore define holomorphic functions agreeing with  $\Psi, D$  on the real interval. The complex recurrence supplies the extension: we do not require the principal expression for  $g$  or  $h$  to be defined at every later complex iterate. The same Binet formula makes  $\beta^n z_n$  converge locally uniformly to  $\Psi$ , so its derivatives also converge locally uniformly.

Strict positivity of  $\Psi'$  requires an additional argument. Put  $p_n = dx_n/dx > 0$  and  $\chi_n = (1 + e^{-x_{n-1}})^{-1}$ . Differentiating the recurrence gives  $p_{n+1} = p_n + \chi_n p_{n-1}$ . For  $q_n = p_n + \beta p_{n-1}$ ,

$$q_{n+1} = \varphi q_n - (1 - \chi_n) p_{n-1} \geq \varphi \chi_n q_n,$$

because  $p_{n-1} \leq q_n/\beta$  and  $\varphi\beta = 1$ . It follows that

$$(1 + \beta^2)\Psi'(x) = \lim_{n \rightarrow \infty} \beta^n q_n \geq (1 + \beta g'(x)) \prod_{j=0}^{\infty} \chi_j > 0.$$

The infinite product is positive since  $\sum_j (1 - \chi_j) \leq \sum_j e^{-x_{j-1}} < \infty$ . Finally, the series for  $\Psi$  and  $d(x) \rightarrow 0$  show that  $\Psi(x) = x + o(1)$ . Strict increase and this equivalent establish the claimed range and homeomorphism property.  $\square$

### 38.3 Effective remainder bounds and the profile

The series above also supply explicit error control. Suppose that  $a = g(x) > \log \varphi$  and  $x > a$ . For  $j \geq 1$  the recurrence implies  $x_{j-1} \geq x + (j-1)a$ . Consequently

$$|D(x) - d(x)| \leq e^{-a} + \frac{\varphi e^{-x}}{1 - \varphi e^{-a}}, \quad (38.25)$$

$$\left| \Psi(x) - x - \frac{D(x)}{\sqrt{5}} \right| \leq \frac{e^{-x}}{\sqrt{5}} \left( \frac{\varphi}{1 - \varphi e^{-a}} + \frac{\beta}{1 - \beta e^{-a}} \right). \quad (38.26)$$

The second bound uses the cancellation of the  $j = 0$  terms in (38.21) and (38.22). More explicitly, the expression inside the absolute value is

$$E(x) = \frac{1}{\sqrt{5}} \sum_{j=1}^{\infty} (\beta^j - (-\varphi)^j) \varrho(x_{j-1}). \quad (38.27)$$

For truncation after the terms  $j = 0, \dots, J-1$ ,  $J \geq 1$ , the two series have respective tail bounds

$$\left| \sum_{j=J}^{\infty} (-\varphi)^j \varrho(x_{j-1}) \right| \leq \frac{\varphi^J e^{-x_{J-1}}}{1 - \varphi e^{-x_{J-2}}}, \quad (38.28)$$

$$0 \leq \frac{1}{\sqrt{5}} \sum_{j=J}^{\infty} \beta^j \varrho(x_{j-1}) \leq \frac{\beta^J e^{-x_{J-1}}}{\sqrt{5}(1 - \beta e^{-x_{J-2}})}. \quad (38.29)$$

The denominators are positive on the ray just specified. These bounds follow by using  $x_{J-2}$  as a lower bound for each subsequent orbit increment. They require no estimate of an unspecified asymptotic constant.

Define  $P$  first on a sufficiently large ray in its argument by

$$P(\log_{\varphi} \Psi(x)) = \Psi(x) D(x). \quad (38.30)$$

Proposition 38.2, including  $\Psi' > 0$ , makes this well-defined and real analytic by the holomorphic inverse-function theorem. The two conjugacy identities immediately give

$$P(\log_{\varphi} \Psi(x) + 1) = -P(\log_{\varphi} \Psi(x)).$$

This identity extends  $P$  uniquely to a real-analytic function on all of  $\mathbb{R}$  satisfying (38.4). In particular  $P$  and all its derivatives are bounded. The normalization  $\Psi(x)/x \rightarrow 1$  fixes the phase coordinate used here.

Since  $\Psi(x) \sim x$ , boundedness of  $P$  gives  $D(x) = O(x^{-1})$ . The inherited relation  $g(x) = \beta x + o(1)$  and (38.25) then yield

$$\begin{aligned} d(x) &= \frac{P(\log_{\varphi} \Psi(x))}{\Psi(x)} + O(e^{-\beta x}), \\ \Psi(x) &= x + \frac{D(x)}{\sqrt{5}} + O(e^{-x}) = x + O(x^{-1}). \end{aligned} \quad (38.31)$$

Thus an exact change of variable makes the leading profile accurate up to an exponentially small error in  $x = \log s$ . The preceding displayed bounds give explicit versions of both errors.

The companion transform has a slightly stronger error in this same coordinate. Its defining identity gives

$$\log(e^{\beta x} G(e^x)) = -d(x) - \varrho(g(x)).$$

The  $j = 0$  term in the series for  $D - d$  cancels, so

$$\left| \log(e^{\beta x} G(e^x)) + D(x) \right| \leq \frac{\varphi e^{-x}}{1 - \varphi e^{-g(x)}}. \quad (38.32)$$

The coordinate error for the logarithm of the normalized  $F$  is  $O(s^{-\beta})$ ; for the logarithm of the normalized  $G$  it is  $O(s^{-1})$ .

### 38.4 Proof of the expansion to every order

We now prove Theorem 38.1, except for nonvanishing, which is certified in the next subsection. Let  $y = \Psi(x)$ ,  $t = \log_\varphi x$ , and  $v = y/x - 1$ . Equations (38.30) and (38.27) give

$$v = \eta H_t(v) + \frac{E(x)}{x}, \quad \eta = \frac{1}{\sqrt{5}x^2}, \quad H_t(v) = \frac{P(t + \log_\varphi(1+v))}{1+v}. \quad (38.33)$$

Here  $v = O(x^{-2})$  and  $E(x) = O(e^{-x})$ . Real analyticity and periodicity of  $P$  give a holomorphic extension to some fixed strip around the real phase axis: local extensions glue by uniqueness, and compactness of a period provides a common positive width. The analytic implicit-function theorem therefore produces a unique small solution  $v_0 = v_0(\eta, t)$  of  $v_0 = \eta H_t(v_0)$ , holomorphic in  $\eta$  near zero with a convergence radius uniform for real  $t \bmod 2$ . Since  $|\eta \partial_v H_t(v)| \leq 1/2$  there, subtracting the two equations shows

$$v - v_0 = O(e^{-x}/x). \quad (38.34)$$

For every fixed  $N$ , Taylor's theorem in  $\eta$ , uniformly on the phase circle, gives the Lagrange expansion

$$v_0(\eta, t) = \sum_{n=1}^N \frac{\eta^n}{n!} \left. \frac{d^{n-1}}{dv^{n-1}} H_t(v)^n \right|_{v=0} + O(\eta^{N+1}). \quad (38.35)$$

The coefficient identity can be established at the level of finite Taylor jets: substitution into  $v_0 = \eta H_t(v_0)$  determines the coefficients recursively and gives the displayed Lagrange coefficient. The analyticity just proved additionally makes the infinite Taylor series converge absolutely, uniformly for real phase and for  $\eta$  in a sufficiently small closed disk. For an explicit majorant, choose a common circle  $|v| = \rho$  in the domain of  $H_t$ , and let  $|H_t(v)| \leq M_H$  there for every real phase. Cauchy's coefficient bound gives

$$|[\eta^n]v_0(\eta, t)| = \frac{1}{n} |[v^{n-1}]H_t(v)^n| \leq \frac{M_H^n}{n\rho^{n-1}}.$$

In particular the increasing derivative order in (38.7) introduces no hidden factorial divergence.

Writing  $u = 1 + v$ , each differentiation of  $u^{-n}P(t + \log_\varphi u)^n$  replaces its current power  $u^{-j}$  by  $u^{-j-1}$  and applies  $\mathcal{D} - j$  to the function of phase. Thus

$$\left. \frac{d^{n-1}}{dv^{n-1}} H_t(v)^n \right|_{v=0} = \left[ \prod_{j=n}^{2n-2} (\mathcal{D} - j) \right] P(t)^n. \quad (38.36)$$

Finally,  $d(x) = D(x) + O(e^{-\beta x}) = \sqrt{5}xv_0 + O(e^{-\beta x})$ . Multiply (38.35) by  $\sqrt{5}x$  and use (38.36). This gives (38.5) with the remainder  $O(x\eta^{N+1}) = O(x^{-2N-1})$  and the coefficients (38.7). Formula (38.32) gives (38.6). Absolute convergence of the full Taylor series proves (38.8). Equations (38.34) and (38.26) give, more precisely,  $D(x) = \sqrt{5}xv_0 + O(e^{-x}) = S(x) + O(e^{-x})$ . Combine this with (38.25) and (38.32) to obtain (38.9)–(38.10). Finally the exact identity

$$s^\alpha F(s) + s^{-\beta} = (s^\beta G(s))^{-1}$$

and (38.10) prove (38.11), since  $S(x)$  is bounded and tends to zero. This proves every asserted expansion, without assuming differentiable asymptotics for  $F$  or  $G$  in advance.

### 38.5 A certified nonzero profile value

The nonvanishing assertion is a finite computation followed by a rigorous infinite-tail estimate. It can be reproduced from the source code in this package; no externally supplied decimal value of  $F$  is needed.

Generate the exact integers  $a_n$  by the inherited triangular recurrence

$$\begin{aligned} c_{0,k} &= 1, & c_{n,0} &= 0 \quad (n \geq 1), & a_n &= c_{n,1}, \\ c_{n,k} &= c_{n,k-1} + \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1}. \end{aligned} \quad (38.37)$$

With  $s_0 = 1/100$ , the exact Stieltjes remainder gives rational bounds

$$\sum_{j=0}^{31} a_j (-s_0)^j \leq F(s_0) \leq \sum_{j=0}^{30} a_j (-s_0)^j. \quad (38.38)$$

Set  $s_{-1} = s_0 F(s_0)$  and  $s_{k+1} = s_k(1 + s_{k-1})$ . The inverse identity proves inductively that  $U(s_k) = s_{k-1}$ . Hence the true, generally unknown seed in (38.38) defines an orbit of the actual transform. Put  $x_* = \log s_{115}$ , so  $g(x_*) = \log s_{114}$ .

The included script `code/04-densities-profile-verify_dynamics.py` uses rational arithmetic for (38.37) and (38.38), then 100-digit Decimal intervals for the orbit and invariant series. Basic arithmetic is rounded outwards. Each correctly rounded exponential, logarithm and square root is enlarged to the adjacent representable number on both sides, using the documented Decimal operations [12]. The full endpoints are recorded in `data/04-densities-profile-dynamics_certificate.json`. The following widened decimal enclosures are among the script's exact assertions:

$$26.4807143957 < x_* < 26.4807143959, \quad (38.39)$$

$$16.366 < g(x_*) < 16.367, \quad (38.40)$$

$$0.00003580 < d(x_*) < 0.00003582, \quad (38.41)$$

$$|D(x_*) - d(x_*)| < 0.000000079. \quad (38.42)$$

The last inequality is the explicit bound (38.25). In particular  $D(x_*) > 0$ , proving  $P \neq 0$ . For a stronger numerical enclosure, the script takes the first five terms of both invariant series and bounds the remaining terms by (38.28)–(38.29). It proves

$$0.00095032738531560 < P(\log_\varphi \Psi(x_*)) < 0.00095032738531562. \quad (38.43)$$

This interval encloses one value of  $P$ , not its global maximum.

Every decimal statement just displayed concerns the same true orbit point admitted by the exact rational seed bounds. It does not claim an enclosure for all points in the small interval (38.39). This distinction is essential when transporting a functional identity from an interval-valued seed.

**Corollary 38.3** (Sharp rate and genuine alternation). *Let  $M = \max_{t \in [0,2]} |P(t)|$ . Then  $M > 0.00095032738531560$  and*

$$\limsup_{s \rightarrow \infty} (\log s)(s^\alpha F(s) - 1) = M, \quad \liminf_{s \rightarrow \infty} (\log s)(s^\alpha F(s) - 1) = -M, \quad (38.44)$$

$$\limsup_{s \rightarrow \infty} (\log s)(s^\beta G(s) - 1) = M, \quad \liminf_{s \rightarrow \infty} (\log s)(s^\beta G(s) - 1) = -M. \quad (38.45)$$

Thus the  $O(1/\log s)$  relative error is optimal: neither error is  $o(1/\log s)$ , and neither is  $O(s^{-\delta})$  for any  $\delta > 0$ . Along the inverse orbit starting at  $s_{115}$ ,

$$\lim_{n \rightarrow \infty} (-1)^n (\log s_{115+n}) (s_{115+n}^\alpha F(s_{115+n}) - 1) = P(\log_\varphi \Psi(x_*)) > 0. \quad (38.46)$$

Both normalized transforms cross their leading power laws infinitely often.

*Proof.* Equations (38.13)–(38.14) show that the two scaled errors have limiting profiles  $P$  and  $-P$ . Their arguments  $\log_\varphi \log s$  pass through every phase infinitely often. Antiperiodicity makes the maximum and minimum opposite, giving the four limits. On the specified orbit,  $\Psi(h^n(x_*)) = \varphi^n \Psi(x_*)$  and  $D(h^n(x_*)) = (-\beta)^n D(x_*)$  exactly. Since  $h^n(x_*)/\Psi(h^n(x_*)) \rightarrow 1$ , the coordinate estimate (38.31) proves (38.46). Eventual alternating signs on this orbit, followed by continuity, prove the crossing assertions. The other claims follow from  $M > 0$ .  $\square$

### 38.6 Scope and further questions

This resolves Part II's second-order golden-ratio question (Research question 23.3), including its suggested orbit oscillation, and gives a convergent inverse-logarithmic expansion with smaller power corrections. It is a result about the real positive Stieltjes-transform axis. It does not by itself prove the unresolved fine Bell-number normalization of the coefficients, nor a second-order distribution-function or density asymptotic for the measures. [Batch 86, 4 October 2026. The normalization, Conjecture 8.1, is proved in Part V (manuscript 02 of the same batch, Theorem 46.1), by an argument that has not yet been independently reviewed. Manuscript 01, from which Part IV is taken, was written without knowledge of manuscript 02, and calls the normalization open throughout. Its statement here remains true in the sense intended: nothing in this section bears on the normalization.]

Several concrete follow-up problems remain.

1. Determine the maximum  $M$  with certified bounds over an entire phase interval, and count the zeros of  $P$  in a period. The certificate above proves only one nonzero value and the existence of zeros forced by antiperiodicity.
2. Locate the complex singularities of the analytic continuation of  $P$ , and obtain explicit widths for its holomorphic strip. This would make the convergence radius and constants in the inverse-logarithmic expansion effective.
3. Develop certified Fourier bounds for  $P$ . Antiperiodicity permits only odd frequencies in a period-two Fourier representation, and analyticity gives exponential decay. Effective decay rates and bounds on the individual coefficients remain to be determined.
4. Recover sharp second-order behavior of the representing measures near zero. A suitable Tauberian theorem must preserve the slowly varying oscillation and control its remainder; differentiation of the leading distribution-function equivalent alone is insufficient.
5. Use the explicit invariant series to design efficient evaluation over all phases. The forward inverse recurrence is especially accurate once  $g(x) > 0$ ; obtaining uniformly efficient certified seeds across a fundamental phase interval is a separate numerical problem.
6. Investigate analogous profiles for the family of nonlinear inverse relations  $U^{-1}(t) = t(1+U(t))^q$ . The stable characteristic root, its sign and its modulus should determine whether the leading correction alternates and which power of  $\log s$  replaces  $1/\log s$ . Existence of the appropriate positive-measure solutions must be established before importing the present argument.

### 39 The companion sequence: an expansion to every fixed shift

The comparison of A088713 with A088714 is governed by a large component in a weighted composition. We prove a general statement because its hypotheses are exactly those already available for this pair. Throughout this section,  $O_K$  constants may depend on a fixed nonnegative integer  $K$  and on the sequence, but not on  $n$ .

**Theorem 39.1** (Renewal expansion for rapidly growing log-convex weights). *Suppose  $a_0 = 1$ ,  $a_n > 0$ , the sequence  $(a_n)$  is log-convex, and  $a_n/a_{n-1} \sim n/\log n$ . Define the formal series  $A(z) = \sum a_n z^n$ ,  $C(z) = (1 - zA(z))^{-1} = \sum c_n z^n$ , and*

$$\gamma_j = [z^j]C(z)^2. \quad (39.1)$$

For every fixed  $K \geq 0$ ,

$$c_n = \sum_{j=0}^K \gamma_j a_{n-1-j} + O_K(a_{n-K-2}) \quad (n \rightarrow \infty). \quad (39.2)$$

The error is nonnegative when  $n > 2K$ . This is an asymptotic expansion in consecutive shifts:  $a_{n-j-1}/a_{n-j} \sim \log n/n$  for fixed  $j$ . No smooth expansion of the consecutive ratios is assumed.

#### 39.1 A convolution estimate

Set  $\omega_n = a_{n-1}$  for  $n \geq 1$ , and  $q_n = \omega_{n+1}/\omega_n$  for  $n \geq 1$ . Then  $\omega_1 = 1$ ,  $(q_n)$  is nondecreasing, and  $q_n \sim n/\log n$ . For  $1 \leq j \leq n-1$  put  $T_{n,j} = \omega_j \omega_{n-j}$ .

**Lemma 39.2.** *For each fixed integer  $J \geq 1$ ,*

$$\sum_{j=J}^{n-J} \omega_j \omega_{n-j} = O_J(\omega_{n-J}). \quad (39.3)$$

There is  $\eta > 0$  such that

$$\sum_{n/4 \leq j \leq 3n/4} \omega_j \omega_{n-j} = O(e^{-\eta n} \omega_{n-1}). \quad (39.4)$$

In particular

$$S_n := \frac{1}{\omega_n} \sum_{j=1}^{n-1} \omega_j \omega_{n-j} = O(1/q_{n-1}) \rightarrow 0. \quad (39.5)$$

Integer endpoints in these sums are rounded inward.

*Proof.* Let  $L = \lfloor n/4 \rfloor$ . For  $j \leq L$ ,

$$\frac{T_{n,j+1}}{T_{n,j}} = \frac{q_j}{q_{n-j-1}} \leq \frac{q_L}{q_{n-L-1}} \rightarrow \frac{1}{3}.$$

Thus the ratio is at most  $1/2$ , uniformly on this range, for all sufficiently large  $n$ . From  $L$  to the middle, log-convexity makes  $T_{n,j}$  nonincreasing. The sum from  $J$  to  $L$  is at most  $2T_{n,J}$ ; the remainder to the middle is at most  $n2^{-(L-J)}T_{n,J}$ . Reflecting across the middle proves (39.3), since  $\omega_J$  is fixed. Starting the same bound at  $J = 1$  shows that the central sum is at most a constant times  $n2^{-L}\omega_1\omega_{n-1}$ , which implies (39.4). Taking  $J = 1$  in (39.3) proves (39.5).  $\square$

### 39.2 Uniform comparison and the unique large part

The renewal recurrence and composition expansion are

$$c_n = \omega_n + \sum_{j=1}^{n-1} \omega_j c_{n-j}, \quad (39.6)$$

$$c_n = \sum_{k=1}^n \sum_{\substack{s_1 + \dots + s_k = n \\ s_i \geq 1}} \prod_{i=1}^k \omega_{s_i}. \quad (39.7)$$

Choose  $N$  so that  $S_n \leq 1/2$  for  $n \geq N$ , and choose  $M \geq 2$  with  $c_n \leq M\omega_n$  for  $1 \leq n < N$ . Induction using (39.6) gives

$$c_n \leq \omega_n + MS_n \omega_n \leq M\omega_n \quad (n \geq N).$$

Therefore  $c_n = O(\omega_n)$  uniformly. Since  $\gamma_n = 2c_n + \sum_{j=1}^{n-1} c_j c_{n-j}$  for  $n \geq 1$ , the same estimate and (39.5) give  $\gamma_n = O(\omega_n)$ .

A composition has at most one part larger than  $n/2$ . If that part has size  $n - j$ , the parts on its left and right are arbitrary weighted compositions of combined size  $j$ . Their generating function is  $C(z)^2$ . Consequently the exact contribution of all compositions with such a part is

$$\sum_{0 \leq j < n/2} \omega_{n-j} \gamma_j. \quad (39.8)$$

For  $j \geq K + 1$ , its remaining contribution is bounded by a constant times

$$\sum_{j=K+1}^{\lfloor n/2 \rfloor} \omega_j \omega_{n-j} = O_K(\omega_{n-K-1}).$$

For a composition with every part at most  $n/2$ , split it at its first partial sum at least  $n/4$ . The partial sum lies in  $[n/4, 3n/4)$ . The split is uniquely specified by the composition, so the total weight of this class is at most

$$\sum_{n/4 \leq j \leq 3n/4} c_j c_{n-j} \leq M^2 \sum_{n/4 \leq j \leq 3n/4} \omega_j \omega_{n-j} = O(e^{-\eta n} \omega_{n-1}).$$

For fixed  $K$ , the quotient  $\omega_{n-1}/\omega_{n-K-1}$  grows at most polynomially, as follows from the ratio equivalent. This last contribution is therefore  $o(\omega_{n-K-1})$ . Combining the two classes proves (39.2), with  $\omega_{n-K-1} = a_{n-K-2}$ . Every omitted composition has nonnegative weight, which proves the sign assertion and completes the proof of Theorem 39.1.

### 39.3 Consequences for A088713

Applying Proposition 36.1 gives a positive answer to Part II's Research question 23.6, together with all its fixed-shift corrections. For this pair,

$$(\gamma_0, \dots, \gamma_6) = (1, 2, 5, 16, 64, 308, 1716).$$

For example,

$$c_n = a_{n-1} + 2a_{n-2} + 5a_{n-3} + 16a_{n-4} + O(a_{n-5}), \quad (39.9)$$

$$\frac{c_n}{a_{n-1}} = 1 + \frac{2}{r_{n-1}} + \frac{5}{r_{n-1}r_{n-2}} + O\left(\frac{(\log n)^3}{n^3}\right). \quad (39.10)$$

In particular,

$$\frac{c_n}{a_{n-1}} \rightarrow 1, \quad \frac{c_n}{a_{n-1}} = 1 + \frac{2 \log n}{n} + o\left(\frac{\log n}{n}\right). \quad (39.11)$$

One must retain the actual ratios in (39.10) to interpret it as a second-order statement. Replacing all of them by  $n/\log n$  would require more information about their errors than is assumed here.

**Corollary 39.3** (Additive comparison of shifted consecutive ratios). *For  $r_n = a_n/a_{n-1}$ ,*

$$\frac{c_{n+1}}{c_n} = r_n + 2\left(1 - \frac{r_n}{r_{n-1}}\right) + O(1/r_n), \quad \frac{c_{n+1}}{c_n} - \frac{a_n}{a_{n-1}} \rightarrow 0.$$

*Proof.* Use (39.2) with  $K = 1$  at  $n$  and  $n + 1$ , divide, and expand the denominator. Adjacent ratios are asymptotic, so the resulting quadratic relative errors contribute  $O(1/r_n)$  after multiplication by  $r_n$ . Finally  $r_n/r_{n-1} \rightarrow 1$ .  $\square$

The proof uses only ordinary weighted compositions and the leading ratio law. It does not prove the finer additive expansion of  $r_n$  needed for the Bell-number normalization. The script `code/04-densities-profile-verify_renewal.py` checks the exact composition identity independently, and records numerical diagnostics for the consecutive-shift expansion; finite diagnostics are not used in its proof.

## 40 Verification, reproducibility, and interpretation of Part IV

Manuscript 01 was delivered with four verification programs, their exact or interval output, and a short reproduction guide. Two of the programs concern Part IV and are shipped with this report as `code/04-densities-profile-verify_dynamics.py` and `code/04-densities-profile-verify_renewal.py`, with their recorded outputs `data/04-densities-profile-dynamics_certificate.json` and `data/04-densities-profile-renewal_checks.json`; the other two belong to the parts of the manuscript printed in other reports (Section 35). The programs do not access the network. The stored OEIS prefixes and the repository pin make their finite reference data explicit. Standard-library integer and rational arithmetic suffice for the renewal checks. The dynamics certificate additionally uses correctly rounded decimal elementary functions with outward enclosure. The report's README gives the commands for rerunning both programs on a copy.

### 40.1 What was checked

| Program                                                   | Verification performed                                                                                                                                                                                                                              |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>code/04-densities-profile-verify_dynamics.py</code> | Generates $a_0, \dots, a_{31}$ exactly, encloses $F(1/100)$ by its finite Stieltjes remainders, propagates 115 inverse steps with outward rounding, and encloses both invariant-series tails.                                                       |
| <code>code/04-densities-profile-verify_renewal.py</code>  | Compares two independent formal coefficient constructions through degree 25; checks the stored OEIS prefixes; enumerates all 16,383 compositions of sizes 1 through 14 and checks the large-part identity; computes diagnostics through $n = 260$ . |

Separate mathematical reviews checked the global-inversion hypotheses, boundary limits, and zero descent in the density theorem; the invariant coordinates, analytic reversion, and interval

enclosure in the golden-ratio expansion; and the convolution bounds in the renewal theorem. (The manuscript's reviews of its parts on maps and on partitions are reported with those parts.) These reviews are internal checks of a new manuscript. They are not external refereeing or machine-checked formal proofs.

## 40.2 A certificate with a mathematical role

The interval calculation in Section 38.5 has a stronger role than a plot or an agreement of finite prefixes. The proof reduces nonvanishing of the analytic profile to the strict inequality  $d(x) > |D(x) - d(x)|$  at one actual point of the OEIS solution. The enclosure establishes

$$0.00003580 < d(x) < 0.00003582, \quad |D(x) - d(x)| < 7.9 \cdot 10^{-8}.$$

It also encloses the corresponding profile value by

$$0.00095032738531560 < P(\log_{\varphi} \Psi(x)) < 0.00095032738531562.$$

Both endpoints are deliberately rounded outwards from the more precise enclosures recorded in the certificate file named above. The coefficient truncation error, elementary-function rounding, orbit propagation, and invariant-series tails are all included. The narrow interval stored for the explicit error *bound* does not claim to enclose the actual absolute error.

This proves  $P \not\equiv 0$ . Antiperiodicity then proves the occurrence of both signs at arbitrarily large arguments. It does not locate the maximum of  $P$ , determine its least positive period, or establish how many zeros it has in a period.

The seed  $s_0 = 1/100$ , its rational bounds (38.38) and the inverse orbit are those of Part II's enclosures, (22.2) and (22.1); Part II used them to illustrate the constant-one law, whereas here the same kind of enclosure, carried to the invariant series, is part of a proof.

## 40.3 Diagnostics that do not enter the proofs

The renewal expansion implies that, for fixed  $K$ ,

$$E_{K,n} := \frac{c_n - \sum_{j=0}^K \gamma_j a_{n-1-j}}{a_{n-K-2}} \longrightarrow \gamma_{K+1}.$$

This follows by applying Theorem 39.1 with  $K+1$ . The exact-integer calculations give the following rounded values.

| $n$   | $c_n/a_{n-1}$ | $E_{0,n}$ | $E_{1,n}$ | $E_{2,n}$ | $E_{3,n}$  |
|-------|---------------|-----------|-----------|-----------|------------|
| 20    | 1.235499      | 2.692517  | 7.716257  | 29.465201 | 142.036296 |
| 50    | 1.118554      | 2.315888  | 6.089803  | 20.729599 | 88.744437  |
| 100   | 1.069135      | 2.178820  | 5.594920  | 18.478299 | 76.410619  |
| 200   | 1.039769      | 2.101282  | 5.330955  | 17.352619 | 70.645714  |
| 260   | 1.032154      | 2.081579  | 5.265425  | 17.079934 | 69.282715  |
| Limit | 1             | 2         | 5         | 16        | 64         |

The slow approach explains why finite ratios alone would be a poor substitute for the large-part proof.

## 41 A proposed research program for Part IV

The following directions are ordered by how directly they continue the proved structures. Each states a concrete objective and the obstacle that remains. The local questions at the end of Sections 37 and 38 supply additional variants. (The manuscript's research program also has directions for its parts on maps and on partitions; they are printed with those parts.)

### 41.1 Finish the finer Bell comparison

The leading coefficient ratio and the new fixed-shift renewal expansion leave the principal coefficient-normalization problem open. In the notation of Part I, let  $B_n$  be the Bell number and let  $W(n) > 0$  satisfy  $W(n)e^{W(n)} = n$ . The conjecture is that a finite constant  $C > 0$  exists with

$$a_n \sim C B_n \exp(W(n)^2 + 3W(n)). \quad (41.1)$$

A proved sufficient condition from Part I (Proposition 8.2) is, for some fixed  $p \geq 0$ ,

$$\frac{a_n}{a_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1 + W(n))^2} + O\left(\frac{W(n)^p}{n}\right). \quad (41.2)$$

After division by the leading ratio, the indicated error is summable; this is the precision needed to produce a nonzero multiplicative constant.

The new endpoint analysis concerns the measures near zero. High moments instead probe their remote tails, so it does not directly imply (41.2). A useful next target is uniform control of the moment ratios on windows  $n - j$  with  $j = O(W(n))$ , together with the complementary part of the coefficient recurrence. Part III formulates an exact scalar defect (30.8) whose bound  $O(W(n)^q/n^2)$  would suffice. Its definition and precision should be retained when attempting this step. Proving only an  $o(1)$  error in a formal local approximation will not settle (41.1).

[Batch 86, 4 October 2026. Part V, from manuscript 02 of the same batch, proves (41.1) (Theorem 46.1) without proving (41.2): it proves the complementary estimate (30.7) (Theorem 44.1) and then sums a linearized Poisson equation instead of controlling individual ratios. That proof has not yet been independently reviewed. The ratio expansion (41.2) remains open (Research question 49.2).]

### 41.2 Quantify the analytic oscillation profile

The construction supplies an analytic, canonical, nonzero function  $P$ , with a rigorous evaluator along inverse orbits. It leaves several well-posed numerical and analytic problems:

1. Produce a certified enclosure of  $P$  and  $P'$  on an entire period, and use it to determine the number and multiplicity of its zeros and extrema.
2. Bound the width of a complex strip of analyticity explicitly. This would turn existence of exponentially decreasing Fourier coefficients into effective truncation bounds.
3. Determine whether the first allowed Fourier harmonic dominates. Antiperiodicity allows only the frequencies  $\exp(\pi i(2k + 1)t)$ ; it does not by itself prove such dominance.
4. Bound an explicit threshold for convergence of the inverse-log series and describe the exponentially smaller correction terms. The present series is convergent for sufficiently large  $\log s$ , but it is not an exact representation of the transform without its stated exponentially small remainder.

These are refinements of a proved oscillation, not a renewed question about whether oscillation exists.

### 41.3 From positive analytic densities to effective tail laws

The density theorem removes possible gaps, atoms, and internal zeros, and gives precise behavior at the origin. The large- $x$  behavior of  $p(x)$  and  $q(x)$  remains the natural bridge to the finer moment asymptotics. One can ask for a tail equivalent with an error uniform under the exponential tilts that localize moments near their saddle. Such a result would supply information that positivity and real analyticity alone cannot provide.

A more immediate computational objective is a certified solver for the boundary relation

$$\int \frac{t^2 d\nu(t)}{(u-t)^2 + v(u)^2} = 1$$

on compact positive intervals, coupled to the density formulas of Section 37.4. Finite moments and transform identities should be used to bound the truncated tails. An algorithm with an explicit enclosure for the discarded measure would allow quantitative plots and tests of global monotonicity or global log-convexity. The derivative equivalents already give  $(\log p)''(x) \sim \beta/x^2$  and  $(\log q)''(x) \sim \alpha/x^2$  at zero, so both densities are strictly log-convex sufficiently near zero. The two global shape properties remain unresolved.

### 41.4 Broaden the two general mechanisms

Two proofs invite abstraction beyond these particular OEIS entries. First, the renewal theorem uses a uniform contraction of  $q_j/q_{n-j-1}$  away from the endpoints. One can seek sharp hypotheses on positive log-convex weights, including ratio laws regularly varying with an arbitrary positive index, under which the same full fixed-shift expansion holds. One should also identify what replaces the exponential central bound at the boundary of that regime.

Second, inverse functional equations can produce a recurrence with one expanding eigenvalue and one negative contracting eigenvalue. The present Fibonacci recurrence turns this spectral structure into an analytic antiperiodic correction. A general theorem should state which perturbations preserve analytic conjugacies, what the resonant corrections are when additional eigenvalues are present, and how to certify that a particular combinatorial solution has a nonzero stable coordinate.

### 41.5 Formalization and an independent submission path

The renewal convolution estimates are a comparatively self-contained target for formalization (the manuscript names, besides them, the exact bridge decomposition and positive coefficient formulas of its part on maps). The interval certificate can be ported to a small formally checked rational-enclosure kernel. The density arguments, like the Mellin arguments of the manuscript's part on partitions, require a broader analytic foundation, making them later targets.

For external mathematical review, the manuscript's parts can be separated into articles with their dependencies preserved. No OEIS update or submission has been made as part of this work. Independent refereeing remains appropriate before asserting historical priority.

## Part V

# The Bell normalization of A088714 and A088713

## 42 Source, scope and notation of Part V

**Summary.** Part V proves Part I’s Conjecture 8.1, Part II’s Research question 23.2:

$$a_n = C_* \mathcal{N}_n \left( 1 + O \left( \frac{W(n)^5}{n} \right) \right), \quad 0 < C_* < \infty, \quad \text{hence} \quad a_n \sim C_* B_n \exp(W(n)^2 + 3W(n))$$

(Theorem 46.1). The proof has four steps.

1. The complementary endpoint to every fixed order (Theorem 44.1): decomposing by the position of the unique large inner index,  $T_n([CW(n)]) = \sum_{h \leq P} \delta_h a_{n-1-h} + O(a_{n-P-2})$  whenever  $C \log 2 > P + 5$ . At Part III’s window this is  $J_n = 2/r_n + 5/(r_n r_{n-1}) + O(W(n)^3/n^3)$ : it proves Part III’s missing estimate (30.7) with  $q = 2$ , and turns the one-sided bound of Section 32 into an asymptotic equality.
2. Positive barriers (Theorem 45.1):  $c \mathcal{N}_n \leq a_n \leq C \mathcal{N}_n$ .
3. An unconditional bounded additive ratio error (Lemma 46.2):  $R_n = n/W(n) + O(1)$ , where Part III proves only  $\liminf(r_n - n/W(n)) \geq 3/2$  (Theorem 29.1).
4. A linearized Poisson equation with summable error (Lemma 46.3) and a column-stability summation lemma for slowly varying Poisson-tail kernels (Lemma 46.4), which turn the bounded normalized sequence into a convergent one.

Neither Part III’s local-curvature estimate (31.1) nor its tilted-variance estimate (31.4) is used. For the companion sequence,  $c_n \sim a_{n-1}$ , every fixed-order expansion  $c_n = \sum_{h \leq P} \gamma_h a_{n-1-h} + O(a_{n-P-2})$ , and  $c_n = C_*(W(n)/n) \mathcal{N}_n(1 + O(W(n)^5/n))$  with the same constant (Theorem 47.1). Not proved: the value of  $C_*$ , the ratio expansion (8.7), or the optimality of the rate.

**Review status of the main theorem.** Theorem 46.1 settles a conjecture that Parts I–III left open, and it comes from a single AI-generated manuscript. Its proof has not yet been independently reviewed. At placement the argument of Sections 44–46 was followed step by step without finding a gap, and its finite consequences were checked against the data of Parts I–III (Section 48); neither is a review. Until one is made, Conjecture 8.1 should be cited as proved in Part V, which is unrefereed, and not as settled.

**Provenance.** Part V is the part on A088714 and A088713 of manuscript 02 of batch 86 of the repository’s incoming-reports intake: *Signs, Factorial Divergence, and Bell Normalization. New results for OEIS A321941, A088714, and A088713*, “Building on selected research reports in ProveIt”, by “ChatGPT”, dated 4 October 2026 (a 2,537-line source and a 36-page PDF). It arrived as `oeis_research_bundle.zip` in commit `ae9baa422` and was placed in commit `0f084afa9`. The manuscript pins the repository at commit `b7e4f25b6e79cc669fb245cba520017a0f4228b7` (3 October 2026, 17:53:40 PDT) and identifies this report by the Git blob `3a4ade894005d824a4ae3ead0b769253a178e3cf` of its source, which was still the source at the placement commit, so the manuscript read Parts I–III as printed here. Its Part II (Sections 8–12) is printed here as Sections 43–47, and the corresponding passages of its Sections 1, 13–15 and Appendix B as this section and Sections 48 and 49. Its other part (Sections 2–7 and Appendix A:  $r_k < 0$  for every  $k \geq 3$ , the

large-order expansion and the Borel singularity of the asymptotic coefficients behind A321941) is Part II of `congruences-and-valuations/a321941-asymptotic-coefficient-integrality` (Sections 13–21 and Appendices C–D there, label prefix `bgg:sg:`). [Batch 86 reciprocal notes, 4 October 2026.] Its Theorem 13.1, `bgg:sg:thm:sign`, proves  $r_k < 0$  for every  $k \geq 3$ , and its Theorem 13.2, `bgg:sg:thm:largeorder`, the large-order expansion. Part V’s programs and data are shipped with the prefix `05-bell-norm-`; the manuscript’s verification driver, the two records it writes, and its source audit cover both parts and are shipped once, here; the A321941 programs are shipped with that report. The manuscript’s  $\text{\LaTeX}$  source, delivery README, PDF and checksum manifest are not shipped; they survive in the archive of the arrival commit.

**What is not reprinted.** The manuscript’s displays of its three inherited inputs are Part III’s (26.1)–(26.3) and are replaced by references; its outline of the secant argument for the ratio law is kept. Its Lemma 8.1 restates Part III’s Lemma 28.1 and “include[s] its proof”; the proof is Part III’s, step for step, so the statement is recalled and the proof is not reprinted. Repeated displays of the defining equation (1.1), the recurrence (28.1), the Bell asymptotic (6.6) and, in two later sections, the normalizer are replaced by references. Everything else in the manuscript’s Sections 8–12 is printed. From Sections 13–15, the A088714 material is printed in Sections 48 and 49: the A321941 rows of its first table and its Research questions 15.6–15.8 are printed with the A321941 part, its Research question 15.9 (formalization) in both places with the targets of each, and its Section 14.3, which tells how to compile the delivered source, is replaced by the README.

**Editorial changes.** Every label carries the prefix `bnc:`; the manuscript’s labels are unchanged after it. (Its labels `eq:normalizer` and `eq:localcoeff` coincided with labels of Part I.) References to “the source report” and labels quoted as text now point to the cited statements of Parts I–III; file names are the shipped ones. Four symbols are renamed (see the notation paragraph); no normalization changed. The bibliography adds nothing: the manuscript’s entries for A088714, A088713, the Lambert function, the Bell numbers and Moser–Wyman are already in it, and its entries for this report and the repository are replaced by internal references. Dated notes in Section 48 compare its diagnostics with Parts III and IV. Nothing else in the mathematics was changed.

**Status.** Part V is AI-assisted and unrefereed, and it is not formalized: no Lean or Rocq development checks any statement of this report, and placement in this collection confers no formal status. The manuscript reports independent internal reviews (its source audit, shipped as `data/05-bell-norm-SOURCE_AUDIT.json`, records: “Two independent reviews passed the full convergence proof”); it states that they are not external refereeing or a proof-assistant certificate, and the audit records `external_peer_review`, `proof_assistant_formalized` and `worldwide_priority_exhaustively_verified` all as false. Its claim of novelty concerns “the inspected reports and the primary literature cited here” and “is not an exhaustive claim of worldwide priority”. The supplied computations check finite certificates and identities; the infinite claims are proved by the estimates in the text, not inferred from finite tests. The value of  $C_*$  is not certified numerically, and agreement of the diagnostics at two precisions is a stability check, not interval arithmetic. The manuscript’s statement that no OEIS entry or repository commit is changed described the delivery.

**Relation to Part IV.** See the corresponding paragraph of Section 35: the fixed-order companion expansion of Theorem 47.1 is also Theorem 39.1, proved independently and more generally; Part V’s

proof is printed in Section 47 as a second route, and its normalization of  $c_n$  with  $C_*$  is new. Manuscript 01, the source of Part IV, calls the finer normalization open.

**Notation.** Part V uses Part I's  $R_n = a_n/a_{n-1}$ ,  $B_n$ ,  $\mathcal{N}_n$  and  $w = W(n)$ , the solution of  $we^w = n$ , and Part III's  $b_{n,m} = [z^m]A(z)^{n+1-m}$ ,  $d = w/(1+w)$  and window  $L = \lceil 16W(n) \rceil$ . Its other symbols are local to Part V; Table 6 lists those that also occur in Parts I–IV. Four symbols are renamed:

1.  $\vartheta(w) = 2 + w/(2(1+w)^2)$  is the manuscript's  $\delta(w)$ , since the manuscript also writes  $\delta_h$  for the endpoint coefficients, and Parts I and III write  $\delta$  for  $R_n - n/w$ ; thus  $n/w + \vartheta(w)$  is Part III's candidate  $R(n)$ ;
2.  $\Xi_n$ , the linearized Poisson defect of Lemma 46.3, is the manuscript's  $\mathcal{E}_n$ , and  $\bar{\xi}_n = \Xi_n/(1+W(n))$  is its  $e_n$ : Part III's  $\mathcal{E}_n$  (30.8) is a different, though related, scalar defect;
3. in Section 44 the manuscript abbreviates  $A_N(z_n)$  as  $S_n$ ; it is written out, since  $S_n$  is the model row sum of Section 45;
4.  $\mathcal{Q}_n(u)$ , the comparison operator of Section 45, is the manuscript's  $Q_n(u)$ , since  $Q_n = R_n/R_{N+2}$  in Section 44.

The tempting false readings:  $T_n(L)$  is Part III's  $a_n J_n$ , not Part I's Touchard polynomial;  $s_n = \mathcal{N}_n/\mathcal{N}_{n-1}$  is a model ratio, not Part III's  $s_n = n/w + c(w)$ ;  $y_n = \log(a_n/\mathcal{N}_n)$  is Part III's diagnostic  $D_n^N$ ; and  $\delta_h$ , the coefficients of  $D(z) = z^{-1}(1 - 1/(zA(z)))'$ , are not Part I's increment  $\delta$ .

Table 6: Letters of Part V that also occur in Parts I–IV.

| letter                 | elsewhere in the report                                                                                                  | Part V                                                                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $R, r$                 | Part I $R_n$ ; Part II forward $r_n$ ; Part III $r_n$ , candidate $R(n)$ ; Part IV $r_n$                                 | $R_n = a_n/a_{n-1}$ (Part I's); $r_n$ only where Part III is quoted                                                                                                         |
| $T, J$                 | Part I $T_n(\lambda)$ Touchard; Part III $J_n$                                                                           | $T_n(L) = a_n J_n$ , the complementary sum                                                                                                                                  |
| $\delta, \vartheta, D$ | Parts I, III $\delta = R_n - n/w$ ; Part I $D_n(k)$ , $D_n^{\text{num}}$ ; Part III $D_{n,m}$ , $D_n^N$ ; Part IV $D(x)$ | $\delta_h = [z^h]D(z)$ , endpoint coefficients; $\vartheta(w)$ as above                                                                                                     |
| $\mathcal{E}, \Xi, E$  | Part III $\mathcal{E}_n$ (30.8); Part IV $E(x)$ , $E_{K,n}$                                                              | $\Xi_n$ , $\bar{\xi}_n$ as above; $E_n$ a region's contribution (Sections 44 and 47); $E_{n,m} = \prod_{j \leq m} s_n/s_{n-j}$ , the model analogue of Part III's $D_{n,m}$ |
| $s, S$                 | Part III $s_n = n/w + c(w)$ , $s$ a degree; Part IV $S_n$ , $S(x)$                                                       | $s_n = \mathcal{N}_n/\mathcal{N}_{n-1}$ ; $S_n$ the model row sum                                                                                                           |
| $\gamma, c, C$         | Part II $c_n$ (A088713), $C(z)$ ; Part IV $\gamma_j$ ; Part I $C$ in Conjecture 8.1                                      | $\gamma_h = [z^h]C(z)^2$ (Part IV's $\gamma_h$ ); $C_*$ the limit constant; $c < C$ barrier constants; $C$ also a window constant                                           |
| $P, Q$                 | Part I $Q_n$ ; Part IV profile $P$ , $Q_n(t)$                                                                            | $P$ a fixed truncation order; $Q_n = R_n/R_{N+2}$ ; $\mathcal{Q}_n(u)$                                                                                                      |
| $N, \mathcal{N}, B$    | Part I $\mathcal{N}_n$ , $B_n$ , $B(x) = xA(x)$                                                                          | $N = \lfloor n/2 \rfloor$ , $A_N$ the truncation; $\mathcal{N}(x)$ for real $x$ ; $B(z) = zA(z)$ ; $B_N$ the column boundary (46.10)                                        |
| $p, q, k, \kappa$      | Part III $p_t(m)$ ; Part IV densities $p, q$                                                                             | $p_w(m)$ Poisson weights at mean $w$ ; $q_w(j)$ Poisson tails; $k_w(j)$ , $\kappa_i$ column quantities                                                                      |
| $u, v, y, \xi, \ell$   | Part IV $u$ , $v(u)$ ; Part III $\eta$                                                                                   | $u_n = a_n/\mathcal{N}_n$ , $y_n = \log u_n$ , $\xi_n = y_n - y_{n-1}$ , $\ell_{n,m} = y_n - y_{n-1-m}$ ; $v_{n,m}$ weights, and $v = n + \lceil 20W \rceil$ in one proof   |

## 43 Inherited inputs and the new target

In Part V  $a_n$  denotes A088714 as defined by (1.1). We use Part I's  $R_n = a_n/a_{n-1}$  for its consecutive ratio (Part III writes  $r_n$ ). The manuscript chose  $R_n$  to avoid the symbol  $r_k$  of its other part, on A321941, which is not printed here. All series involving  $A$  are formal unless a finite polynomial or an explicitly convergent expression is specified.

### 43.1 The precise earlier theorems

We use three results of Parts I and II, in the form in which Part III states them as (26.1)–(26.3) (with  $r_n$  there for  $R_n$  here): the growth law (26.1), which is Theorem 1.1; the moment representation (26.2) by a probability measure  $\mu$ , Theorem 12.1; and  $R_n \sim n/\log n \sim n/W(n)$  with  $R_{n+1} \geq R_n$ , (26.3), which is Theorem 12.3 together with the log-convexity of Theorem 15.3.

The logical origin of these inputs is useful. The first uses an exact triangular recurrence and upper comparison with shifted Poisson moments. The second constructs moment-preserving positive operators, then passes to the coefficientwise attracting fixed point of the functional equation. Cauchy–Schwarz applied to (26.2) gives log-convexity, hence monotonicity of  $R_n$ . Finally, secant bounds for the convex sequence  $\log a_n$ , combined with (26.1), prove the ratio equivalent; an  $n$ -th-root equivalent by itself would not justify that step.

For clarity, the secant argument is short. Fix  $0 < \varepsilon < 1$  and bracket  $\log R_n$  between slopes over the intervals  $[(1 - \varepsilon)n, n]$  and  $[n, (1 + \varepsilon)n]$ , with integer endpoints. Insert (26.1), first let  $n \rightarrow \infty$ , and then let  $\varepsilon \downarrow 0$ . This gives  $\log R_n = \log n - \log \log n + o(1)$ . Two consequences used repeatedly below are

$$a_n^{1/n} \sim \frac{n}{e \log n}, \quad \frac{a_n^{1/n}}{R_n} \rightarrow e^{-1}.$$

We do not need the stronger total-positivity or endpoint-transform theorems of Part II. The new results below follow from (26.1)–(26.3) and the coefficient estimate recalled next.

### 43.2 The logarithmic-window coefficient estimate

Extracting coefficients from (1.1) gives Part III’s exact recurrence (28.1),

$$a_n = \sum_{m=0}^{n-1} a_{n-1-m} b_{n,m}, \quad b_{n,m} = [z^m] A(z)^{n+1-m}.$$

Every summand is nonnegative. Lemma 28.1 of Part III gives, for each fixed  $C > 0$  and uniformly over integers  $0 \leq m \leq C \log n$ , the expansion (28.2),

$$b_{n,m} = \frac{n^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} + O_C \left( \frac{(1+m)^6}{n^2} \right) \right).$$

(The manuscript restates that lemma with its proof “to make the subsequent uniform estimates directly checkable”; the proof is Part III’s, step for step, and is not reprinted.)

### 43.3 The normalization to be proved

For real  $x > 0$ , put  $w_x = W(x)$  and define

$$\mathcal{N}(x) = \frac{\exp(x(w_x - 1 + 1/w_x) - 1 + w_x^2 + 3w_x)}{\sqrt{1 + w_x}}, \quad \mathcal{N}_n = \mathcal{N}(n), \quad \mathcal{N}_0 = 1. \quad (43.1)$$

The principal real Lambert function has

$$W'(x) = \frac{W(x)}{x(1 + W(x))} \quad [4].$$

At integers  $n \geq 1$  this is Part I’s normalizer (8.4). The classical Bell saddle equivalent is Part I’s (6.6) [9, 3]. Thus  $\mathcal{N}_n \sim \mathbf{B}_n e^{w^2 + 3w}$ .

The main theorem of Part V, proved in Section 46, is

$$a_n = C_* \mathcal{N}_n \left( 1 + O\left(\frac{W(n)^5}{n}\right) \right), \quad 0 < C_* < \infty. \quad (43.2)$$

The quantitative error in (43.2) is stated relative to the explicit  $\mathcal{N}_n$ . Using only the Bell equivalent (6.6) then gives  $a_n \sim C_* \mathbf{B}_n e^{w^2+3w}$ .

The proof has three stages. First, a decomposition by the position of the unique large inner index resolves the complementary endpoint. Second, a smooth comparison sequence gives positive upper and lower barriers for  $a_n/\mathcal{N}_n$ . Third, a summation identity for slowly varying Poisson-tail kernels proves convergence of the bounded normalized sequence and gives its rate. The final step requires only  $R_n = n/W(n) + O(1)$ , which we prove separately.

## 44 The complementary endpoint to every fixed order

This section proves the complementary-endpoint estimate left open in Part III, the hypothesis (30.7) of Theorem 30.1, and extends it to every fixed order.

### 44.1 Statement

Let  $A(z) = \sum_{n \geq 0} a_n z^n$  be the unique formal solution of (1.1). Thus  $a_0 = a_1 = 1$ ,  $a_2 = 3$ ,  $a_3 = 13$ . The exact coefficient recurrence is (28.1), whose first form  $a_n = \sum_{k=0}^{n-1} a_k [z^{n-1-k}] A(z)^{k+2}$  is Part I's (2.1); there  $b_{n,m} = [z^m] A(z)^{n+1-m}$ . For a fixed constant  $C > 0$ , put

$$w = W(n), \quad L = \lceil Cw \rceil, \quad T_n(L) = \sum_{m=L+1}^{n-1} a_{n-1-m} b_{n,m}.$$

All assertions below concern sufficiently large  $n$ , so that  $L < n/2 - 2$ .

Define a formal series  $D(z) = \sum_{h \geq 0} \delta_h z^h$  by

$$D(z) = A(z)(2A(zA(z)) + zA(z)A'(zA(z))).$$

Its coefficients are nonnegative. More explicitly,

$$\delta_h = \sum_{k=0}^h (k+2) a_k [z^{h-k}] A(z)^{k+1}.$$

The initial values are

$$\delta_0 = 2, \quad \delta_1 = 5, \quad \delta_2 = 24, \quad \delta_3 = 148, \quad \delta_4 = 1052, \quad \delta_5 = 8226.$$

**Theorem 44.1** (Complementary endpoint to any fixed order). *Fix a nonnegative integer  $P$  and a constant  $C$  satisfying*

$$C \log 2 > P + 5.$$

*Then*

$$T_n(\lceil CW(n) \rceil) = \sum_{h=0}^P \delta_h a_{n-1-h} + O_P(a_{n-P-2}).$$

The implied constant may also depend on  $C$ . In particular, for Part III's cutoff  $L = \lceil 16W(n) \rceil$ ,

$$\boxed{\frac{T_n(L)}{a_n} = \frac{2}{R_n} + \frac{5}{R_n R_{n-1}} + O\left(\frac{W(n)^3}{n^3}\right)}.$$

Consequently the previously unproved estimate

$$\frac{T_n(L)}{a_n} = \frac{2}{R_n} + O\left(\frac{W(n)^2}{n^2}\right)$$

holds unconditionally under the inherited theorems.

In Part III's notation  $T_n(L) = a_n J_n$ , with  $J_n$  as in (30.4). The theorem supplies the complementary hypothesis (30.7) of Part III with  $q = 2$ . It also proves the numerical prediction from Part III's endpoint table (Table 4):

$$\frac{n^2}{W(n)^2} \left( \frac{T_n(L)}{a_n} - \frac{2}{R_n} \right) \rightarrow 5.$$

This is a precise improvement on Part III's one-sided lower bound  $J_n - 2/r_n \geq 5/(r_n r_{n-1})$  (Section 32). For  $C = 16$  the general theorem applies to every integer  $P \leq 6$ . For arbitrary fixed  $P$ , one convenient sufficient choice is  $C = 2P + 16$ .

#### 44.2 A truncated-series estimate

Throughout the proof set  $N = \lfloor n/2 \rfloor$  and  $A_N(z) = \sum_{j=0}^N a_j z^j$ . No convergent interpretation is given to the divergent formal series  $A$ .

**Lemma 44.2** (Uniform finite-polynomial bound). *For  $z_n = 4/R_n$ ,*

$$A_N(z_n) = 1 + z_n + O(z_n^2), \quad A_N(z_n)^{n+3} = \exp((4 + o(1))W(n)).$$

*Proof.* Log-convexity says that the graph of  $\log a_j$  lies below the chord joining its values at any two indices. For  $2 \leq j \leq N$ , put

$$q_n = \left( \frac{a_N}{a_2} \right)^{1/(N-2)}.$$

Then  $a_j \leq a_2 q_n^{j-2}$ . The inherited root and ratio equivalents give

$$\frac{q_n}{R_n} \rightarrow \frac{1}{2e}.$$

Indeed  $q_n/a_N^{1/N} \rightarrow 1$ , because  $\log a_N = O(N \log N)$ , and  $a_N^{1/N} \sim N/(e \log N)$ , while  $R_n \sim n/\log n$ . It follows that  $z_n q_n \rightarrow 2/e < 1$ . Hence

$$0 \leq A_N(z_n) - 1 - z_n \leq \frac{a_2 z_n^2}{1 - z_n q_n} = O(z_n^2).$$

Also  $n/R_n \sim W(n)$ , and  $n/R_n^2 \rightarrow 0$ . Taking logarithms proves the second assertion.  $\square$

The strict inequality  $2/e < 1$  is why the evaluation point  $4/R_n$  works. This estimate uses only a finite polynomial of degree  $\lfloor n/2 \rfloor$ .

### 44.3 The region with a large outer index

Write  $H = n - N - 2$ , so that  $H \sim n/2$ . The terms whose outer index is  $k > N$  correspond exactly to  $0 \leq m \leq H$ . For  $m \leq H \leq N$ , nonnegativity gives

$$b_{n,m} \leq A_N(z_n)^{n+1} z_n^{-m}.$$

Monotone ratios give

$$\frac{a_{n-1-m}}{a_n} = \prod_{j=n-m}^n R_j^{-1} \leq R_{N+2}^{-(m+1)}.$$

Put  $Q_n = R_n/R_{N+2} \rightarrow 2$ . Therefore, for any  $L < H$ ,

$$\frac{1}{a_n} \sum_{m=L+1}^H a_{n-1-m} b_{n,m} \leq \frac{Q_n}{R_n} A_N(z_n)^{n+1} \sum_{m>L} (Q_n/4)^m.$$

The geometric series is valid for all sufficiently large  $n$ . Since  $R_n \sim e^w$ , for  $L = \lceil Cw \rceil$  this gives

$$\frac{1}{a_n} \sum_{m=L+1}^H a_{n-1-m} b_{n,m} \leq \exp((3 - C \log 2 + o(1))w). \quad (44.1)$$

For a fixed  $P$ ,

$$\frac{a_{n-P-2}}{a_n} = \prod_{j=0}^{P+1} R_{n-j}^{-1} \sim e^{-(P+2)w}.$$

The strict inequality  $C \log 2 > P + 5$  makes the right side of (44.1)  $o(a_{n-P-2}/a_n)$ . Thus this entire region is negligible at the claimed precision.

### 44.4 The region with no large inner index

Now consider  $k \leq N$ . Every such term has  $m = n - 1 - k \geq n - 1 - N > L$ , so this entire region belongs to the complementary sum  $T_n(L)$ . Expand the coefficient of  $A(z)^{k+2}$  as a sum over ordered tuples  $(j_1, \dots, j_{k+2})$  with

$$k + j_1 + \dots + j_{k+2} = n - 1.$$

Let  $E_n$  be the contribution for which every  $j_i \leq N$ . Using the finite-polynomial bound and positivity,

$$\begin{aligned} E_n &= \sum_{k=0}^N a_k [z^{n-1-k}] A_N(z)^{k+2} \\ &\leq z_n^{-(n-1)} \sum_{k=0}^N a_k z_n^k A_N(z_n)^{k+2} \\ &\leq z_n^{-(n-1)} A_N(z_n)^{N+3}. \end{aligned}$$

Here the sum is at most  $A_N(z_n)^{N+2} \cdot A_N(z_n)$ , the last factor being  $\sum_{k \leq N} a_k z_n^k$ . Since  $a_n^{1/n}/R_n \rightarrow e^{-1}$ ,

$$\log(E_n/a_n) \leq n(1 - \log 4 + o(1)).$$

In particular  $E_n = O(a_n e^{-cn})$  for some fixed  $c > 0$ . This is  $o(a_{n-P-2})$  for every fixed  $P$ .

#### 44.5 The unique large inner index

Every remaining configuration has an inner index  $j_i > N$ . There can be at most one, since  $j_1 + \dots + j_{k+2} \leq n - 1 < 2(N + 1)$ . Write that unique index as  $n - 1 - h$ . Then  $0 \leq h \leq H$ , the outer index satisfies  $0 \leq k \leq h$ , and the remaining inner indices sum to  $h - k$ . All these remaining indices are automatically at most  $N$ . There are  $k + 2$  choices for the location of the large inner index. Consequently this region is exactly

$$\sum_{h=0}^H a_{n-1-h} \sum_{k=0}^h (k+2) a_k [z^{h-k}] A(z)^{k+1} = \sum_{h=0}^H \delta_h a_{n-1-h}. \quad (44.2)$$

The coefficientwise inequality  $A^{k+1} \leq A^{k+2}$  and the defining recurrence imply

$$0 \leq \delta_h \leq (h+2) a_{h+1}. \quad (44.3)$$

We need the following elementary convolution consequence of the inherited ratio equivalent.

**Lemma 44.3** (A weighted endpoint tail). *For every fixed integer  $p \geq 0$ ,*

$$\sum_{h=p}^H (h+2) a_{h+1} a_{n-1-h} = O_p(a_{n-1-p}). \quad (44.4)$$

*Proof.* Let  $V_h = a_{h+1} a_{n-1-h}$ . Then

$$\frac{V_{h+1}}{V_h} = \frac{R_{h+2}}{R_{n-1-h}} \leq 1 \quad (0 \leq h < H).$$

For  $K = \lfloor n/3 \rfloor$ , uniformly over  $0 \leq h \leq K$ ,

$$\frac{V_{h+1}}{V_h} \leq \frac{R_{K+2}}{R_{n-1-K}} \rightarrow \frac{1}{2}.$$

Thus the ratio is at most  $3/4$  on that range for large  $n$ . Starting at the fixed index  $p$ , the part up to  $K$  is bounded by

$$V_p \sum_{u \geq 0} (p+u+2)(3/4)^u = O_p(V_p).$$

Monotonicity bounds the rest of the sum by

$$O(n^2)(3/4)^{K-p} V_p = O_p(V_p).$$

Finally  $V_p = a_{p+1} a_{n-1-p}$ , and  $a_{p+1}$  is fixed. □

Apply (44.4) with  $p = P + 1$ , use (44.3), and remove the first  $P + 1$  terms from (44.2). The remaining sum is  $O_P(a_{n-P-2})$ . Combining this with the two negligible regions proves the theorem.

#### 44.6 An identity for the endpoint coefficients

The coefficients  $\delta_h$  need not be computed by a nested composition. Set  $B(z) = zA(z)$ . The original equation is

$$B(z) = z + B(z)B(B(z)).$$

Differentiate, divide by  $B'(z)$ , and use  $B'(z) = 1 + O(z)$ :

$$1 - \frac{1}{B'(z)} = B(B(z)) + B(z)B'(B(z)).$$

The right side equals  $zD(z)$ , so

$$D(z) = \frac{1}{z} \left( 1 - \frac{1}{(zA(z))'} \right).$$

This identity is entirely formal. It yields the triangular exact-integer recurrence

$$\delta_h = (h+2)a_{h+1} - \sum_{i=1}^h (i+1)a_i\delta_{h-i}.$$

The subtraction in this computational formula does not obscure positivity, which is already proved by the composition formula.

#### 44.7 Consequence for the remaining Bell-normalization problem

At the existing cutoff  $L = \lceil 16W(n) \rceil$ , the complete exact recurrence now has the rigorously controlled form

$$a_n = \sum_{m=0}^L b_{n,m}a_{n-1-m} + 2a_{n-1} + 5a_{n-2} + O(a_{n-3}).$$

Part III's uniform coefficient theorem (Lemma 28.1) supplies

$$b_{n,m} = \frac{n^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} + O\left(\frac{(1+m)^6}{n^2}\right) \right) \quad (0 \leq m \leq L).$$

Thus the large-inner-coefficient obstruction listed in Part III (Section 32) is removed. Its curvature estimate (31.1) remains unproved:

$$\log R_{n-j} - \log R_n = -\frac{W(n)}{1+W(n)} \frac{j}{n} + O\left(\frac{W(n)^q}{n^2}\right), \quad 0 \leq j \leq L.$$

The next sections establish convergence for the normalized positive recurrence, whose lags are concentrated near  $1 + \text{Poisson}(W(n))$ . This approach proves multiplicative Bell normalization without first deriving every pointwise curvature estimate. The following sections develop that alternative: positive barriers first give bounded-factor normalization, and a summation argument then proves convergence.

## 45 Positive barriers for the Bell normalizer

The complementary-endpoint estimate also yields a stronger asymptotic conclusion without the still-open pointwise curvature lemma. It pins the conjectured normalization to within two fixed positive multiplicative constants.

Let  $\mathcal{N}(x)$ ,  $\mathcal{N}_n = \mathcal{N}(n)$  for  $n \geq 1$ , and  $\mathcal{N}_0 = 1$  be as in (43.1). The classical Bell-number saddle asymptotic gives

$$\mathcal{N}_n \sim B_n \exp(W(n)^2 + 3W(n)).$$

Here  $B_n$  denotes the Bell number (OEIS A000110). The same explicit normalizer is Part I's (8.4), and the Bell asymptotic is Part I's (6.6).

**Theorem 45.1** (Bounded-factor Bell normalization). *There exist constants  $0 < c < C < \infty$  such that*

$$c\mathcal{N}_n \leq a_n \leq C\mathcal{N}_n \quad (n \geq 0).$$

*Equivalently,*

$$a_n = \Theta\left(\mathcal{B}_n \exp(W(n)^2 + 3W(n))\right).$$

*In logarithmic form, with  $w = W(n)$ ,*

$$\log a_n = n(w - 1 + 1/w) - 1 + w^2 + 3w - \frac{1}{2} \log(1 + w) + O(1).$$

This does not assert that  $a_n/\mathcal{N}_n$  converges. It proves that the ratio is bounded and bounded away from zero. The theorem improves Part I's  $n$ -th-root asymptotic to a bounded additive error in the logarithm. Its proof uses a Taylor expansion of the explicit comparison function  $\mathcal{N}$ , and makes no smoothness assumption about the actual sequence  $a_n$ .

### 45.1 Local ratios of the explicit comparison function

Write

$$f(x) = \log \mathcal{N}(x), \quad s_n = \frac{\mathcal{N}_n}{\mathcal{N}_{n-1}}, \quad w = W(n), \quad d = \frac{w}{1+w}.$$

Using  $W'(x) = W(x)/(x(1+W(x)))$ , direct differentiation gives

$$f'(x) = W(x) + \frac{1}{x} \frac{W(x)}{1+W(x)} \left( 2W(x) + 3 - \frac{1}{2(1+W(x))} \right).$$

Consequently

$$f''(x) = \frac{W(x)}{x(1+W(x))} + O\left(\frac{W(x)}{x^2}\right), \quad f'''(x) = O(x^{-2}). \quad (45.1)$$

These bounds hold uniformly when  $x$  varies over  $[n - O(W(n)), n]$ .

Taylor's theorem applied to  $f(n) - f(n-1)$  gives

$$\log s_n = w + \frac{d}{n} \left( 2w + \frac{5}{2} - \frac{1}{2(1+w)} \right) + O(w/n^2).$$

Thus, putting

$$\vartheta(w) = 2 + \frac{w}{2(1+w)^2},$$

so that  $n/w + \vartheta(w)$  is Part III's candidate  $R(n)$ , we obtain

$$s_n = \frac{n}{w} + \vartheta(w) + O(w/n). \quad (45.2)$$

The identity behind the constant correction is

$$\frac{d}{w} \left( 2w + \frac{5}{2} - \frac{1}{2(1+w)} \right) = \vartheta(w).$$

Let  $L = \lceil 16w \rceil$ . For  $0 \leq j \leq L$ , apply (45.1) to  $g(x) = f(x) - f(x-1)$ . Since

$$g'(x) = \frac{W(x)}{x(1+W(x))} + O(W(x)/x^2),$$

integration over  $[n-j, n]$  gives, uniformly,

$$\log s_{n-j} - \log s_n = -\frac{dj}{n} + O(w^2/n^2). \quad (45.3)$$

There is no claim that (45.3) holds for  $R_n = a_n/a_{n-1}$ .

For  $0 \leq m \leq L$ , define the model correction

$$E_{n,m} = s_n^{m+1} \frac{\mathcal{N}_{n-1-m}}{\mathcal{N}_n} = \prod_{j=0}^m \frac{s_n}{s_{n-j}}.$$

Summing (45.3), then exponentiating, gives

$$E_{n,m} = 1 + \frac{d m(m+1)}{2n} + O(w^4/n^2), \quad (45.4)$$

uniformly on the same window.

## 45.2 The comparison recurrence has summable row error

We now use the already-proved coefficient expansion of Part III (Lemma 28.1):

$$b_{n,m} = \frac{n^m}{m!} \left( 1 + \frac{3m(m-1)}{2n} + O\left(\frac{(1+m)^6}{n^2}\right) \right), \quad 0 \leq m \leq L. \quad (45.5)$$

It is important that (45.5) is a theorem about coefficients of a finite degree; it assumes no neighboring-ratio asymptotic beyond the inherited root-growth bound.

Define the model row sum

$$S_n = \sum_{m=0}^L b_{n,m} \frac{\mathcal{N}_{n-1-m}}{\mathcal{N}_n} + 2 \frac{\mathcal{N}_{n-1}}{\mathcal{N}_n}.$$

**Lemma 45.2** (Summable model row error). *For  $L = \lceil 16W(n) \rceil$ ,*

$$S_n = 1 + O\left(\frac{W(n)^6}{n^2}\right). \quad (45.6)$$

*Proof.* Put  $t = n/s_n$ . Multiplying (45.4) and (45.5) yields

$$\begin{aligned} & \sum_{m=0}^L b_{n,m} \frac{\mathcal{N}_{n-1-m}}{\mathcal{N}_n} \\ &= \frac{1}{s_n} \sum_{m=0}^L \frac{t^m}{m!} \left( 1 + \frac{3m(m-1) + d m(m+1)}{2n} + O(w^6/n^2) \right). \end{aligned}$$

Here  $t \sim w$  and  $e^t/s_n \rightarrow 1$ . A Poisson Chernoff bound shows that the discarded tails after  $L = \lceil 16w \rceil$ , including all fixed polynomial moments, are  $O(n^{-10})$ . Therefore

$$S_n = \frac{e^t}{s_n} \left( 1 + \frac{(3+d)t^2 + 2dt}{2n} + O(w^6/n^2) \right) + \frac{2}{s_n}. \quad (45.7)$$

Equation (45.2) implies

$$t = w - \frac{\vartheta(w)w^2}{n} + O(w^3/n^2), \quad \frac{e^t}{s_n} = 1 - \frac{\vartheta(w)w(1+w)}{n} + O(w^4/n^2).$$

Also  $2/s_n = 2w/n + O(w^2/n^2)$ . The coefficient of  $1/n$  in (45.7) is

$$-\vartheta(w)w(1+w) + \frac{(3+d)w^2 + 2dw}{2} + 2w = 0.$$

The equality is an exact rational identity in  $w$ . This proves (45.6).  $\square$

### 45.3 Positive barriers

From the complementary-endpoint theorem and positivity, there is a fixed  $K < \infty$  such that, for every sufficiently large  $n$ ,

$$T_n(L) = 2a_{n-1} + \mathcal{R}_n, \quad 0 \leq \mathcal{R}_n \leq K a_{n-2}. \quad (45.8)$$

The lower inequality can also be read directly from the  $k = 0$  term of the defining recurrence. For the upper inequality, the theorem gives  $\mathcal{R}_n = 5a_{n-2} + O(a_{n-3})$ , and  $a_{n-3}/a_{n-2} \rightarrow 0$ .

Let  $u_n = a_n/\mathcal{N}_n > 0$ , and set

$$\mathcal{Q}_n(u) = \sum_{m=0}^L b_{n,m} \frac{\mathcal{N}_{n-1-m}}{\mathcal{N}_n} u_{n-1-m} + 2 \frac{\mathcal{N}_{n-1}}{\mathcal{N}_n} u_{n-1}.$$

Dividing the exact coefficient recurrence by  $\mathcal{N}_n$  and using (45.8) gives

$$\mathcal{Q}_n(u) \leq u_n \leq \mathcal{Q}_n(u) + K \frac{\mathcal{N}_{n-2}}{\mathcal{N}_n} u_{n-2}. \quad (45.9)$$

All coefficients are nonnegative, and all referenced indices are strictly smaller than  $n$ . By (45.2),  $\mathcal{N}_{n-2}/\mathcal{N}_n = O(w^2/n^2)$ . Together with (45.6), this supplies constants  $K_1, n_0$  such that, for  $n \geq n_0$ , both the lower and upper row sums in (45.9) lie between  $1 - \varepsilon_n$  and  $1 + \varepsilon_n$ , where

$$0 \leq \varepsilon_n = K_1 W(n)^6/n^2 \leq \frac{1}{2}, \quad \sum_{n \geq n_0} \varepsilon_n < \infty.$$

Set

$$m_n = \min_{0 \leq j \leq n} u_j, \quad M_n = \max_{0 \leq j \leq n} u_j.$$

Since every coefficient in (45.9) is nonnegative,

$$m_n \geq m_{n-1}(1 - \varepsilon_n), \quad M_n \leq M_{n-1}(1 + \varepsilon_n) \quad (n \geq n_0).$$

Iteration gives

$$m_n \geq m_{n_0-1} \prod_{j=n_0}^n (1 - \varepsilon_j), \quad M_n \leq M_{n_0-1} \prod_{j=n_0}^n (1 + \varepsilon_j).$$

The first infinite product is positive and the second finite, because  $\sum \varepsilon_j < \infty$  and  $0 \leq \varepsilon_j \leq 1/2$ . The finite initial segment has positive minimum and finite maximum. This proves the bounded-factor normalization theorem.

## 45.4 From bounded factors to a convergent normalization

The new conclusions are:

$$\frac{T_n(\lceil 16W(n) \rceil)}{a_n} = \frac{2}{R_n} + \frac{5}{R_n R_{n-1}} + O(W(n)^3/n^3), \quad 0 < \inf_n \frac{a_n}{\mathcal{N}_n} \leq \sup_n \frac{a_n}{\mathcal{N}_n} < \infty.$$

Thus both divergence to infinity and decay to zero of the conjectured normalized sequence have been excluded.

The positive-barrier argument by itself does not prove existence of a limit

$$\lim_{n \rightarrow \infty} \frac{a_n}{\mathcal{N}_n}.$$

The next section proves that limit using a summation lemma for the slowly varying Poisson averages. The sharper individual-ratio expansion and identification of the normalization constant remain further research questions.

## 46 Convergence of the finer Bell normalization

This section completes the normalization conjecture by summing a slowly varying averaging equation. No pointwise second-difference estimate for the actual ratios is assumed.

**Theorem 46.1** (Finer Bell normalization). *There is a finite strictly positive constant  $C_*$  such that*

$$a_n = C_* \mathcal{N}_n \left( 1 + O\left( \frac{W(n)^5}{n} \right) \right).$$

Consequently

$$a_n \sim C_* B_n \exp(W(n)^2 + 3W(n)).$$

Here  $\mathcal{N}_n$  is Part I's normalizer (8.4). The proof establishes the existence of  $C_*$ , without identifying its numerical value or an exact expression in familiar constants. It does not establish the stronger conjectured pointwise expansion  $R_n = n/W(n) + 2 + W(n)/(2(1 + W(n))^2) + O(W(n)^p/n)$ .

### 46.1 Constant-scale control of the actual ratios

**Lemma 46.2** (An unconditional bounded additive ratio error). *With  $R_n = a_n/a_{n-1}$  and  $w = W(n)$ ,*

$$R_n = \frac{n}{w} + O(1). \quad (46.1)$$

*Proof.* The inherited ratio equivalent gives  $R_n \sim n/w$ . Put  $L_n = \lceil 16W(n) \rceil$  and  $t = n/R_n$ . The coefficient estimate on this window implies

$$\sum_{m=0}^{L_n} b_{n,m} R_n^{-m-1} = \frac{e^t}{R_n} \left( 1 + O(w^2/n) \right). \quad (46.2)$$

The uniform estimate before Poisson summation is  $b_{n,m} = n^m/m! (1 + O(w^2/n))$ ; the discarded Poisson tail is  $O(n^{-10})$ . This reasoning is valid whenever the trial denominator is asymptotic to  $n/w$ .

Monotonicity of the actual ratios gives  $a_{n-1-m}/a_n \geq R_n^{-m-1}$ . The exact recurrence and positivity therefore imply

$$1 \geq \frac{e^{n/R_n}}{R_n} (1 + O(w^2/n)).$$

The parenthesized factor is positive for large  $n$ . Taking logarithms yields

$$\frac{n}{R_n} - \log R_n \leq O(w^2/n).$$

The function  $F_n(r) = n/r - \log r$  is strictly decreasing, has zero  $n/W(n)$ , and has derivative of magnitude comparable to  $w^2/n$  on any interval whose points are asymptotic to  $n/w$ . The mean value theorem yields  $R_n \geq n/w - O(1)$ .

For the upper bound, set

$$v = n + \lceil 20W(n) \rceil.$$

For sufficiently large  $n$ ,  $v - L_v \geq n$ , since  $W(v) - W(n) = O(W(n)/n)$ . Thus every denominator  $R_{v-j}$  in  $a_{v-1-m}/a_v = \prod_{j=0}^m R_{v-j}^{-1}$ , for  $0 \leq m \leq L_v$ , is at least  $R_n$ . The complementary-endpoint theorem gives  $T_v(L_v)/a_v = O(w/n)$ . Apply (46.2) at  $v$ , with trial denominator  $R_n \sim v/W(v)$ . The exact recurrence now implies

$$1 \leq \frac{e^{v/R_n}}{R_n} (1 + O(w^2/n)) + O(w/n).$$

It follows that  $F_v(R_n) \geq -O(w^2/n)$ . The same mean-value argument gives  $R_n \leq v/W(v) + O(1)$ . Since

$$\frac{d}{dx} \frac{x}{W(x)} = \frac{1}{1 + W(x)}$$

and  $v - n = O(w)$ , we have  $v/W(v) = n/W(n) + O(1)$ . This proves (46.1).  $\square$

Put

$$u_n = \frac{a_n}{\mathcal{N}_n}, \quad y_n = \log u_n, \quad \xi_n = y_n - y_{n-1} \quad (n \geq 1).$$

The bounded-factor theorem says  $y_n = O(1)$ . Combining (46.1) with the model-ratio expansion gives

$$\xi_n = O(W(n)/n). \quad (46.3)$$

For convenience extend  $y_n = y_0$  when  $n \leq 0$ , and set  $\xi_n = 0$  when  $n \leq 0$ .

## 46.2 A Poisson averaging equation with summable error

Let

$$p_w(m) = e^{-w} \frac{w^m}{m!} \quad (m \geq 0), \quad w = W(n).$$

Define

$$\Xi_n = y_n - \sum_{m \geq 0} p_w(m) y_{n-1-m}.$$

**Lemma 46.3** (Linearized Poisson equation). *For the actual normalized sequence,*

$$\Xi_n = O(W(n)^6/n^2). \quad (46.4)$$

*Proof.* Write  $L = \lceil 16w \rceil$ ,

$$v_{n,m} = b_{n,m} \frac{\mathcal{N}_{n-1-m}}{\mathcal{N}_n}, \quad s_n = \frac{\mathcal{N}_n}{\mathcal{N}_{n-1}},$$

and

$$\ell_{n,m} = y_n - y_{n-1-m}.$$

By (46.3), uniformly for  $m \leq L$ ,

$$|\ell_{n,m}| \leq K(m+1)w/n = O(w^2/n). \quad (46.5)$$

The exact recurrence and  $T_n(L) = 2a_{n-1} + O(a_{n-2})$  give, after division by  $a_n$ ,

$$1 = \sum_{m=0}^L v_{n,m} e^{-\ell_{n,m}} + \frac{2}{s_n} e^{-\xi_n} + O(w^2/n^2). \quad (46.6)$$

Here the remainder is  $O(a_{n-2}/a_n) = O(w^2/n^2)$ ; no unknown normalization constant occurs.

The already-proved model row sum is

$$\sum_{m=0}^L v_{n,m} + \frac{2}{s_n} = 1 + O(w^6/n^2).$$

Taylor's estimate  $e^{-x} = 1 - x + O(x^2)$ , (46.5), and the bounded row sum transform (46.6) into

$$\sum_{m=0}^L v_{n,m} \ell_{n,m} + \frac{2}{s_n} \xi_n = O(w^6/n^2). \quad (46.7)$$

Uniformly for  $m \leq L$ , the explicit model calculations give

$$v_{n,m} = p_w(m) \left(1 + O(w^2/n)\right). \quad (46.8)$$

To see this directly, put  $t = n/s_n$ . Then

$$v_{n,m} = \frac{e^t}{s_n} p_t(m) E_{n,m} \frac{m! b_{n,m}}{n^m},$$

where  $E_{n,m}$  is the model correction from (45.4). Every factor other than  $p_t(m)$  is  $1 + O(w^2/n)$ ; also  $t - w = O(w^2/n)$ , whence

$$\log \frac{p_t(m)}{p_w(m)} = w - t + m \log(t/w) = O(w^2/n).$$

This proves (46.8).

Replacing  $v_{n,m}$  by  $p_w(m)$  in (46.7) costs  $O(w^4/n^2)$  by (46.5). The term  $2\xi_n/s_n$  is  $O(w^2/n^2)$ . Finally,  $y_n$  is bounded, so the omitted Poisson tail after  $L$  contributes  $O(n^{-10})$ . The result is (46.4).  $\square$

### 46.3 A summation lemma for slowly varying Poisson kernels

For  $w > 0$ , define

$$q_w(j) = \mathbb{P}(\text{Poisson}(w) \geq j) \quad (j = 0, 1, \dots), \quad k_w(j) = \frac{q_w(j)}{1+w}.$$

Since

$$\sum_{j \geq 0} q_w(j) = 1 + w,$$

the  $k_w(j)$  form a probability distribution. For an integer  $n \geq 1$ , abbreviate  $k_n(j) = k_{W(n)}(j)$ .

**Lemma 46.4** (Column stability and vanishing boundary). *Define*

$$\kappa_i = \sum_{j \geq 0} k_{i+j}(j) \quad (i \geq 1).$$

Then  $\kappa_i < \infty$ , and

$$\kappa_i = 1 + O(W(i)/i). \quad (46.9)$$

If a sequence  $\xi_i$  satisfies  $\xi_i = O(W(i)/i)$ , then

$$B_N := \sum_{i=1}^N \xi_i \sum_{j > N-i} k_{i+j}(j) = O(W(N)^2/N). \quad (46.10)$$

*Proof.* First consider the derivative in the Poisson parameter. For  $j \geq 1$ ,

$$\frac{\partial}{\partial w} q_w(j) = p_w(j-1),$$

whereas  $q_w(0) = 1$ . Consequently, for every  $j \geq 0$ ,

$$\left| \frac{\partial}{\partial w} k_w(j) \right| \leq \frac{2}{1+w}. \quad (46.11)$$

Fix large  $i$ , write  $w = W(i)$ , and set  $J = \lceil 16w \rceil$ . For  $0 \leq j \leq J$ , the inequality  $W'(x) \leq 1/x$  gives

$$0 \leq W(i+j) - W(i) \leq j/i.$$

By (46.11),

$$\sum_{j=0}^J |k_{i+j}(j) - k_i(j)| \leq \frac{2}{i(1+w)} \sum_{j=0}^J j = O(w/i). \quad (46.12)$$

The tails are much smaller. For  $J < j \leq i$ , stochastic monotonicity of Poisson laws and  $W(i+j) \leq W(2i) \leq w + \log 2$  give

$$k_{i+j}(j) \leq \frac{q_{w+\log 2}(j)}{1+w}.$$

For a Poisson variable  $M$  of mean  $\lambda = w + \log 2$ ,

$$\sum_{j > J} q_\lambda(j) = \mathbb{E}(M - J)_+ \leq \lambda \mathbb{P}(M \geq J).$$

The usual exponential Markov bound at  $J \sim 16w$  shows that, after division by  $1 + w$ , this is  $O(i^{-10})$ . The same estimate applies to  $\sum_{j>J} k_i(j)$ .

For  $j > i$ ,  $W(i + j) \leq \log(2j)$  for sufficiently large  $i$ . The Poisson Chernoff inequality gives

$$q_{W(i+j)}(j) \leq \left( \frac{e \log(2j)}{j} \right)^j \leq 2^{-j},$$

uniformly for such  $j$  and all sufficiently large  $i$ . Thus  $\sum_{j>i} k_{i+j}(j) = O(2^{-i})$ . The same bound for each fixed  $i$ , after ignoring a finite prefix, proves  $\kappa_i < \infty$ . Combining these tail estimates with (46.12) and  $\sum_j k_i(j) = 1$  proves (46.9).

For (46.10), set  $J = \lceil 16W(N) \rceil$ . For indices  $N - J < i \leq N$ , (46.9) gives  $\kappa_i \leq 2$  for large  $N$ . Their contribution in absolute value is at most

$$2J \max_{N-J < i \leq N} |\xi_i| = O(W(N)^2/N).$$

For the remaining indices  $i \leq N - J$ , rearrange the nonnegative upper bound using  $n = i + j > N$ . The sequence  $\xi_i$  is bounded, so it suffices to bound

$$\sum_{n>N} \sum_{i \leq N-J} k_n(n-i).$$

For  $N < n \leq 2N$ , every index  $n - i$  is at least  $J$ , and  $W(n) \leq W(N) + \log 2$ . The same Poisson-tail argument gives

$$\sum_{i \leq N-J} k_n(n-i) = O(N^{-10})$$

uniformly. Summing over these  $n$  gives  $O(N^{-9})$ . For  $n > 2N$ , all indices satisfy  $n - i \geq n/2$ . Since  $W(n) \leq \log n$ ,

$$k_n(n-i) \leq \left( \frac{2e \log n}{n} \right)^{n/2} \leq 2^{-n/2}$$

for large  $N$ . The sum over at most  $N$  indices  $i$ , then over  $n > 2N$ , is exponentially small. This proves (46.10).  $\square$

#### 46.4 Completion of the convergence proof

Since

$$y_n - y_{n-1-m} = \sum_{j=0}^m \xi_{n-j},$$

the Poisson equation (46.4) is exactly

$$\Xi_n = \sum_{j \geq 0} q_{W(n)}(j) \xi_{n-j}.$$

All nonzero  $\xi_{n-j}$  in this identity have  $j < n$ . Define

$$\bar{\xi}_n = \frac{\Xi_n}{1 + W(n)} = \sum_{j \geq 0} k_n(j) \xi_{n-j}.$$

Equation (46.4) implies

$$\bar{\xi}_n = O(W(n)^5/n^2), \quad \sum_{n \geq 1} |\bar{\xi}_n| < \infty. \quad (46.13)$$

Summing the finite triangular sums for  $1 \leq n \leq N$  gives the exact identity

$$\sum_{n=1}^N \bar{\xi}_n = \sum_{i=1}^N \xi_i \sum_{j=0}^{N-i} k_{i+j}(j) = \sum_{i=1}^N \kappa_i \xi_i - B_N. \quad (46.14)$$

By (46.3) and (46.9),

$$\sum_{i \geq 1} |(\kappa_i - 1)\xi_i| < \infty, \quad \sum_{i > N} |(\kappa_i - 1)\xi_i| = O(W(N)^2/N). \quad (46.15)$$

The boundary  $B_N$  tends to zero by (46.10). Equations (46.13)–(46.15) show that  $\sum_{i=1}^N \xi_i$  converges. Since  $y_N = y_0 + \sum_{i=1}^N \xi_i$ , there is a finite real  $\ell$  such that  $y_N \rightarrow \ell$ .

More precisely,

$$\ell = y_0 + \sum_{n \geq 1} \bar{\xi}_n - \sum_{i \geq 1} (\kappa_i - 1)\xi_i,$$

and both displayed series converge absolutely. The tails in (46.13) and (46.15), together with (46.10), give

$$y_N = \ell + O(W(N)^5/N).$$

Exponentiation proves

$$a_n = e^\ell \mathcal{N}_n \left( 1 + O(W(n)^5/n) \right).$$

Thus  $C_* = e^\ell$  is finite and strictly positive. The classical Bell saddle equivalent completes the claimed Bell-number formulation.

**Logical dependency.** The proof uses the inherited root-growth, monotone-ratio, and logarithmic-window coefficient theorems; the new complementary-endpoint estimate; explicit differentiation of the proposed normalizer; positive barriers; and the elementary column-stability argument. Neither Part III's unproved local-curvature estimate (31.1) nor its sufficient tilted-variance estimate (31.4) enters the argument. The existence portion of the finer Bell comparison is thereby proved by a different route.

## 47 The companion sequence A088713 has the same constant

Let

$$C(z) = \frac{1}{1 - zA(z)} = \sum_{n \geq 0} c_n z^n.$$

Its coefficients are OEIS A088713 [5]:

$$1, 1, 2, 6, 24, 118, 674, \dots$$

Part II proves only  $\log c_n = \log a_{n-1} + O(\log n)$  (Proposition 20.2), and explicitly leaves  $c_n \sim a_{n-1}$  unproved (Research question 23.6). The same decomposition by a unique large inner index proves this equivalent and a full fixed-order endpoint expansion.

**A second route.** The fixed-order expansion in the first display of Theorem 47.1 below is Theorem 39.1 of Part IV for this pair (the coefficients  $\gamma_h$  are those of (39.1)). Manuscripts 01 and 02 proved it independently. Part IV's theorem is more general: it holds for every log-convex sequence with  $a_n/a_{n-1} \sim n/\log n$ , and its consequence (39.11) gives the first correction  $2 \log n/n$ . The proof below uses the composition structure of A088714 and the bound  $c_h \leq a_h$ ; it is printed as a second proof of that expansion. The normalization of  $c_n$  with the constant  $C_*$  of Theorem 46.1, the last two displays of the theorem, is new in Part V.

Put

$$C(z)^2 = \sum_{h \geq 0} \gamma_h z^h.$$

The first endpoint coefficients are

$$\gamma_0 = 1, \quad \gamma_1 = 2, \quad \gamma_2 = 5, \quad \gamma_3 = 16, \quad \gamma_4 = 64, \quad \gamma_5 = 308.$$

**Theorem 47.1** (Companion endpoint expansion and normalization). *For every fixed nonnegative integer  $P$ ,*

$$c_n = \sum_{h=0}^P \gamma_h a_{n-1-h} + O_P(a_{n-P-2}).$$

In particular,

$$c_n = a_{n-1} + 2a_{n-2} + 5a_{n-3} + O(a_{n-4}), \quad c_n \sim a_{n-1}.$$

With the same constant  $C_*$  as in the A088714 normalization theorem,

$$c_n = C_* \frac{W(n)}{n} \mathcal{N}_n \left( 1 + O\left( \frac{W(n)^5}{n} \right) \right).$$

Consequently

$$c_n \sim C_* \frac{W(n)}{n} B_n \exp(W(n)^2 + 3W(n)).$$

*Proof.* The formal renewal identity gives

$$c_n = \sum_{k=1}^n [z^{n-k}] A(z)^k.$$

Again write  $N = \lfloor n/2 \rfloor$ ,  $H = n - N - 2$ , and  $A_N(z) = \sum_{j=0}^N a_j z^j$ . Configurations in which every selected inner index is at most  $N$  contribute

$$E_n = [z^n] \frac{z A_N(z)}{1 - z A_N(z)}.$$

At  $z_n = 4/R_n$ , the finite-polynomial estimate gives  $A_N(z_n) = 1 + z_n + O(z_n^2)$ , so  $z_n A_N(z_n) < 1$ . Nonnegative coefficients therefore yield

$$E_n \leq z_n^{-n} \frac{z_n A_N(z_n)}{1 - z_n A_N(z_n)} = O(z_n^{-n+1}).$$

The inherited root growth implies  $E_n = O(a_{n-1} e^{-\eta n})$  for some fixed  $\eta > 0$ . Only the rational function built from the finite polynomial  $A_N$  has been evaluated; no convergence of  $A$  or  $C$  is assumed.

Any other configuration has one and only one inner index greater than  $N$ , since the sum of all inner indices is at most  $n - 1 < 2(N + 1)$ . Write its value as  $n - 1 - h$ , where  $0 \leq h \leq H$ . If the original product had  $k$  factors, the remaining  $k - 1$  inner indices have total degree  $h - (k - 1)$ , and there are  $k$  possible locations for the large index. Thus these configurations contribute exactly

$$\sum_{h=0}^H a_{n-1-h} \sum_{k=1}^{h+1} k[z^{h-k+1}]A(z)^{k-1}.$$

The inner coefficient equals  $\gamma_h$ , because

$$\sum_{k \geq 1} k(zA(z))^{k-1} = \frac{1}{(1 - zA(z))^2} = C(z)^2.$$

Hence

$$c_n = E_n + \sum_{h=0}^H \gamma_h a_{n-1-h}. \quad (47.1)$$

For the tail estimate, first note that the original functional equation is equivalent to  $A(z) = C(zA(z))$ . Since every coefficient is nonnegative and  $A(0) = 1$ , this implies coefficientwise  $C(z) \leq A(z)$ , or  $c_h \leq a_h$ . Log-convexity and  $a_0 = 1$  give  $a_i a_j \leq a_{i+j}$ . Therefore

$$\gamma_h = \sum_{i=0}^h c_i c_{h-i} \leq \sum_{i=0}^h a_i a_{h-i} \leq (h+1)a_h.$$

For fixed  $p$ , the same geometric-ratio argument as in the weighted endpoint-tail lemma gives

$$\sum_{h=p}^H (h+1)a_h a_{n-1-h} = O_p(a_{n-1-p}).$$

Indeed the consecutive product ratio is  $R_{h+1}/R_{n-1-h}$ ; it is at most  $3/4$  up to  $n/3$ , for large  $n$ , and at most  $1$  for the rest of the range. Taking  $p = P + 1$  in (47.1) proves the fixed-order expansion.

The first two terms give  $c_n = a_{n-1}(1 + O(W(n)/n))$ . The A088714 theorem gives  $a_{n-1} = C_* \mathcal{N}_{n-1}(1 + O(W(n)^5/n))$ . Finally the explicit model ratio satisfies

$$\mathcal{N}_{n-1} = \frac{W(n)}{n} \mathcal{N}_n(1 + O(W(n)/n)).$$

Combining these three relations proves the companion normalization with the same constant  $C_*$ .  $\square$

## 48 What has changed mathematically, and the computations of Part V

The source boundary and the new conclusions for A088714 and A088713 are summarized in Table 7. (The manuscript's table also has two rows on A321941, printed with its part on that sequence.) In this problem an  $n$ -th-root asymptotic leaves room for multiplicative factors whose logarithms diverge; the new proofs supply exactly the missing global control.

For A088714 the important distinction is between an explicit *comparison* sequence and the actual sequence. Differentiating  $\log \mathcal{N}(x)$  is legitimate and gives a smooth model for the local weights. The actual ratios are initially used only through monotonicity and their proved equivalent.

Table 7: Earlier results of Parts I–III and the conclusions of Part V.

| Topic            | Inspected earlier boundary                                                                           | Part V                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| A088714 endpoint | The local coefficient expansion is proved; an upper bound for the complementary endpoint is missing. | Every fixed-order endpoint expansion, beginning $2a_{n-1} + 5a_{n-2} + 24a_{n-3}$ .       |
| A088714 scale    | $a_n^{1/n} \sim n/(e \log n)$ and $R_n \sim n/\log n$ ; full Bell normalization is conjectural.      | $a_n = C_* \mathcal{N}_n(1 + O(W(n)^5/n))$ , with $C_*$ finite and positive.              |
| A088713 scale    | $\log c_n = \log a_{n-1} + O(\log n)$ ; the equivalent remains open.                                 | $c_n \sim a_{n-1}$ , a fixed-order expansion, and the same normalization constant $C_*$ . |

Positive barriers first bound the normalized sequence; a separate bounded-additive-error argument then makes linearization valid. The column-sum identity converts a summable averaging defect into convergence. This separation avoids assuming the sought local regularity in order to prove the normalization.

[Batch 86, 4 October 2026. The A088713 row overlaps Part IV: the fixed-order expansion and  $c_n \sim a_{n-1}$  are also Theorem 39.1 and (39.11), proved independently (Section 47); the normalization of  $c_n$  with  $C_*$  is Part V’s alone.]

#### 48.1 What is checked exactly

The A088714 verification programs use the Python standard library. The included checks, in `code/05-bell-norm-verify_complement.py` with its recorded output `data/05-bell-norm-endpoint_verification.json`, cover:

1. The exact A088714 recurrence, endpoint-coefficient identities, and 209 three-region decompositions through  $n = 32$ .
2. The companion prefix, its endpoint coefficients, and 27 exact companion decompositions.

(The manuscript’s list begins with four items on A321941, checked by the programs shipped with its part on that sequence; the shared driver `code/05-bell-norm-run_checks.py` runs all of them, see the README.) The tests are finite certificates and checks of algebraic identifications. They do not establish the infinite asymptotic remainder bounds; those are proved in Sections 44 to 47.

#### 48.2 Finite normalization diagnostics

The A088714 coefficients through 600 were generated as exact integers. The normalization table uses Decimal arithmetic at precisions 80 and 120, each with 20 guard digits. The displayed values agree at the two precisions. The files are:

- the program: `code/05-bell-norm-normalization_diagnostics.py`;
- the coefficients: `data/05-bell-norm-coefficients_600.txt`;
- the run at 80 digits: `data/05-bell-norm-normalization_80.json`;
- the run at 120 digits: `data/05-bell-norm-normalization_120.json`.

The Lambert equation  $we^w = n$  is solved by Newton iteration, and logarithms are taken only after the integer coefficients have been computed.

The normalization theorem proves that the normalized sequence has a positive finite limit. It does not make the last finite value in Table 8 a numerical approximation with a certified error bound for that limit. The constants hidden in the asymptotic error have not been made explicit

Table 8: Finite diagnostics for the two Bell-normalized sequences. Decimals are rounded; they are not certified intervals for  $C_*$ .

| $n$ | $a_n/\mathcal{N}_n$ | $R_n - n/W(n)$ | $c_n/a_{n-1}$ |
|-----|---------------------|----------------|---------------|
| 25  | 0.032465579673      | 2.539032789    | 1.199848754   |
| 50  | 0.038089854602      | 2.312570363    | 1.118554273   |
| 100 | 0.041763346109      | 2.205967626    | 1.069134920   |
| 200 | 0.044282054708      | 2.146342106    | 1.039768649   |
| 400 | 0.045986198156      | 2.110948266    | 1.022599759   |
| 600 | 0.046702103047      | 2.096967019    | 1.016155103   |

here. Likewise, the apparent approach of  $R_n - n/W(n)$  to two is consistent with the remaining pointwise-ratio conjecture; the proved statement at that precision is only  $O(1)$ .

[Batch 86, 4 October 2026. The column  $a_n/\mathcal{N}_n$  is  $\exp D_n^N$  in the notation of Part III: at all six displayed indices it equals the exponential of the column `log_smooth_normalization` of `data/03-local-estimates-diagnostics.csv` to every printed digit, and the coefficients agree with `data/02-golden-moments-coefficients.csv` for  $n \leq 401$ . The column  $R_n - n/W(n)$  is Part I's diagnostic of Table 1. Part III's table reaches  $a_n/\mathcal{N}_n = 0.04785$  at  $n = 1600$ ; none of these values is an estimate of  $C_*$  with an error bound.]

## 49 A proposed further-research program for Part V

The following questions are deliberately more precise than a request for additional numerical terms. Each identifies a mathematical target that is not established by the present proofs. (The manuscript's questions on A321941, its Research questions 15.6–15.8, are printed with its part on that sequence.)

**Research question 49.1** (An effective value of  $C_*$ ). Can the implicit constants and starting indices in the normalization proof be made explicit enough to give a certified numerical interval for  $C_*$ ?

The convergence proof already represents  $\log C_*$  as the difference of two absolutely convergent auxiliary series. An effective version would bound their tails, together with the column boundary term, using computable constants. The finite values in Table 8 can guide acceleration, but interval arithmetic and explicit inequalities are needed for a certified answer. An expression in familiar constants is a separate question; numerical recognition alone would not settle it.

**Research question 49.2** (The next term of the actual ratio). Does the expansion proposed in Part I, (8.7), hold in the form

$$R_n = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1 + W(n))^2} + O\left(\frac{W(n)^p}{n}\right)$$

for some fixed  $p$ , and what is the sharp exponent or leading remainder?

We proved this shape for the explicit model ratio  $s_n$ , but only  $R_n = n/W(n) + O(1)$  for the actual sequence. The full normalization does not justify differentiating its unknown error term. A quantitative smoothing argument for the normalized recurrence, or control of tilted moment variances, could supply the missing pointwise information.

**Research question 49.3** (Higher-order Bell corrections). Is there a complete expansion of

$$\log \frac{a_n}{C_* \mathcal{N}_n}$$

in powers of  $1/n$ , with coefficients that are explicit functions of  $W(n)$ ?

The fixed-order endpoint expansion supplies every required large-inner-index coefficient. The local coefficient extraction can also be continued to higher orders. The remaining task is to propagate these expansions through the averaging equation with a uniform error. The present  $O(W(n)^5/n)$  bound is sufficient for convergence and is not claimed to be optimal.

**Research question 49.4** (A general normalization theorem for growing lags). Which slowly varying lag laws admit a column-sum argument of the kind used in Section 46?

For a general triangular averaging kernel  $k_n(j)$ , the exact summation identity suggests three quantities to control: the completed column sums, their errors weighted by the increments of the unknown sequence, and the boundary left by truncating the triangular sum. A useful theorem would translate regularity of the lag law and its mean into checkable bounds for these quantities. Poisson laws with logarithmic mean are one tractable instance. This could apply to other self-compositional generating functions without a separate local-curvature theorem for each sequence.

**Research question 49.5** (Endpoint expansions of growing order). How far may the truncation order  $P$  grow with  $n$  in the endpoint formulas for A088714 and A088713?

The current theorems hold for every *fixed*  $P$ . Allowing  $P = P(n) \rightarrow \infty$  requires estimates uniform in the rapidly growing endpoint coefficients  $\delta_h$  and  $\gamma_h$ . One aim is an optimally truncated expansion with a quantified small remainder. Another is a limiting description of the defect left outside the unique large inner index. Substituting a positive number into the divergent endpoint series would not answer either question.

**Research question 49.6** (A formal proof with a small trusted certificate). Which portions can be formalized first while keeping the analytic dependency chain explicit?

For Part V, the coefficientwise majorants and the positive-kernel summation identity are good initial targets (the manuscript names also the finite rational certificate and the exact spectral bridge of its part on A321941). A useful staged formalization would state the analytic inputs as explicit theorems, verify every finite truncation algebraically, and then discharge the limiting estimates. This would also make the distinction between formal series identities and convergent analytic identities mechanically visible.

*Appendix A belongs to Part II; it follows Parts III–V only because the appendix closes the report.*

## A Dependency and novelty audit

The main logical dependencies are as follows. The formal equation and the two compact positive-operator constructions prove existence of a limiting moment measure. The inherited growth bound supplies exponential integrability and determinacy, and also rules out bounded support. Moment positivity proves the strict Hankel and log-convexity statements. Weak convergence of compact approximants proves the real functional equation. Its inverse form, independently of any ratio asymptotics, proves the golden-ratio transform law. The self-contained Tauberian lemma then proves the small- $t$  distribution-function laws. Separately, log-convexity plus the inherited growth bound proves the consecutive ratios, and these prove the far tails by size-biasing.

This separation avoids several invalid shortcuts: no divergent ordinary series is substituted at a positive real argument; no neighboring ratio is deduced just from an  $n$ th-root equivalent; no infinite zero or positivity statement is inferred from finite determinants; no cumulative mass asymptotic is differentiated to invent a density; and no numerical agreement is used to exchange the compact-approximation and large-argument limits.

The repository snapshot inspected for the source article (Part I) was commit

`1f1981f682b2878bde51a6ad40c22777f362fc05`

The selected prior TeX file has Git blob identifier

`ccb50ef335ac4de27bd98a5a152fb48c978c7a96`

and is located under

`SetTheory/Cardinals/docs/reports/  
generating-functions-and-asymptotics/  
oeis-sequence-asymptotics/a088714-bell-scale-growth/`

The paper’s own source and tests are independent deliverables, not changes to that repository. (Placement note, 2026-10-01: they now form Part II of that report, beside the unchanged Part I; see Section 11.) The accompanying source audit, shipped as `02-golden-moments-SOURCE_AUDIT.md`, records the scope of the literature check and the distinction between prior general machinery and the sequence-specific deductions made here.

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